Credit, Fundamentals, and the Price–Rent Disconnect in Australian Housing Markets: Evidence from Five Capital Cities, 2006–2025*

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Abstract

Why have dwelling prices and rents in Australia's largest cities come apart? We model a city housing market in which collateral-constrained investors determine benchmark valuations, transaction prices adjust gradually, and rents respond to renter incomes, lease repricing and tenure switching. Using quarterly data for five capital cities over 2006–2025, we find that rents track local fundamentals, while investor credit co-moves strongly with prices within the quarter and hardly with CPI rents. The dynamic rent equation implies that about a third of the gap to market rents closes each quarter, with long-run pass-through of wage and employment growth jointly close to one. Panel local projections show prices moving first and CPI rent responses building after about a year, including when the current price change is held fixed. New-lease rents respond earlier and approach the price response within two years. Matching the model's impulse responses to these projections, we estimate that transaction prices close about a quarter of the gap to benchmark valuation each quarter. Instantaneous adjustment fits far worse, and gradual adjustment remains supported under alternative investor purchase-volume dynamics. In the model, lease repricing and tenure switching both delay rents, but their relative contributions remain uncertain. A simulated four-year credit tightening lowers prices within a year and rents over about three years; CPI rent inflation lags house-price inflation by about a year. The predicted amplification of credit effects at low interest rates fits the data only with sufficiently impatient investors.

Keywords: Price–rent disconnect, Credit conditions, Collateral constraints, Staggered leases, Local projections, Impulse-response matching

1. Introduction

Dwelling prices and rents in Australia's major cities have moved apart. From the early 2010s to 2025, the ratio of dwelling prices to rents rose in all five of the largest capitals, peaking at close to twice its 2011–12 level in Sydney (Figure 3). Rental affordability was also a significant concern. In 2019–20, 51 percent of lower-income renter households in New South Wales spent more than 30 percent of their gross income on housing costs, the highest share of any state or territory (Australian Bureau of Statistics, 2022).

In the user-cost framework, the price of a dwelling is the present value of the rents it earns, and the cost of capital links the two (Poterba, 1984). A common explanation for their divergence is that credit expansion and tax incentives such as negative gearing raised the demand for housing as an asset (Hulse et al., 2020; Stephens, 2025). That account explains why prices might rise. It

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does not explain why rents lag behind prices, or why rents keep rising even as they come apart from prices. The question is one of transmission: how does a change in credit conditions reach the asset price and the rent, and how fast? Macro-housing models with collateral constraints make asset prices sensitive to credit (Iacoviello, 2005; Kaplan et al., 2020; Kiyotaki and Moore, 1997), and Greenwald and Guren (2025) show that credit moves the price–rent ratio when the rental and owner-occupied markets are segmented. Little is known about the dynamics of the two responses: whether rents respond at all, with what delay, and through which frictions.

We study these dynamics in Australia's five largest capital cities. They share a national policy framework, including monetary policy, negative gearing, and migration settings, but differ in investor concentration, geographic constraints, and exposure to migration. National financial conditions therefore reach all five cities at once, while the local fundamentals that rents track follow each city's own cycle. This makes the five capitals a natural panel for asking how credit, and local fundamentals, reach prices and rents over time within each city.

The setting has a further advantage: national lending rules changed on known dates. In December 2014 the Australian Prudential Regulation Authority (APRA) asked banks to keep growth in investor lending below 10 percent a year, and in March 2017 it capped interest-only loans at 30 percent of new mortgage lending. NSW announced a housing affordability package in June 2017 (NSW Government, 2017). These dates give breaks around which to ask whether the drivers of prices and rents changed, and the APRA measures also serve as instruments for investor credit. We treat the dates as markers of the period, not as the start of a shift (Section 4.1).

Research questions and approach. We ask a structural question, not a policy-evaluation one: what drives rents and what drives prices in these five cities, and did those drivers change over the sample? We ask three questions. First, do the same forces govern rents and dwelling prices, or different ones? Second, did the drivers of either shift around the mid-2010s, when national lending rules changed and the 2017 NSW package arrived? Third, if a disconnect opened between prices and rents, is that because the two respond to different drivers, because they adjust at different speeds, or both?

We organise the analysis around a semi-structural dynamic model of a city housing market (Section 3). A collateral-constrained investor provides the benchmark valuation of dwellings, so looser credit raises that valuation at once. Transaction prices follow over several quarters, through an adjustment rule motivated by search and gradual seller repricing. Rents are set in a separate market for housing services, where the market rent rises with renter incomes and with the number of renter households per rental dwelling. Credit reaches this market only indirectly. Higher prices push some would-be buyers into renting, measured rents reprice only as leases roll over, and new dwellings arrive after a construction lag. The model yields testable predictions: rents load on income and market tightness; prices load on credit conditions and the cost of capital; the price loading on credit rises when interest rates are low; and a credit innovation moves prices on impact and rents only after a delay.

Our strategy puts the inferential weight where a five-city quarterly panel can bear it: on variation over time within each city, 74 quarters per city, rather than on the thin five-city cross-section. We estimate two reduced-form equations. A rent equation relates rent growth to market tightness (net migration relative to dwelling completions) and to wage growth and employment growth. A price equation relates price growth to the cost of capital (the cash rate), to investor credit, and to the same market tightness. We trace the timing with panel local projections of prices and rents on an investor-credit innovation, and we check that timing by instrumenting credit with two measures by which APRA restrained investor lending in 2015–2018. Finally, we estimate the model's parameters by matching its impulse responses to the local projections, decompose the delay in rents into the contributions of each friction, and simulate a credit tightening.

Contributions. This paper makes four contributions. First, we develop a tractable model that explains why prices and rents respond to credit at different speeds. Standard user-cost reasoning ties rents to prices through the cost of capital. Our model instead links the asset market to a collateral-based benchmark valuation and the service market to rents that clear against renter incomes and the rental stock. It predicts the sign and timing of the two responses, and it shows why measured investor credit and prices move together within the quarter.

Second, we estimate the model by matching its impulse responses to panel local projections of prices, rents, and investor credit, carrying sampling uncertainty through a block bootstrap. The baseline model fits all three paths. Transaction prices close about 23 percent of the gap to benchmark valuation each quarter (90 percent bootstrap interval 12 to 38 percent), with a half-life of about two and a half quarters. Instantaneous adjustment fits substantially worse under the baseline commitments mapping. We test that mapping by allowing investor purchase volumes to absorb part of the credit response, with both proportional and separate-decay paths (Section 6.3). Across the separate-decay scenarios, the estimated speed ranges from 21 to 27 percent a quarter, and instantaneous adjustment fits worse in every case. This separates sampling precision within the baseline from sensitivity to the interpretation of measured commitments.

On the rent side, the early response of rents on new leases, set out below, supports a central role for lease repricing in the measured delay. The baseline model also assigns a role to gradual tenure switching, but the data do not sharply distinguish its speed from its long-run size. We therefore use the friction decomposition to quantify the mechanisms within the model, without treating their baseline ranking as an empirical finding. An illustrative four-year tightening lowers prices within a year and rents over about three years. In this simulation CPI rent inflation lags house-price inflation by about a year. The model's prediction of stronger credit effects at low interest rates fits the data only with sufficiently impatient investors.

Third, we show that different forces govern rents and dwelling prices in these cities, as the model predicts. Rents track local fundamentals (market tightness, wage growth, and employment growth) throughout the sample and city by city. Estimating the rent equation in the dynamic form that staggered leases imply, we find that measured rents close about a third of the gap to market rents each quarter. Rents on new leases, which we observe for four of the cities, follow the same fundamentals without the persistence of CPI rents, and CPI rents follow them with a lag of two to five quarters. In the long run, wage and employment growth together pass through to rents almost one for one, as the model predicts, although wage growth alone passes through only about half. Investor credit co-moves far more tightly with prices than with rents, before and after 2017, and lagged credit predicts price growth once price momentum is controlled for. The cost of capital carries its expected sign only when the cash rate is lagged one quarter, and even then its estimate is not significant.

Fourth, we characterise the timing of the contrast with panel local projections. Prices take up most of their response to a credit innovation within three quarters. Rents respond little in the first year and build up gradually after that, so the gap between the two responses narrows over three years. Rents on new leases respond from the first quarter and approach the price response within two years, so credit reaches the rents that new tenants pay well before it shows in CPI rents. The pattern holds with controls for wage and employment growth, and when the current price change is held fixed. Instrumenting credit with the two APRA measures, and testing them with the correlation-restriction test of Dzhumashev and Tursunalieva (2026), supports the delay in rents but does not settle how fast credit reaches prices. The disconnect is therefore a difference in transmission speed and a medium-run wedge, not a permanent insulation of rents from financial conditions.

The paper proceeds as follows. Section 2 reviews the literature, Section 3 sets out the model, and Section 4 describes the data and methods. Section 5 reports the reduced-form evidence that the model must match, and Section 6 estimates the model, discusses what identifies its parameters, and uses it to decompose the delay in rents, to simulate a credit tightening, and to trace the implications for measured inflation. Section 7 discusses the implications, and Section 8

concludes. An online appendix reports the slope and break tests, the instrumented projections, further robustness checks, and the tests with rents on new leases.

2. Literature and Theoretical Background

The paper sits at the intersection of housing economics and macroeconomics. Housing economics treats a dwelling as both a consumption good, whose rent is the price of housing services, and an asset, whose price is the present value of those services and of expected capital gains (Poterba, 1984). In the user-cost framework, the cost of capital links the two, so prices and rents move together when both respond to broad housing demand and can diverge when the asset side loads on financial conditions that the service side does not. Applied work uses this relation to ask whether prices are justified by rents and financing costs (Himmelberg et al., 2005), and in Australia to assess whether dwellings are overvalued and how prices respond to interest rates and supply (Fox and Tulip, 2014; Saunders and Tulip, 2019). The cointegration literature treats rents and prices as possibly $I(1)$ series whose long-run relationship can weaken (Gallin, 2008; Kenny, 1999). Macroeconomics supplies the other half: models in which credit conditions move asset prices through collateral constraints, and methods for tracing how a shock propagates over time. We combine the two. From housing economics we take the split between the asset market and the market for housing services, the frictions specific to rental markets, and the local fundamentals that drive rents. From macroeconomics we take the collateral channel, impulse responses estimated by local projections, and estimation by matching a model's impulse responses to them. The paragraphs below review four literatures in turn: credit and house prices, the dynamics of the price–rent ratio, rental-market frictions and housing demand, and the regional transmission of national policy.

Our model draws on macro-housing work in which borrowing limits tied to asset values make asset prices sensitive to credit conditions (Iacoviello, 2005; Kiyotaki and Moore, 1997). Favilukis et al. (2017) and Kaplan et al. (2020) study how credit conditions and expectations move house prices and the price–rent ratio in general equilibrium, and Favara and Imbs (2015) show that exogenous credit expansions raise house prices. Closest to our question, Greenwald and Guren (2025) show that credit conditions move the price–rent ratio when the rental and owner-occupied markets are segmented, and that the strength of this effect depends on how easily dwellings move between the two markets. Our model shares the emphasis on segmentation, but it works at quarterly frequency and adds the frictions that govern how fast credit reaches rents. We keep the collateral channel and pair it with a deliberately simple rental market, so that the model maps directly into the equations and impulse responses we estimate.

A second line of work asks what moves the price–rent ratio over time. Kishor and Morley (2015) decompose the ratio in US metropolitan areas into expected rent growth, expected returns and a residual, and find the residual large where booms and busts are frequent and sensitive to mortgage rates. Granziera and Kozicki (2015) show that a rational-expectations asset-pricing model matches the average price–rent ratio but not its swings, which require extrapolative expectations, and Bolt et al. (2019) estimate a model of trend-following and mean-reverting beliefs on eight countries. Dieci and Westerhoff (2016) link prices, the housing stock and rents in a stock-flow model with such beliefs, and Tomura (2013) shows that optimism among credit-constrained borrowers can generate boom–bust cycles. These papers explain the swings in the ratio mainly through expectations. A related literature documents momentum in house prices (Case and Shiller, 1989; Glaeser et al., 2014) and explains it through search frictions and strategic complementarity in sellers' pricing (Guren, 2018; Head et al., 2014), while Fève and Moura (2024) show that momentum can also arise without frictions when the driver of value itself builds up gradually. Our model keeps expectations rational and traces the gap to the financing conditions of investors, to the gradual adjustment of transaction prices, and to slow frictions on the rental side. Because we measure the path of credit conditions directly, we can separate slow price adjustment from a slowly building driver.

On the rental side, two frictions delay the response of rents. Rents on existing leases adjust infrequently (Genesove, 2003; Shin, 2025), which we capture with staggered repricing in the spirit of Calvo (1983). New supply responds to prices through adjustment costs (Hayashi, 1982) and arrives only after a construction lag (Kydland and Prescott, 1982), and regulatory and geographic constraints shape how large that response is (Glaeser and Gyourko, 2003; Saiz, 2010). Recent work suggests that financial conditions also shape the supply response: Greenaway-McGrevy and Phillips (2023) study the construction response to Auckland's upzoning, and Egan and McQuinn (2023) identify regime shifts in which the usual relationship between supply and prices breaks down. Migration enters as a demand fundamental. Work on immigration and house prices is well developed (Sanchis-Guarner, 2023), but the effect on rents is less studied, although renters are over-represented among recent migrants (Nygaard, 2011). Sharpe (2019) attributes a substantial share of the variation in US rents to immigration. In Australia the evidence concerns prices: Moallemi and Melser (2020) find that an immigrant inflow of 1 percent of a postcode's population raises housing prices by about 0.9 percent a year, and Li et al. (2024) find that housing costs themselves shape internal migration. Through this sorting, rents feed back into the regional distribution of labour (Haas and Osland, 2014). We capture the demand side through market tightness, net migration relative to dwelling completions, rather than through migration alone.

National policy also reaches regional housing markets unevenly. Koeniger et al. (2022) and Calza et al. (2013) show that the housing effects of monetary policy depend on regional and institutional features of housing and mortgage markets, and in Australia Graham and Read (2023) find that interest-rate changes affect metropolitan and regional markets very differently. As Stephens (2025) notes, this literature has concentrated on homeownership and the mortgage channel. Recent work on macroprudential policy studies its effect on house price expectations (Kuang et al., 2026) and on the size of the house price cycle (Carro, 2023). The literature says relatively little about rental markets, even as rental housing accommodates a growing share of urban populations. Tracing how a common credit innovation reaches prices and rents is one way to fill that gap.

A large literature outside economics documents the transformation of housing into a financial asset and the resulting disconnect between prices and rents (Aalbers et al., 2021; Fernandez and Aalbers, 2016; Hulse et al., 2020). It establishes the phenomenon but usually leaves the disconnect as a documented correlation rather than a mechanism.

Our contribution brings these strands together. We model the asset and rental markets of a city, derive testable predictions, estimate the reduced forms on the time dimension within five cities, and then estimate the model's structural parameters by matching its impulse responses to the empirical ones. For housing economics, this turns the price–rent disconnect from a documented correlation into a dynamic mechanism with measured speeds of adjustment. For macroeconomics, it shows that the collateral channel reaches the price of housing services as well as the asset price, but slowly, through frictions in the rental market. We can therefore say not only that prices move first and rents follow, but how fast prices adjust and how much of the delay in rents each friction can account for.

3. A Dynamic Model of Prices and Rents

This section sets out a semi-structural model of one city's housing market in quarter t. It has three blocks: renter households who buy housing services, investors who own the rental stock and borrow against it, and builders who add to that stock. A collateral-constrained investor supplies a frictionless benchmark valuation, while observed transaction prices adjust gradually towards it. On the rental side, slow tenure switching and staggered lease repricing delay the transmission of price changes to measured rents. New housing supply responds to prices with a construction lag. We omit the city subscript j where this causes no confusion.

The model is partial equilibrium by design. Wages, employment, migration, lending limits, the policy rate, and therefore credit conditions are set outside the city housing market, which

suits a city that is small relative to the national economy and subject to national credit policy. Households do not solve a tenure-choice problem: the tenure response to prices is a reduced-form rule, governed by its long-run size and its speed. Both choices are deliberate. The question is how a given change in credit conditions reaches prices and rents, and over what horizon, not why credit conditions change or what they do to welfare. A partial-equilibrium model whose responses map directly into the local projections answers that question with parameters the data can estimate. Fully specified general-equilibrium models with tenure choice, such as Kaplan et al. (2020) and Greenwald and Guren (2025), endogenise credit and tenure, but they have many more parameters than a five-city quarterly panel can estimate. The cost of our choice is that the model cannot say how credit conditions themselves respond to house prices, and that its tenure parameters describe behaviour in the sample rather than deep preferences. Appendix A gives the derivations.

3.1. Renters and the market rent

A renter household chooses consumption c_t and housing services q_t to solve

$$\max_{c_t,\, q_t} \ln c_t + \gamma \ln q_t \quad \text{s.t.} \quad c_t + R_t^* q_t \le W_t e_t, \tag{1}$$

where R_t^* is the rent on a lease signed in quarter t, W_t is the wage, and e_t is the number of employed persons per household. Each household spends the share $\sigma \equiv \gamma/(1+\gamma)$ of its income on housing services. Let H_t^R be the number of renter households and K_t the rental stock, with one dwelling supplying one unit of services. Clearing the rental market, $H_t^R q_t = K_t$, gives the market rent

$$R_t^* = \sigma\, W_t e_t\, \frac{H_t^R}{K_t}. \tag{2}$$

The market rent rises with renter income and with the number of renter households per rental dwelling. Neither dwelling prices nor credit enter equation (2); they can move rents only by changing H_t^R or K_t.

The number of renter households changes through two channels:

$$\ln H_t^R - \ln H_{t-1}^R = \nu\, \frac{M_t}{H_{t-1}^R} + \Delta \hat{s}_t. \tag{3}$$

The first term is net migration M_t, with ν renter households formed per migrant; recent migrants are over-represented among renters (Nygaard, 2011). The second term is tenure switching. Writing $\hat{x}_t$ for the log deviation of a variable x_t from its steady-state path, the switching component follows

$$\hat{s}_t = (1-\kappa)\, \hat{s}_{t-1} + \kappa\, \eta\, (\hat{p}_{t-1} - \hat{w}_{t-1}), \tag{4}$$

where $p_t = \ln P_t$ is the log dwelling price and $w_t = \ln W_t$ the log wage. When prices rise relative to wages, more households who would have bought keep renting ($0 < \eta \le 1$). They act on last quarter's prices and adjust gradually ($0 < \kappa \le 1$), because saving a deposit, searching for a dwelling, and ending a lease all take time. We need $\eta \le 1$ because, with a larger η and no supply response, the model has no unique bounded solution (Section 6.1).

Measured rents differ from the market rent. The CPI rent series we use covers rents paid on all current leases (Section 4.1). Each quarter a fraction ω of leases reprices to the market rent and the rest keep their rent, in the spirit of Calvo (1983); rents on sitting tenancies are known to adjust infrequently (Genesove, 2003; Shin, 2025). The log measured rent r_t therefore follows

$$r_t = (1-\omega)\, r_{t-1} + \omega\, r_t^*, \qquad 0 < \omega \le 1, \tag{5}$$

with $r_t^* = \ln R_t^*$.

3.2. Investors and the collateral channel

We first derive a frictionless benchmark in which investors own the rental stock and trade at a common price V_t, which also determines collateral values. They are risk neutral with discount factor Λ, and $\Lambda(1+i_t)<1$, so they are impatient relative to the policy rate i_t and borrow as much as lenders allow. Their dividend in quarter t is

$$d_t = R_t K_t - V_t\left[K_{t+1}-(1-\delta)K_t\right] + B_t - (1+i_{t-1})B_{t-1},$$

where $R_t=\exp(r_t)$ is the rent collected on current leases, V_t the common trading price, δ the depreciation rate, and B_t debt. Lenders limit debt to a fraction of the value of the stock:

$$B_t \le \phi_t V_t K_{t+1}, \tag{6}$$

where $\phi_t\in(0,1)$ is the effective loan-to-value limit set by lending standards and macroprudential policy, such as APRA's investor-lending benchmarks. Investors maximise $\mathbb{E}_t\sum_{k\ge0}\Lambda^k d_{t+k}$ subject to (6). The first-order condition for debt gives the shadow value of the constraint,

$$\lambda_t = 1-\Lambda(1+i_t) > 0, \tag{7}$$

so the constraint binds. The first-order condition for the stock gives the asset-pricing equation

$$V_t\left(1-\lambda_t\phi_t\right) = \Lambda\,\mathbb{E}_t[R_{t+1}+(1-\delta)V_{t+1}]\,. \tag{8}$$

Each dwelling an investor buys also raises the amount the investor can borrow. The term $\lambda_t\phi_t$ is the value of that extra borrowing capacity, for which investors are willing to pay. Iterating (8) forward gives

$$V_t = \mathbb{E}_t\sum_{k=1}^{\infty}(1-\delta)^{k-1}\left(\prod_{m=0}^{k-1}\chi_{t+m}\right)R_{t+k}, \qquad \chi_t\equiv\frac{\Lambda}{1-\lambda_t\phi_t}, \tag{9}$$

so the price capitalises expected rents and the current and expected path of credit conditions. In this benchmark, investors are the marginal buyers and Equation (8) determines their common trading and collateral value. We use this valuation equation as the forward-looking block of the estimated system below.

Two features of this pricing block deserve comment. First, investors account for roughly a third of new housing commitments, and owner-occupiers also bid for dwellings. The empirical price equation supports focusing on investor finance: adding owner-occupier commitments leaves the investor-credit coefficient positive and significant, while the owner-occupier coefficient is small and insignificant (Online Appendix E). This motivates the benchmark's investor-finance channel. Second, investors discount at Λ rather than at the policy rate, so the policy rate affects prices only through the shadow value of the collateral constraint, λ_t. With $\phi_t=0$ the policy rate would have no effect on prices. This differs from the user-cost framework, in which the interest rate is the opportunity cost of holding housing (Poterba, 1984). We keep the simpler form because it isolates the collateral channel and delivers the state dependence in Proposition 3; adding a user-cost term would strengthen the effect of the policy rate on prices without changing the predictions for rents. The assumption $\Lambda(1+i_t)<1$ must hold throughout, so that the constraint binds.

The benchmark valuation V_t, which we call the asset value, capitalises expected rents and financing conditions. We distinguish it from the observed transaction price P_t, introduced below; in steady state the two coincide. We assume $(1-\delta)\Lambda<1-\bar\lambda\bar\phi$, so that $\varpi<1$ below and the present value in (9) is finite. Log-linearising the benchmark valuation equation around a steady state gives

$$\hat v_t = \varepsilon\left(\hat\phi_t+\hat\lambda_t\right)+(1-\varpi)\,\mathbb{E}_t\hat r_{t+1}+\varpi\,\mathbb{E}_t\hat v_{t+1}, \qquad \varepsilon\equiv\frac{\bar\lambda\bar\phi}{1-\bar\lambda\bar\phi}, \qquad \varpi\equiv(1-\delta)\bar\chi\in(0,1), \tag{10}$$

where $\hat{\lambda}_t = -(\Lambda/\bar{\lambda})(i_t - \bar{\imath})$ and bars denote steady-state values. The parameter ε is the elasticity of the asset value with respect to credit conditions, holding expected rents and values fixed. The weight on next quarter's rent, $1 - \varpi = \bar{\chi}\bar{R}/\bar{P}$, is small when the price–rent ratio is high.

Transaction prices do not jump to the asset value. Housing is traded in a search market: sellers set asking prices and revise them infrequently, and buyers take time to find a dwelling, so transaction prices adjust to news gradually and display momentum (Case and Shiller, 1989; Glaeser et al., 2014; Guren, 2018; Head et al., 2014). We capture this with a partial-adjustment rule that mirrors the staggered repricing of rents:

$$\hat{p}_t = (1 - \theta)\,\hat{p}_{t-1} + \theta\,\hat{v}_t, \qquad 0 < \theta \leq 1, \tag{11}$$

where θ is the share of the gap between the asset value and the transaction price that closes each quarter. With $\theta = 1$, transaction prices equal the asset value and capitalise the whole expected path of credit conditions at once.

Equation (11) is a semi-structural extension of the frictionless benchmark. Its parameter θ measures the speed at which observed prices absorb changes in the asset value. Search models motivate such gradual adjustment through marketing time, infrequent seller repricing and strategic complementarity (Guren, 2018; Head et al., 2014). We do not solve for transaction allocation or investor optimality away from the benchmark, and the investor's Euler equation is not imposed at P_t when $P_t \neq V_t$. Thus θ summarises adjustment in the trading process; it does not separately identify those underlying frictions. Builders and tenure decisions respond to P_t, while the valuation equation evaluates the expected rent path generated by these responses.

We hold θ fixed across the credit innovations in the sample. This provides a parsimonious measure of adjustment speed; Section 6.2 examines the implications of faster adjustment for the illustrative tightening. Observed commitments finance purchases at transaction prices, so their mapping uses P_t, as follows.

Investor credit in the data is the flow of new loan commitments. For the empirical commitments mapping, new loans finance investor purchases at an effective loan-to-value limit, so log investor credit is

$$\iota_t \equiv \ln(\phi_t P_t X_t) = \ln\phi_t + p_t + \ln X_t, \qquad \Delta\iota_t = \Delta\ln\phi_t + \Delta p_t + \Delta\ln X_t, \tag{12}$$

where X_t is the number of dwellings investors buy in quarter t. Measured credit growth thus combines the change in credit conditions with the current change in prices.

3.3. Builders and the construction lag

Builders take the price and the stock as given and choose housing investment I_t to maximise

$$P_t I_t - I_t - \frac{\alpha}{2}\left(\frac{I_t}{K_t} - \delta\right)^2 K_t,$$

where the replacement cost of a dwelling is normalised to one and $\alpha > 0$ scales adjustment costs (Hayashi, 1982). The first-order condition is

$$\frac{I_t}{K_t} = \delta + \frac{1}{\alpha}\,(P_t - 1). \tag{13}$$

Investment starts as soon as prices rise, but dwellings started in quarter t are completed only in quarter $t + J$, with $J \geq 1$ (Kydland and Prescott, 1982). A share ψ of completions C_t enters the rental stock:

$$C_{t+J} = I_t, \qquad K_t = (1 - \delta)K_{t-1} + \psi\,C_t. \tag{14}$$

3.4. Market tightness and rent growth

Combining (2), (3), and (14) links the growth of the market rent to the tightness measure we construct from the data, $\text{TIGHT}_t = M_t/C_t$.

Lemma 1. *Suppose each renter household occupies one rental dwelling, so that $K_{t-1} \approx H^R_{t-1}$, and the completion rate $g_t \equiv C_t/K_{t-1}$ moves slowly relative to tightness, so that $g_t \approx \bar{g}$. Then, to a first-order approximation,*

$$\Delta r^*_t \approx (\delta - \psi\bar{g}) + \nu\bar{g}\, TIGHT_t + \Delta w_t + \Delta \ln e_t + \Delta \hat{s}_t. \tag{15}$$

Rent growth thus depends on the level of tightness, not on its change. When migration is high relative to completions, renter households grow faster than the rental stock, and the market rent keeps rising for as long as that holds. This is why tightness enters the rent equation in levels, and why differencing it removes its explanatory power (Online Appendix E). The tightness coefficient, $\nu\bar{g}$, is the product of renter households formed per migrant and the quarterly completion rate, so the model predicts it to be small and positive. Without the assumption that the completion rate moves slowly, the tightness and supply terms in (15) read $g_t(\nu\,\text{TIGHT}_t - \psi)$: the effect of tightness then scales with the completion rate, which rose in the building boom of 2016–2018, so the coefficient we estimate is an average over the sample.

3.5. Predictions

The model yields four propositions and a corollary. Proofs are in Appendix A.

Proposition 1 (Rents and fundamentals)**.** *Measured rent growth satisfies*

$$\Delta r_t = \omega\left[\Delta r^*_t + (r^*_{t-1} - r_{t-1})\right], \tag{16}$$

with Δr^*_t given by (15)*. Hence: (i) rent growth rises with wage growth, with growth in employment per household, and with the level of tightness; (ii) given tightness, prices affect rents only through tenure switching, with at least a one-quarter lag, and credit has no direct effect; (iii) if Δr^*_t moves one for one with a single regressor x_t that follows a first-order autoregression with coefficient $\rho_x \in [0,1)$, the coefficient from regressing Δr_t on x_t converges to $\omega/[1-(1-\omega)\rho_x]$, which lies between ω and one and rises with ρ_x; (iv) equivalently,*

$$\Delta r_t = (1-\omega)\,\Delta r_{t-1} + \omega\,\Delta r^*_t, \tag{17}$$

so the coefficient on lagged rent growth identifies $1-\omega$, and the long-run coefficient of Δr_t on each driver of Δr^*_t equals that driver's coefficient in (15)*.*

Parts (iii) and (iv) matter for estimation. The gap $r^*_{t-1} - r_{t-1}$ is not observed. In a static regression it falls into the error term, and a persistent regressor picks up part of its effect, so the coefficient on a persistent driver such as wage growth is pushed towards one and the coefficient on a less persistent or noisier driver such as employment growth towards ω. Part (iv) removes the problem. Lagged rent growth summarises the unobserved gap, so the dynamic form estimates ω directly, and its long-run coefficients can be compared with the structural ones. In particular, (15) implies a long-run wage coefficient of one.

Proposition 2 (Prices and credit)**.** *Solving* (10) *forward,*

$$\hat{v}_t = \sum_{k=0}^{\infty} \varpi^k\, \mathbb{E}_t\Big[\varepsilon\,(\hat{\phi}_{t+k} + \hat{\lambda}_{t+k}) + (1-\varpi)\,\hat{r}_{t+k+1}\Big], \qquad \hat{p}_t = \theta \sum_{k=0}^{\infty} (1-\theta)^k\, \hat{v}_{t-k}. \tag{18}$$

Hence: (i) looser credit conditions raise the asset value in the same quarter, and the transaction price by the share θ of that rise, with the rest following in later quarters; (ii) a higher policy rate lowers the price, since $\partial\hat{\lambda}_t/\partial i_t = -\Lambda/\bar{\lambda} < 0$; (iii) rental fundamentals, including tightness and wages, affect prices only through expected rents, with a weight of $1-\varpi$ per quarter.

Corollary 1 (Measured credit)**.** *By* (12)*,* $\Delta\iota_t$ *contains* Δp_t*, so a regression of* Δp_t *on* $\Delta\iota_t$ *does not recover* ε*. Replacing* $\Delta\iota_t$ *with* $\Delta\iota_{t-1}$ *removes the current price change but not last quarter's,* Δp_{t-1}*, which predicts* Δp_t *when price growth is persistent. Neither slope recovers* ε*, and the model does not determine which of the two is larger.*

We therefore report both slopes and read them together. If both are positive and close, the price–credit association does not rest on the same-quarter term alone (Online Appendix C).

Proposition 3 (State dependence)**.** *The credit elasticity* $\varepsilon = \bar{\lambda}\bar{\phi}/(1-\bar{\lambda}\bar{\phi})$ *rises with the loan-to-value limit* $\bar{\phi}$ *and with the shadow value* $\bar{\lambda} = 1-\Lambda(1+\bar{\imath})$*, which falls as the policy rate rises. The price response to a given change in credit conditions is therefore larger when lending limits are loose and interest rates are low. Both are set nationally, although the effective limit can differ across cities with the mix of investors and interest-only loans.*

Proposition 3 gives a reason to expect the price loading on investor credit to change over time without any change in local conditions. For much of the period after 2017, policy rates were lower than in the preceding decade, and APRA lifted its investor-lending benchmark in 2018; both raise ε. APRA's cap on interest-only lending from 2017 and the 2022–23 tightening worked in the other direction, so the prediction concerns the level of interest rates and lending limits, not a break at 2017:Q1. Online Appendix E tests it with the level of the cash rate.

Proposition 4 (Timing)**.** *Let credit conditions loosen persistently at date* t *(*$\hat{\phi}_t > 0$*), with wages, employment, migration, and the policy rate unaffected, and let* $J \geq 2$*. Then: (i) the measured rent does not respond on impact,* $\hat{r}_t = 0$*, while the price rises,* $\hat{p}_t = \theta\hat{v}_t > 0$*, provided the direct collateral effect* $\varepsilon\sum_{k\geq 0}\varpi^k\mathbb{E}_t\hat{\phi}_{t+k}$ *exceeds any discounted fall in expected rents; if* $\theta < 1$*, the price keeps rising in the following quarter as long as* $\hat{v}_{t+1} > \theta\hat{v}_t$*, so a persistent loosening produces a price response that builds after impact; (ii) one quarter later,* $\hat{r}_{t+1} = \omega\kappa\eta\,\hat{p}_t$*, which is small when leases reprice slowly and tenure switching is gradual, and for* $1 \leq h \leq J-1$ *the rent response is positive as long as the price response stays positive; (iii) from* $h = J$*, completions of dwellings started after the shock add to the rental stock and pull the rent response below the path in (ii); if the loosening decays, both responses return to zero.*

With a positive supply response, the model does not imply that looser credit raises rents permanently. With a permanent loosening, building continues until the price returns to its steady-state level, which credit conditions do not affect. By (9), the higher χ then requires lower rents. In the twelve-quarter estimation, however, the effective supply response is estimated at zero, and the data provide little information about adjustment beyond that horizon. The estimated window therefore captures the rising phase of the rent response rather than its long-run equilibrium.

Proposition 4 describes the impulse responses that our local projections estimate. Three features of those projections need comment. First, the build-up of the price response depends on θ. With $\theta = 1$, the price capitalises the whole expected path of credit conditions and jumps on impact; with $\theta < 1$, it builds over several quarters. Section 6 estimates θ and tests $\theta = 1$.

Second, the projection inherits the same-quarter term in Corollary 1. A cumulative response $y_{t+h} - y_{t-1}$ contains the impact quarter at every horizon, so we also estimate post-impact responses $y_{t+h} - y_t$ that hold the current price change fixed.

Third, before new construction is completed, the model routes credit to rents through tenure switching. Once new supply arrives, the construction channel provides an offsetting response. The predictive tests in Section 5.3 examine whether lagged price growth captures the transmission from credit to rents.

Figure 1 summarises the channels. Credit conditions act directly on the asset value, and transaction prices follow it gradually. Prices reach rents through two slow routes that pull in opposite directions: tenure switching, which raises renter demand, and building, which eventually adds to the rental stock and so lowers rents. Migration and completions together determine tightness, which drives rents alongside renter incomes, and staggered leases delay the response of measured rents to all of these forces.

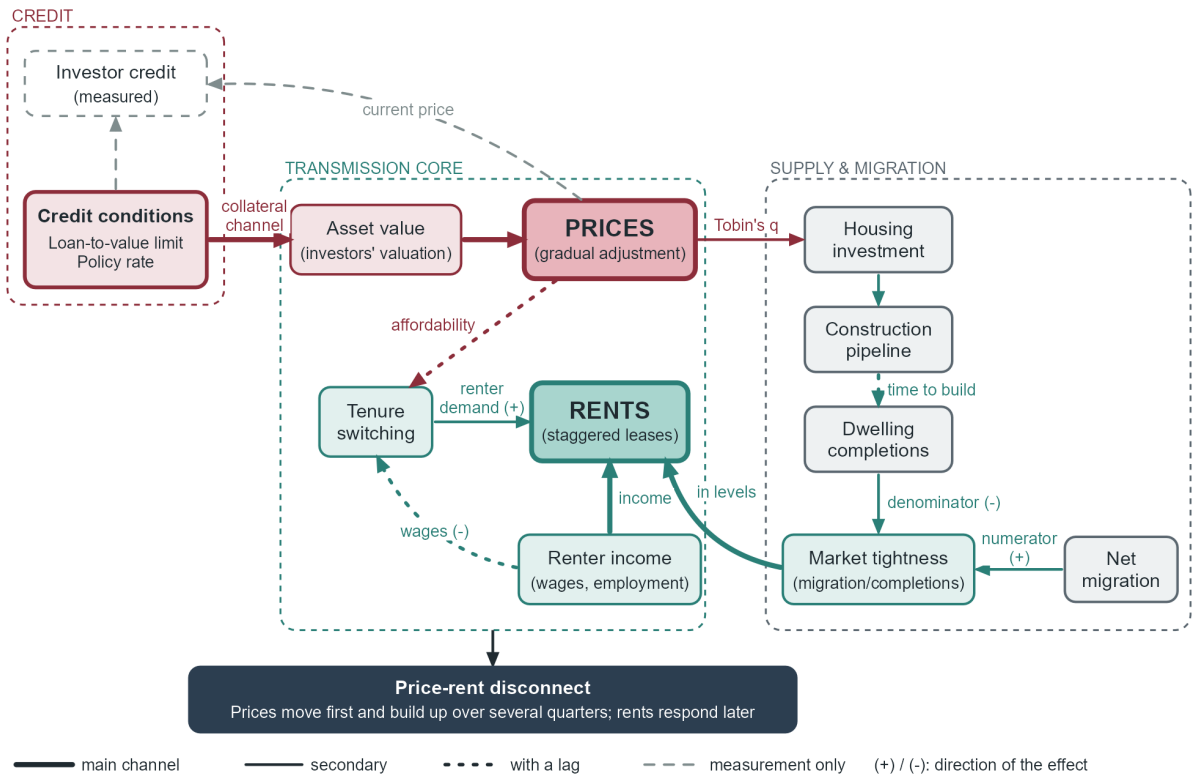


Figure 1: Transmission channels in the model. Credit conditions (the loan-to-value limit and the policy rate) act at once on the asset value through the collateral channel (Section 3.2). Transaction prices close only part of the gap to the asset value each quarter (θ in Equation (11)). Prices then reach rents along two slow routes that pull in opposite directions. Higher prices relative to wages keep would-be buyers renting (tenure switching), which raises renter demand. Higher prices also raise housing investment, which adds to completions after the construction lag; more completions lower market tightness and so rents. Tightness, net migration over completions, drives rent growth in levels alongside renter income. Measured rents reprice only as leases roll over, which delays the response of rents to every force. Measured investor credit reflects both credit conditions and the current price (Equation (12)). Expected rents also feed back into the asset value, with the small weight $1 - \varpi$; the figure omits this link. Thick lines are main channels and thin lines secondary ones, dotted lines act with a lag, and dashed grey lines are measurement only; the signs give the direction of each effect.

4. Data and Methods

4.1. Data sources and description

We use quarterly data for five Australian capital cities (Sydney, Melbourne, Brisbane, Adelaide, and Perth) from 2006:Q3 to 2025:Q4. Most series, including rents, prices, wages, employment, and investor credit, run to 2025:Q4. Net migration is published with a lag and ends one quarter earlier, in 2025:Q3, and so does the market-tightness measure built from it. Because tightness enters lagged one quarter, its 2025:Q3 value predicts the 2025:Q4 outcome, so 2025:Q4 remains the last estimation quarter. The time dimension within each city carries the identification. Each city has up to 78 quarters of data. Differencing and the construction and lag of tightness drop the early quarters, so the estimation uses 74 quarters per city, 370 city-quarters in total. We estimate jointly across the five capitals. We take net migration, dwelling completions and the wage price index from their Australian Bureau of Statistics (ABS) releases by the quarter they describe. Below we also describe the continuous price index we build, because the ABS discontinued its capital-city series in 2021.

Data availability sets the panel window. The ABS capital-city property price indices (Catalogue 6416.0) begin in 2003:Q3, and quarterly state-level overseas migration data begin in 2006:Q3. We chose the five cities on three criteria: population, since each has more than one million residents; complete data on all variables; and variation in housing-market structure.

They span investor-heavy, constrained markets (Sydney, Melbourne), fast-growing markets (Brisbane, Perth), and a less constrained benchmark (Adelaide).

4.1.1. Variable construction and sources

Table B.1 in the appendix gives full definitions, sources, and geographic coverage. In brief, the outcomes are the capital-city rent indices (ABS 6401.0) and residential property price indices (ABS 6416.0). Dwelling completions (ABS 8752.0, Building Activity) measure new housing supply and enter the denominator of the market-tightness measure. Public and private gross fixed capital formation (GFCF; ABS State Final Demand) describe broader investment conditions across cities (Table 1) but do not enter the estimating equations. Employment (ABS 6202.0) and the Wage Price Index (WPI; ABS 6345.0) measure labour-market conditions.

The value of new investor housing loan commitments by state measures investor credit (ABS Lending Indicators, the dataflow that succeeded Catalogue 5601.0 Table 14). The robustness checks in the online appendix also use national investor and owner-occupier housing credit and the net flow of loans switched between purposes, both from the D2 series of the Reserve Bank of Australia (RBA). We call the commitments series investor credit throughout; it is the variable labelled $\Delta \ln$ Investor credit in the estimation tables. The only exception is those checks, where we say when the measure is the RBA D2 stock of credit outstanding or the investor share derived from it. We use the seasonally adjusted value of these commitments, as published by the ABS. The original series is strongly seasonal: regressed on quarter-of-year and city dummies, its quarterly log change has an R^2 of about 0.45, against 0.01 for the adjusted series. As a second credit measure, we use the change in investors' share of new lending, the value of investor commitments over the value of investor and owner-occupier commitments, both seasonally adjusted. Because both values move with dwelling prices, the share does not rise mechanically when prices rise.

Net overseas and interstate migration (ABS 3101.0) measure migration, and the migration-composition check in the online appendix uses net overseas migration by visa and citizenship group (ABS Overseas Migration). The cash rate (RBA) measures monetary policy. State-level variables serve as proxies for the cities, which is reasonable because each capital accounts for 65–79 percent of its state's economic activity.

Two features of the outcome series need stating. The rent measure is the rent component of the ABS Consumer Price Index (Catalogue 6401.0). It is an index of rents actually paid, not a hedonic or repeat-lease rent index. It is the longest consistent capital-city rent series and is standard in Australian housing work, but it tracks average rents paid across the stock, not new-lease asking rents, and we interpret it accordingly: it is the measured rent r_t of equation (5), not the market rent r_t^*. The ABS measures CPI rents net of Commonwealth Rent Assistance, so increases in that assistance, such as those in 2023:Q3 and 2024:Q3, lower measured rent growth for reasons unrelated to the market, and the 2020 eviction and rent moratoria did the same. Indicators for these quarters change the other estimates little (Online Appendix E). For dwelling prices, the ABS discontinued its established residential property price index, so we splice a single continuous capital-city series. For Sydney we use the Bank for International Settlements all-dwellings series. For the other four capitals we extend the ABS 6416.0 index with the ABS Total Value of Dwellings mean-price series, chain-linking at the splice point so that growth rates, not levels, are joined. Chain-linking leaves no break in levels at the splice, but it does change the growth rates after 2021:Q4: for four capitals these come from a mean-price series that is not adjusted for quality or composition, and Sydney's come from a different source. The main results are similar on the sample ending in 2021:Q4 (Online Appendix E). Table B.1 lists each underlying series and its coverage.

For four of the cities we also measure rents on new leases. When a tenancy starts, the landlord lodges the tenant's bond with a state authority, and New South Wales, Victoria, Queensland and South Australia publish quarterly median rents of the bonds lodged, by dwelling type and number of bedrooms (Homes Victoria, 2026; NSW Department of Communities and Justice, 2026; Residential Tenancies Authority, 2026; South Australian Housing Trust, 2026); Western

Australia publishes no comparable series. For each city we chain the changes in these medians within fixed cells of dwelling type and bedrooms, so that shifts in the mix of lettings across these observed categories do not move the index. Changes in unobserved quality or composition within cells can still affect the medians. These series measure the market rent r_t^* of the model more closely than CPI rents do. We use them to test the model rather than in the main estimation, because they cover only four cities and start in 2008 in Adelaide and in 2012 in Brisbane (Online Appendix F).

4.1.2. Cross-city summary statistics

Table 1 summarises each city over the panel period, 2006:Q3–2025:Q4, and shows the range of housing-market conditions in the panel.

Table 1: Cross-city summary statistics, 2006:Q3–2025:Q4

	Sydney	**Melbourne**	**Brisbane**	**Adelaide**	**Perth**
Price Dynamics					
Rent Index, mean	108.6	106.8	106.0	107.5	98.9
Dwelling Price Index, mean	148.3	131.0	128.5	125.2	113.7
Price–rent growth corr. (2016:Q4)	−0.16	−0.35	−0.21	−0.14	0.70
Price–rent growth corr. (2025:Q4)	−0.29	−0.71	−0.38	−0.48	−0.16
Correlation change	−0.13	−0.35	−0.17	−0.35	−0.86
Migration (thousands, quarterly mean)					
Net Overseas	19.9	17.7	9.9	3.6	7.5
Net Interstate	−5.0	≈ 0.0	4.8	−0.7	0.9
Investment (AUD bn, quarterly mean)					
Public GFCF	7.25	5.31	5.25	1.41	2.44
Private GFCF	25.60	21.35	18.24	5.04	15.93
Public/Private ratio	0.28	0.25	0.29	0.28	0.15
Labour Market					
Employment (millions)	3.81	3.10	2.45	0.84	1.34
WPI Index, mean	123.0	123.7	123.4	123.4	123.4
Market Structure					
Geographic constraint[a]	Severe	Moderate	Low	Low	Moderate

Notes: Up to $N = 78$ quarters per city, ending in 2025:Q4. The rent and dwelling price indices have a 2011–12 base, and the WPI a 2008–09 base. Correlations are between quarterly log changes in the dwelling price and rent indices, over a trailing 20-quarter (5-year) window, ending in the quarter shown; they are snapshots of the rolling series, not period averages. Correlations of the index levels are not reported, because both levels trend and their correlation mainly reflects the common trend. [a]Following Saiz (2010): Severe = multiple natural barriers; Moderate = partial; Low = unconstrained.

Three patterns in Table 1 describe the variation in the panel. First, the trailing correlation between price growth and rent growth is lower at 2025:Q4 than at 2016:Q4 in every capital. The fall is largest in Perth, where the end of the mining-investment cycle had moved prices and rents together, and smallest in Sydney. Second, migration pressure is uneven. Sydney and Melbourne receive large overseas inflows. Sydney also loses residents to other states, consistent with displacement driven by affordability (Li et al., 2024), while Melbourne's net interstate flow is close to zero on average. Brisbane gains both overseas and interstate migrants. Third, the ratio of public to private GFCF and the scale of investment differ widely across cities.

Table 2 reports summary statistics for the variables as they enter the estimation: the $I(1)$ series as quarterly log differences, market tightness in levels, and the price–rent gap in log levels. The differenced series are small and roughly symmetric, as the log-difference transformation is designed to make them. Tightness, the ratio of net migration to completions, is the one variable with a wider spread. It takes occasional negative values, mostly in the 2020–2021 border-closure quarters, when net migration itself turned negative.

Figure 2 plots the rent indices. The index variables (rents and the wage price index) are not comparable as levels across cities. Each is normalised to a base period, 2011–12 for rents and 2008–09 for wages, so it measures growth since that base, not the dollar level of rents or wages. A higher index mean therefore reflects faster cumulative growth since the base, not a higher level. The other level variables differ widely with city size. City fixed effects and first-differencing absorb all these level and base differences, so the estimation works from changes

Table 2: Summary statistics for estimation variables, 2006:Q3–2025:Q4

Variable	N	Mean	SD	Min	Max
Δ ln Rent	385	0.008	0.009	−0.025	0.033
Δ ln Price	385	0.013	0.024	−0.057	0.092
Δ ln Investor credit	385	0.013	0.101	−0.349	0.466
Δ Investor share of new lending	385	0.001	0.018	−0.089	0.051
Δ ln WPI (wages)	385	0.007	0.004	−0.001	0.023
Δ ln Employment	385	0.005	0.010	−0.066	0.043
Δ Cash rate	385	−0.030	0.409	−1.887	1.393
Tightness (level)	370	1.430	0.931	−1.354	5.497
Price–rent gap (log level)	390	0.172	0.169	−0.132	0.669

Notes: Pooled across the five capitals. Δ ln is the quarterly log difference (0.010 is about 1 percent growth); Δ Cash rate is the quarterly change in percentage points. Investor credit is the seasonally adjusted value of new investor loan commitments; the unadjusted series has a standard deviation of 0.148. The investor share of new lending is in share units, so 0.010 is one percentage point. Tightness is net migration over the trailing four quarters relative to dwelling completions over the same window; it enters in levels, lagged one quarter. Its negative minimum reflects the 2020–2021 border-closure quarters, when net migration was negative. The price–rent gap is the log of the dwelling price index over the rent index and enters as a lagged level control. N is the number of city-quarters available for each variable after differencing and lagging. Each count reflects its binding constraint: the price–rent gap ($N = 390$) is a level series observed in every quarter; the differenced series ($N = 385$) lose one quarter per city to differencing; and tightness ($N = 370$) loses the early quarters of its trailing-window construction and lag. The estimating sample is the common set of 370 city-quarters, since every reported specification includes lagged tightness.

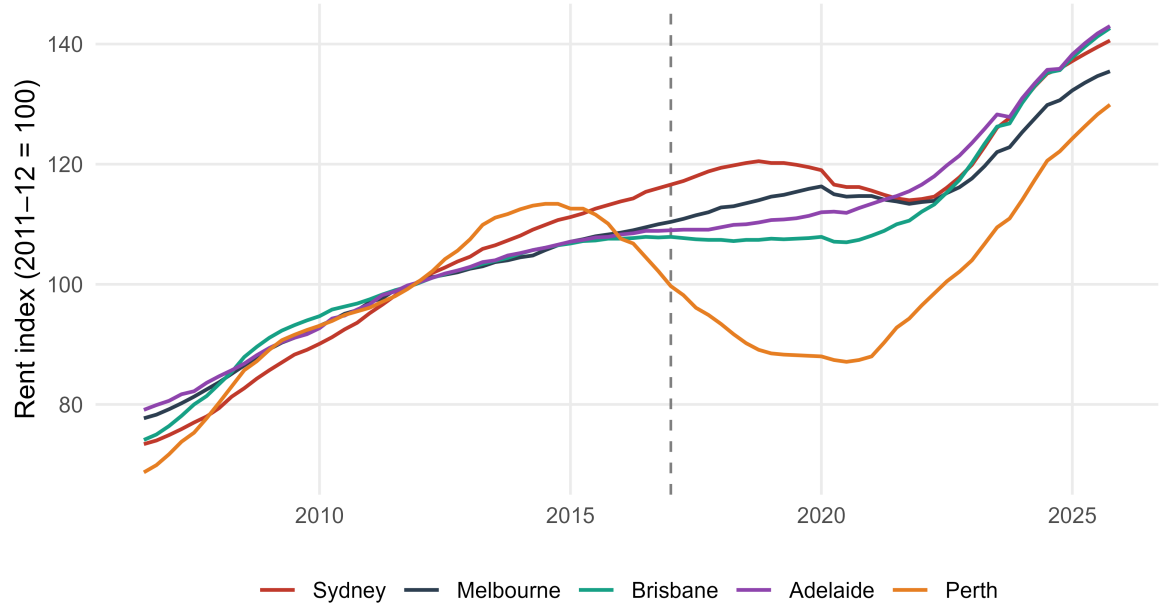


Figure 2: Rent indices, five Australian capital cities, 2006:Q3–2025:Q4 (ABS 6401.0; base: 2011–12 = 100). The dashed line marks 2017:Q1. Perth diverges from about 2014 with the mining investment cycle. The series move broadly together, consistent with common housing-demand fundamentals driving rents across the capitals.

within cities, not from levels across cities. The wide spread of the dwelling price index across cities likewise reflects differences in cumulative growth and city size, not comparable price levels, and the design does not need to balance it.

4.1.3. The disconnect across cities

Prices pulled away from rents over the sample, sharply from 2020. Figure 3 shows the price–rent ratio rising in every capital, and Figure 4 shows that wage-adjusted rents (rents relative to the wage price index) follow a quite different path, driven by local conditions rather than by the price cycle.

The price–rent ratio rose in all five capitals after the early 2010s, peaking at about 1.5 to 2.0 times its 2011–12 value, highest in Sydney (Figure 3). Each ratio equals one in 2011–12 by construction, so the rise is measured from that base; Perth's ratio in the first half of 2025 was close to its 2006 value. Wage-adjusted rents rose until the early 2010s and were broadly

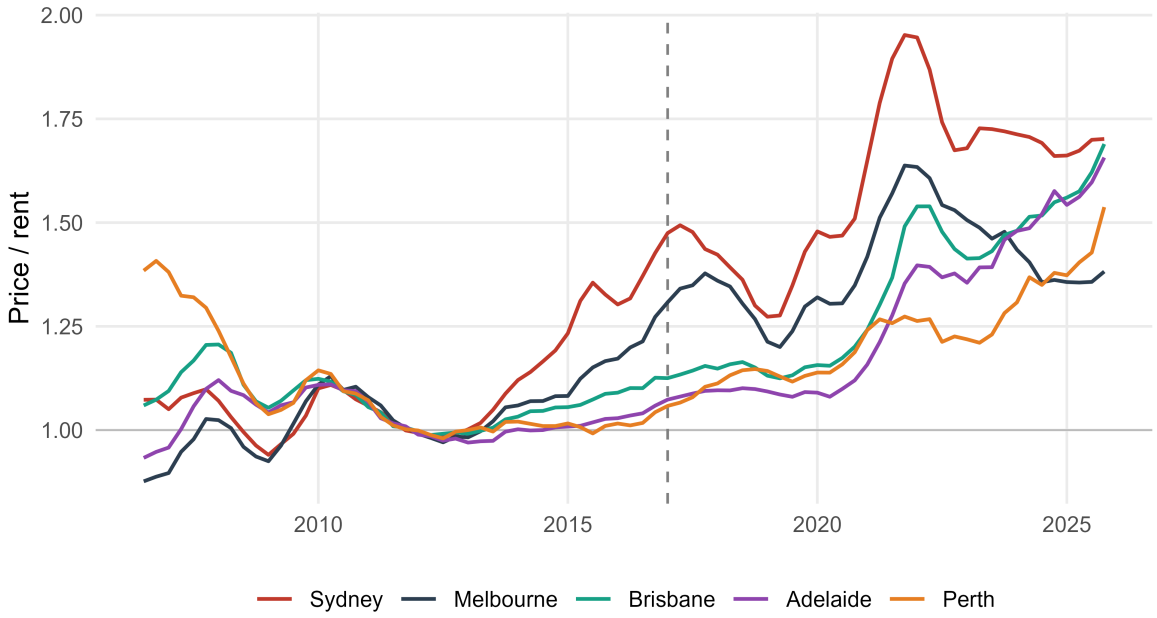


Figure 3: Price–rent ratio by capital city, 2006:Q3–2025:Q4: the dwelling price index over the rent index (both 2011–12 = 100). A rising series means prices are outrunning rents. The dashed line marks 2017:Q1.

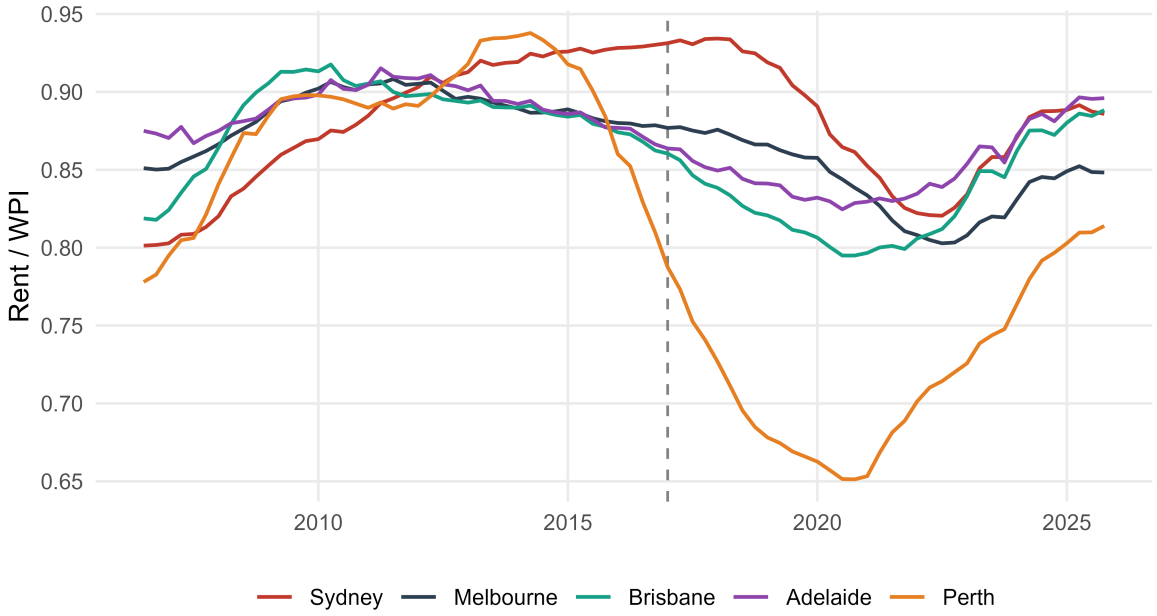


Figure 4: Wage-adjusted rents (rents relative to the state wage price index) by capital city, 2006:Q3–2025:Q4. They rise until the early 2010s and then hold steady or decline. Sydney and Melbourne fall further during the pandemic, to lows in 2022. All five rise again from 2021 or 2022 but stay below their earlier peaks. Perth's mining-cycle collapse and rebound is the main exception. The dashed line marks 2017:Q1.

stable or declined from then until the pandemic (Figure 4). In Sydney and Melbourne they fell further during the pandemic, to lows in 2022. All five have risen since 2021 or 2022, but none has regained its earlier peak. Perth is the exception, with a sharp mining-cycle collapse and rebound in rents. This pattern is consistent with rents responding to local fundamentals.

4.1.4. The 2017 break date

We use 2017:Q1 as a candidate break date, the quarter in which APRA announced its cap on interest-only lending; Online Appendix C explains the choice. It marks a period in which to test whether the drivers of prices and rents change, not a treatment whose effect we estimate,

and we also test for breaks at the two APRA dates and at an unknown date.

4.1.5. Data limitations

The data have five limitations. First, investor credit, wages, employment, migration, and dwelling completions are measured at the state level and proxy city-level measures. Capital cities dominate their state economies, but the proxies add measurement error that may bias the estimated coefficients towards zero. Second, the CPI rent series measures rents on all current leases. It matches the measured rent in our model, but our measure of the market rent, rents on new bonds, covers only four of the cities, starts in 2008 in Adelaide and 2012 in Brisbane, and is built from medians, which move with the quality of the dwellings let. Third, the price series is spliced after 2021, so price growth after 2021:Q4 comes from series that are not quality-adjusted; the main results hold on the sample ending in 2021:Q4. Fourth, the panel begins in 2006:Q3, so it cannot cover the period before the global financial crisis (GFC). Fifth, the estimating equations do not control for construction costs, which may move differently across cities.

4.2. Empirical specification

Sections 5.1–5.5 estimate reduced forms implied by the model; Section 6 then estimates its structural parameters by impulse-response matching. The reduced forms cannot separately identify the tenure parameters κ and η or the adjustment cost α, although the dynamic form of the rent equation identifies the lease-repricing rate ω. The impulse-response matching estimates κ and η only within wide bounds, and Section 6.4 shows which features of the responses identify them. Investor credit is an imperfect, partly endogenous measure of credit conditions (Corollary 1). Table A.1 in Appendix A pairs each object in the model with its counterpart in the data. The CPI rent series measures the rent on all current leases, r_t in (5), not the market rent r_t^*, which we observe only through rents on new bonds in four cities and use only to test the model (Online Appendix F). Wage growth and employment growth measure the growth of renter income, and lagged tightness the growth of renter households relative to the rental stock. Investor loan commitments measure credit conditions, but by (12) they also contain the current price change.

The break indicator is $\text{POST}_t = \mathbb{1}[t \geq 2017{:}Q1]$ for all cities. It marks a candidate structural break around which we test whether the drivers of prices and rents change; it is not a treatment whose causal effect we estimate. Outside Sydney it captures the post-2017 institutional environment broadly; in Sydney it also spans the NSW package of mid-2017. We write a Sydney-only NSW-reform indicator as $D_{jt}^{NSW} = \mathbb{1}[j = \text{Sydney}] \cdot \text{POST}_t$.

4.3. Econometric methods

We estimate two reduced-form equations on the five-city quarterly panel, using variation over time within each city. Most series are trending index levels that are borderline or integrated of order one, so we work in first differences. Market tightness is a ratio of flows, which we treat as stationary, so it enters in levels, lagged one quarter. All models include city fixed effects and use Driscoll–Kraay standard errors (Driscoll and Kraay, 1998), which are robust to heteroskedasticity, autocorrelation, and correlation across cities.

Each method has one limited role. The two reduced-form equations, estimated with within-city fixed effects, establish what co-moves with rents and what co-moves with prices. The mean-group estimator checks that the rent relationship holds city by city, not only on average. Stacked regressions compare the price–credit and rent–credit slopes, and break and Granger tests ask whether either changes over the sample and which way prediction runs (Online Appendix C). The local projections recover the timing that the static slopes compress into one coefficient: how quickly an investor-credit innovation reaches prices and rents. Instrumented local projections check that timing with credit treated as endogenous, and a correlation-restriction test asks whether the instruments, two APRA lending measures, are valid (Online Appendix D). Finally, matching the model's impulse responses to the local projections estimates its parameters and decomposes the delay in rents (Section 6). Everything else, including the rolling partial-R^2

decomposition, is robustness. Each method addresses one link in a single argument: what drives each outcome, how quickly credit transmits to each, and which frictions set that speed.

Stationarity and functional form. We test each series the equations use for a unit root, city by city and with the panel test of Im et al. (2003) (Table E.1). Most behave as $I(1)$ series. Log prices, log employment, log investor credit, and the investor share reject a unit root in at most one city in levels, but in four or five cities in differences. The national cash rate also rejects only in differences. Log wages do not reject in levels either, and wage growth rejects only in the panel test, not in any single city. Log rents are the exception. Their level test rejects in three cities with lags chosen by the Akaike criterion and in two with the Schwarz criterion, but rent growth rejects in no city under either rule. Rent growth is highly persistent, with a first-order autocorrelation of 0.83. A level test allows for only one unit root, and it tends to reject too often when growth itself may have one, so we do not read the rejection as evidence that rents are stationary. Rather than rest the design on a classification of each borderline series, we difference the index variables: log rents, log prices, log wages, log employment, and log investor credit. This is a conservative treatment for series that may be either $I(1)$ or trend-stationary, and it follows common practice for housing-market panels of this kind (Kenny, 1999).

Market tightness, defined as net migration over the trailing four quarters relative to dwelling completions over the same window, is a ratio of flows, which we treat as stationary. The tests support this only in part. Tightness rejects a unit root in three cities with lags chosen by the Schwarz criterion but in none with the Akaike criterion, and its pooled first-order autocorrelation is 0.96. The trailing window adds persistence. The unsmoothed single-quarter ratio rejects in three cities under both rules, with a first-order autocorrelation of 0.55, and it gives a similar positive and significant coefficient in the rent equation (Online Appendix E). We therefore enter tightness in levels. The price–rent gap enters the rent equation in lagged log levels as a level control, not as an error-correction term. The gap would be stationary if log rents and log prices were cointegrated with a unit coefficient, but it does not reject a unit root in any city. A residual-based test, which lets the coefficient differ from one, rejects in no city either; the five city statistics average about -1.86, against a 5 percent critical value of about -3.4. The gap's coefficient is also close to zero and insignificant in every rent specification we estimate, so we do not give it an error-correction interpretation.

The two estimating equations follow from the model. Proposition 1 and Lemma 1 give the rent equation, and Proposition 2 the price equation. Each keeps the terms that the within-city first-difference design can identify, and Figure 1 shows the channels they examine.

Rent equation.

$$\Delta r_{jt} = f_j^R + a\,(p-r)_{j,t-1} + b_1\,\mathrm{TIGHT}_{j,t-1} + b_2\,\Delta w_{jt} + b_3\,\Delta n_{jt} + u_{jt}^R, \tag{19}$$

where r is log rent, $(p-r)$ is the log price–rent gap, TIGHT is market tightness, w is the log wage index, n is log employment, and f_j^R is a city fixed effect. We estimate Equation (19) with a pooled within (fixed-effects) estimator and with a mean-group estimator (Pesaran and Smith, 1995), which averages city-specific regressions. The mean-group estimator checks that the relationship with fundamentals holds city by city, not only on average. The model predicts $b_1 > 0$ and, by Proposition 1(iii), a coefficient b_2 between ω and one, if wages pass through one for one. It predicts a smaller b_3, because employment growth is less persistent than wage growth and only proxies the growth of employment per household. It predicts a small coefficient a, since the price–rent gap stands in for tenure switching, which is slow. We lag tightness one quarter to limit feedback from rents to migration (Li et al., 2024). Proposition 1(iv) also gives the dynamic form

$$\Delta r_{jt} = f_j^R + \varphi_r\,\Delta r_{j,t-1} + a\,(p-r)_{j,t-1} + b_1\,\mathrm{TIGHT}_{j,t-1} + b_2\,\Delta w_{jt} + b_3\,\Delta n_{jt} + u_{jt}^R. \tag{20}$$

In this specification, $\varphi_r = 1-\omega$, so the coefficient on lagged rent growth measures the persistence induced by staggered lease repricing. The implied lease-repricing rate is $\omega = 1-\varphi_r$,

and the long-run coefficients $b_k/(1-\varphi_r)$ correspond to the structural coefficients in Equation (15). The model predicts a long-run wage coefficient of one. With 74 quarters per city, the bias from combining fixed effects with a lagged dependent variable is small (Nickell, 1981).

Price equation.

$$\Delta p_{jt} = f_j^P + c_1 \,\Delta\text{CASH}_t + c_2 \,\Delta\iota_{jt} + c_3 \,\text{TIGHT}_{j,t-1} + u_{jt}^P, \tag{21}$$

where p is the log dwelling price, CASH is the cash rate, ι is log investor credit, and f_j^P is a city fixed effect. First-differencing (10) motivates this form, and for the same reason the policy rate enters as its change. Proposition 2 predicts $c_1 < 0$, $c_2 > 0$, and a relatively small c_3. Both market tightness and wage growth affect prices indirectly through expected rents. We retain tightness to control for local housing-demand pressure relative to new supply, while leaving wage growth to the rent equation in the parsimonious baseline price specification. This is an empirical specification choice rather than a restriction imposed by the model. Following Corollary 1, we estimate the equation twice, with investor credit and the cash rate in the current quarter and with both lagged one quarter. Because price growth is persistent, we also report each version with last quarter's price growth as a control, so that lagged credit cannot stand in for price momentum.

Price and rent slopes on credit. The contrast between the two equations is the gap between the price and rent slopes on investor credit. We test whether the slopes are equal by stacking the two equations with the same controls, and whether either changes after a break by interacting credit with a break indicator. These tests, the break tests at other dates, and Granger tests of which way prediction runs are set out in Online Appendix C.

Local projections. To recover the dynamics that the static slopes compress into a single coefficient, we estimate panel local projections (Jordà, 2005) of the price and rent levels on an investor-credit innovation over horizons $h = 0, \ldots, 12$:

$$\begin{aligned} y_{j,t+h} - y_{j,t-1} = f_{j,h} + b_h \,\Delta\iota_{jt} + \sum_{l=1}^{4} \left(\Gamma_{l,h}^{\iota} \,\Delta\iota_{j,t-l} + \Gamma_{l,h}^{y} \,\Delta y_{j,t-l}\right) \\ + \Xi_h \,\text{TIGHT}_{j,t-1} + \Omega_h \,\Delta\text{CASH}_{t-1} + u_{j,t+h}, \end{aligned} \tag{22}$$

where y is the log price or log rent. The city fixed effects and the lagged controls absorb the predictable component of credit, so b_h traces the cumulative response to the part of credit growth that the controls do not predict. Reading b_h as a response rests on a predictive assumption, the one behind a Granger test: given the city effects and the lagged controls, the credit innovation is unrelated to other shocks to later prices or rents. The controls alone do not establish this orthogonality condition or identify a causal effect. If investors borrow on information that the controls miss, b_h also reflects that information, so Online Appendix D instruments credit with the two APRA measures. We scale b_h to a rise of one standard deviation in investor-credit growth, 0.10 log points in the seasonally adjusted series. Horizon zero retains the within-quarter simultaneity caveat. Because the responses are cumulative, the impact response is part of every later horizon, but the change after impact is less exposed to reverse causality. We focus on the difference in the shape and timing of the two response paths. Proposition 4 predicts a price response on impact, no rent response on impact, and a rent response that builds over the following quarters.

We add two variants. Credit expands when incomes and employment are strong, and any such demand shock, combined with staggered leases, would move prices at once and rents slowly. The first variant therefore adds four lags each of wage growth and employment growth to Equation (22). The second estimates post-impact responses,

$$y_{j,t+h} - y_{j,t} = f_{j,h} + b_h^{\dagger} \,\Delta\iota_{jt} + \zeta_h \,\Delta p_{jt} + (\text{controls}) + u_{j,t+h}, \qquad h = 1, \ldots, 12, \tag{23}$$

with the controls of the first variant. Holding the current price change fixed removes the term that measured credit shares with prices by Equation (12), so $b_h^\dagger$ uses only the variation in credit that reflects credit conditions and purchases. In the rent projections we also hold the current rent change and four lags of price growth fixed.

5. Reduced-Form Evidence

This section establishes the facts the model must match. Sections 5.1 and 5.2 show that fundamentals govern rents and that investor credit co-moves strongly with prices and hardly with rents. Section 5.3 traces the dynamic responses of prices and rents to a credit innovation, which are the targets for estimating the model in Section 6. Section 5.4 locates the delay using new-lease rents, and Section 5.5 summarises what instrumenting credit adds. The reduced forms describe co-movement and timing; the structural interpretation comes from the model estimated in Section 6. The online appendix reports the robustness checks.

5.1. Fundamentals govern rents

Table 3 reports the rent equation. Rent growth responds strongly and significantly to lagged market tightness, to wage growth, and to employment growth. These relationships are stable across cities: the mean-group estimates, which average city-by-city regressions, show the same pattern as the pooled within estimates. Each estimate matches Proposition 1. Tightness enters positively in levels with a small coefficient, as Lemma 1 predicts, and the wage coefficient lies well above the employment coefficient, as part (iii) predicts when wage growth is more persistent. Part (iii) covers a single regressor, however. With several correlated drivers, the omitted gap can load on wage growth by more than part (iii) allows: the static wage coefficient, 0.73, exceeds the long-run coefficient of 0.53 from the dynamic form below. The lagged price–rent gap does not enter rent growth in either specification; its point estimate is close to zero, with a p-value of 0.40 in the pooled model and 0.14 in the mean-group model. Rents track the fundamentals that a standard account of housing demand predicts, and they do not chase the widening gap between prices and rents. This result holds in every specification we report below, and it anchors the paper empirically. Rents on new leases, which we observe for four of the cities, load on the same fundamentals, with long-run coefficients on tightness and wages close to those of CPI rents, and the price–rent gap does not enter their growth significantly either (Online Appendix F).

The dynamic form in Equation (20) gives a sharper test of the model. The coefficient on lagged rent growth is about 0.65, so the implied repricing rate is $\hat{\omega} \approx 0.35$ (s.e. 0.10): measured rents close about a third of the gap to market rents each quarter, which matches a mix of annual and shorter leases. Rents on new leases suggest slower repricing, about 0.12 to 0.15 a quarter, but only if CPI rents absorb them one for one, which the data reject (Section 6.4). The dynamic form fits much better, raising the within-R^2 from 0.55 to 0.74. Its long-run wage coefficient is 0.53 (s.e. 0.20), below the value of one that Equation (15) predicts ($p = 0.02$). The long-run employment coefficient is 0.38 and the long-run tightness coefficient 0.005. In the model, rents move with renter income. The combined long-run coefficient on wage and employment growth is 0.91 (s.e. 0.29), consistent with approximately unit pass-through, although the wage index alone captures only about half of it. One reason is that the wage price index measures pay rates for a fixed mix of jobs, while renter income also moves with hours and with who is employed, which employment growth partly captures. The price–rent gap stays close to zero, but the gap level is a weak proxy for the tenure channel, which in the model depends on recent price growth; Section 5.3 tests that channel directly.

The same fundamentals govern rents throughout the sample, but their weights change: the employment coefficient roughly doubles after 2017 and the wage coefficient falls, and a test of equal coefficients before and after 2017 rejects ($p = 0.004$). Online Appendix E reports these tests and the rolling coefficients. The change concerns the fundamentals, not the investor-credit channel, so it does not affect the results below.

Table 3: Rent equation, 2006:Q3–2025:Q4

	Within (FD) (1)	Mean group (2)
Price–rent gap$_{t-1}$	0.002	0.010
	(0.003)	(0.007)
Tightness$_{t-1}$	0.006***	0.005***
	(0.001)	(0.001)
Δ ln WPI	0.726***	0.718***
	(0.169)	(0.143)
Δ ln Employment	0.144***	0.111***
	(0.031)	(0.023)
Observations	370	370
R^2	0.548	0.605

Dependent variable: Δ ln rent. City fixed effects; Driscoll–Kraay standard errors in parentheses. R^2 is the within-R^2 in column (1) and the average fit of the city-specific regressions in column (2); the two are not directly comparable. $^{*}p < 0.10$, $^{**}p < 0.05$, $^{***}p < 0.01$.

5.2. Prices respond to investor credit

Table 4 reports the price equation, which leaves wage growth to the rent equation (Section 4.3). We report the regressors both in the current quarter and lagged one quarter, because investor credit and prices are plausibly determined within the same quarter. Investor-credit growth enters positively and significantly at both timings. A rise of one standard deviation in credit growth (0.10) goes with price growth about 1.4 percentage points higher in the quarter, and lagging credit one quarter barely changes the coefficient (0.133 against 0.138, columns (1) and (3)). The cash rate shows most clearly why timing matters. In the current quarter it takes a small positive coefficient; lagged one quarter it turns negative, the sign Proposition 2(ii) predicts, but it is not significant (-0.003, standard error 0.003). We read the positive current-quarter association as the familiar endogeneity of a policy rate that rises during booms, when prices also rise. We read the negative lagged coefficient as the better-behaved estimate of how the cost of capital feeds into prices. We therefore do not interpret the current-quarter cash-rate coefficient, and we treat the lagged specification as the more credible of the two. Tightness generally has a small and statistically insignificant coefficient, although it is marginally significant in the specification with lagged credit and price momentum. This is broadly consistent with Proposition 2(iii): in the model, tightness reaches prices only through expected rents.

Price growth is persistent, and columns (2) and (4) add last quarter's price growth, which enters with a coefficient of about 0.5. The current-quarter credit coefficient falls from 0.138 to 0.099, and the lagged coefficient from 0.133 to 0.065. Both remain large and significant ($p < 0.001$). Lagged credit therefore predicts price growth beyond what price momentum predicts, so the lagged association does not rest on its link with last quarter's price change (Corollary 1). The check also supports the seasonally adjusted measure: with the unadjusted series, the lagged coefficient falls to zero once momentum is controlled for, consistent with seasonal noise obscuring the relationship.

Prices versus rents. The contrast between the two equations is large. In the pooled panel, the price–credit slope is 0.11 before 2017 and the rent–credit slope is -0.01, and a stacked regression with the same controls in both equations rejects equal slopes ($p < 0.01$). The gap holds with credit lagged one quarter and on both sides of every candidate break date. Neither slope changes significantly after 2017 or at either APRA date once credit is seasonally adjusted; an apparent post-2017 strengthening in the unadjusted series reflects seasonal noise that faded over the sample. Granger tests show that lagged credit predicts both price and rent growth, but its summed lag coefficients for rents are about a fifth of those for prices, and with two lags it does not predict rents at all. Online Appendix C reports these tests in full (Tables C.1–C.3).

Table 4: Price equation, 2006:Q3–2025:Q4

	Current quarter, x_t		One-quarter lag, x_{t-1}	
	(1)	(2)	(3)	(4)
Δ ln Investor credit	0.138***	0.099***	0.133***	0.065***
	(0.025)	(0.012)	(0.021)	(0.012)
Δ Cash rate	0.007	0.000	−0.003	−0.006
	(0.005)	(0.002)	(0.003)	(0.004)
Tightness$_{t-1}$	0.002	0.002	0.004	0.003*
	(0.002)	(0.001)	(0.002)	(0.002)
Δ ln Price$_{t-1}$		0.524***		0.486***
		(0.048)		(0.057)
Observations	370	370	370	370
Within-R^2	0.355	0.592	0.353	0.505

Dependent variable: Δ ln price. City fixed effects; Driscoll–Kraay standard errors in parentheses. The price equation excludes wage growth. Investor credit is seasonally adjusted. Columns (1) and (2) enter investor credit and the cash rate in the current quarter (x_t), and columns (3) and (4) lagged one quarter (x_{t-1}). Columns (2) and (4) add last quarter's price growth. Market tightness is lagged one quarter in every column. $^{*}p < 0.10$, $^{**}p < 0.05$, $^{***}p < 0.01$.

5.3. The mechanism: prices move first and rents follow slowly

To recover the timing that the static slopes in Table 4 compress into a single coefficient, we estimate the panel local projections of Equation (22) over twelve quarters, using the seasonally adjusted credit series. Figure 5 plots the cumulative response of each log level to a one-standard-deviation credit innovation, with 90 percent Driscoll–Kraay bands. Horizon zero retains the within-quarter simultaneity noted above. Because each response is cumulative from the quarter before the innovation, the impact response is also part of every later horizon. The change after impact is cleaner, since later prices and rents cannot feed back into credit already drawn. The long-horizon estimates are less precise, and their windows overlap the pandemic, so we read the figure for its timing rather than for the exact magnitude at distant horizons.

The two responses differ in speed, not in whether credit reaches them. Prices respond on impact, by about 0.8 percent (0.008 log points), build to roughly 2.9 percent (0.029 log points) by the third quarter, and stay near that level through the second year, with a shallow dip that lies within the bands. They then drift up to about 3.3 percent (0.033 log points) by the twelfth quarter. Rents barely move in the first year. They rise by about 0.4 percent (0.004 log points) by the fourth quarter, only marginally different from zero, and then climb steadily to about 1.8 percent (0.018 log points) by the twelfth. The horizon from which the rent response is significant moves with the number of control lags, so we treat it as approximate; the shape and the twelfth-quarter magnitude of about 0.02 log points do not. In this reduced form, prices take up most of their response within three quarters, consistent with an asset-pricing channel, while rents build up only after the first year.

The shape survives the two main objections to this reading (Online Appendix E, Figure E.1). A demand shock that raised credit together with incomes would also move prices at once and rents slowly, but adding lags of wage and employment growth leaves both responses almost unchanged. And although measured credit contains the current price change, a credit innovation still predicts later price and rent growth when that change is held fixed.

This is the pattern Proposition 4 predicts. Prices respond on impact through the collateral channel. Rents do not respond on impact, and their early response is the product of three slow margins: the share of leases that reprice, the speed of tenure switching, and the sensitivity of renting to prices. The price response builds after impact rather than peaking at once, as Proposition 4(i) predicts when transaction prices adjust gradually; Section 6 estimates the speed of that adjustment and of the rent-side frictions. Within the twelve-quarter window, the supply offset in Proposition 4(iii) is not visible, which suggests either a construction lag close to three

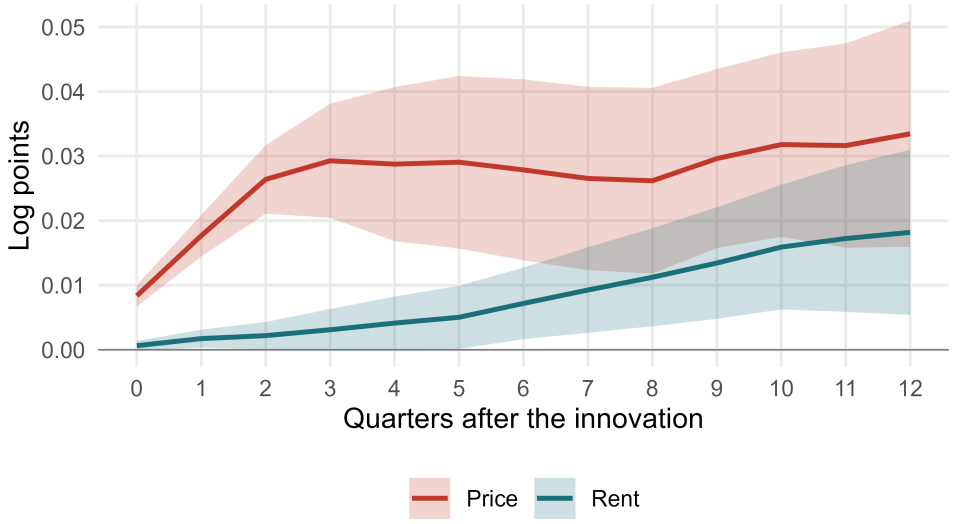


Figure 5: Cumulative response of the price level and the rent level to a one-standard-deviation investor-credit innovation, over twelve quarters, in log points (a response of 0.030 is about 3.0 percent). Panel local projections (Equation (22)) with city fixed effects and four control lags, seasonally adjusted credit series; 90 percent Driscoll–Kraay bands. Horizon zero carries the within-quarter simultaneity caveat, which the later cumulative responses include; the long-horizon estimates are less precise.

years or a small supply response over that horizon.

The model routes credit to rents only through tenure switching, and we test this restriction directly. In the dynamic rent equation, four lags of price growth are jointly significant ($\chi^2_4 = 20.9$, $p < 0.001$), as the tenure channel requires. Four lags of credit growth also add to them ($\chi^2_4 = 10.2$, $p = 0.04$). Past price growth carries part of the link from credit to rents that the model predicts, but it does not absorb the predictive content of credit. The residual association could reflect financing conditions affecting rental supply or landlord decisions, or omitted housing-demand information. The tenure-switching mechanism is therefore a parsimonious account of the observed timing, not an exhaustive explanation of credit transmission to rents.

Over the full horizon the rent response closes part of the gap with the price response. By twelve quarters the rent response is about half the price response, 0.018 against 0.033 log points, and the two bands overlap widely, though the price response is itself imprecisely estimated that far out. The disconnect documented above is therefore a difference in transmission speed and a medium-run wedge, not a permanent insulation of rents from financial conditions: a credit innovation that lifts prices within the first year works through to rents over the following two to three years. The persistent gap in the descriptive series reflects sustained financial conditions and price-specific drivers, rather than rents never responding. The lag also matters for affordability, because the rent consequences of a credit-driven price boom appear only after a year or more.

Section 5.5 summarises how this timing holds up when credit is instrumented.

5.4. New leases reveal where measured rents lag

The distinction between new and existing leases locates an important part of the delay. Figure 6 compares prices, new-lease rents and CPI rents on the same city-quarters for Sydney, Melbourne, Brisbane and Adelaide. The new-lease indices use rental bond records; Online Appendix F documents their construction and the remaining composition differences.

Prices take up most of their response within three quarters. New-lease rents rise from the first quarter after the innovation and reach 0.024 log points at eight quarters, against 0.029 for prices. At twelve quarters the responses are 0.030 and 0.035, respectively, while CPI rents have risen by 0.017. The new-lease response exceeds the CPI response in 98, 95 and 94 percent of block-bootstrap draws at four, eight and twelve quarters. The gap between prices and new-lease rents is significant only at four quarters. These comparisons show that much of the measured

three-year gap lies between new and existing leases, rather than between prices and newly contracted rents.

The exact repricing rate is less clear. With one-for-one long-run transmission from new-lease to CPI rents imposed, filtering the new-lease response gives $\omega \simeq 0.12$; $\omega = 0.15$ fits better than the baseline 0.35 in 96 percent of bootstrap draws. Allowing the two indices to have different long-run loadings makes the rate much less precise. This distinction matters for interpreting the model: the evidence directly supports delayed adjustment of CPI rents relative to new leases, whereas the speed and size of tenure switching remain inferred from the model. Section 6.5 therefore treats the friction decomposition as accounting within the estimated model, not as a direct measurement.

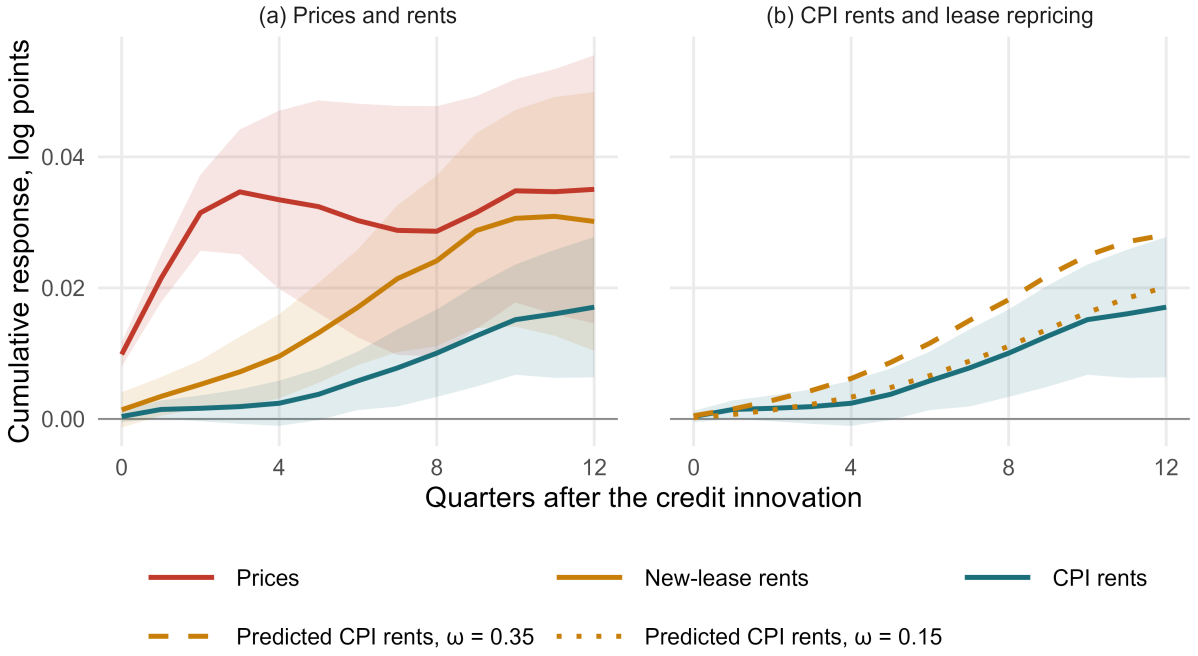


Figure 6: Cumulative responses to a one-standard-deviation investor-credit innovation in Sydney, Melbourne, Brisbane and Adelaide, in log points, from Equation (22) estimated for each outcome on the same city-quarters. Panel (a): prices, rents on new leases and CPI rents, with 90 percent Driscoll–Kraay bands. Panel (b): CPI rents beside the CPI response that Equation (5) predicts from the response of rents on new leases, with $\omega = 0.35$ (Table 5) and with $\omega = 0.15$, the growth-form estimate (Table F.1).

5.5. Credit as an endogenous regressor

The local projections residualise credit growth against observed controls, but investors may borrow ahead of price or rent rises that the controls do not capture. Online Appendix D instruments credit with two measures by which APRA restrained investor lending in 2015–2018, an indicator for the period in which either measure was in force and each city's 2014 investor share times the interest-only cap, and tests the instruments with the correlation-restriction test of Dzhumashev and Tursunalieva (2026). The instrumented projections support the delay in rents: both instruments give rent responses that build over time and lie close to the OLS projections. They do not settle how fast credit reaches prices. The national-timing instrument gives a response that builds over two years and stays below OLS; its Anderson–Rubin tests do not reject zero. The exposure instrument gives a faster price response but fails the correlation-restriction test at four quarters. We therefore use the OLS projections as the quantitative targets and retain the IV results as a limited check on the timing pattern.

6. Estimating the Model

The reduced forms test the signs and timing that the model predicts. This section estimates the model on the dynamic responses of Section 5.3. It asks whether the model can reproduce the size and shape of those responses, how fast transaction prices adjust, which features of the responses identify each parameter, which frictions account for the delay in rents, and what the delay implies for measured inflation.

The projections establish a clear ordering of price, new-lease and CPI rent responses. We interpret that ordering through a model in which lending conditions initiate the adjustment. This interpretation requires a mapping from observed commitments to lending conditions as well as the transmission equations. The APRA exercises provide a limited external check on timing; they do not independently identify the price-adjustment parameter.

6.1. Estimation by impulse-response matching

We choose the model's parameters to minimise the weighted distance between its impulse responses and the local projections of Figure 5. The targets are the responses of the price and rent levels at horizons 0 to 12 and the response of investor credit itself at horizons 1 to 12, 38 moments in all; each is weighted by the inverse of its squared standard error. The credit response is estimated with the same projection as the price response.

An important identifying restriction concerns the mapping from measured investor credit to lending conditions. Equation (12) implies that investor commitments combine credit conditions, transaction prices, and the volume of investor purchases:

$$\Delta \iota_t = \Delta \ln \phi_t + \Delta p_t + \Delta \ln X_t.$$

For baseline impulse-response matching, we hold the response of investor purchases fixed, so that $\hat{\iota}_t = \hat{\phi}_t + \hat{p}_t$. Under this restriction, the joint responses of measured credit and transaction prices discipline the path of credit conditions and allow the price response to inform the asset-valuation and price-adjustment parameters. Movements in measured commitments not accounted for by transaction prices are thus attributed to lending conditions. This mapping is a maintained structural restriction, not a property established by the reduced-form estimates. If investor purchases respond independently to the innovation, some of that response may instead be attributed to lending conditions. The structural estimates and counterfactuals are therefore conditional on this restriction; the reduced-form timing comparison does not require it. Section 6.3 relaxes this restriction while retaining the same empirical targets and weighting matrix.

After a shock, the growth of credit conditions decays at rate ρ_g and their level at rate ρ_d, so credit conditions can keep rising for a few quarters, as measured credit does. The shock is scaled so that measured credit growth on impact equals one standard deviation, as in the projections.

We estimate seven parameters: ε, ρ_g, ρ_d, the speed of price adjustment θ in Equation (11), the tenure parameters κ and η, and the supply response ψ/α. We bound η at 1. With a larger η and no supply response, a rise in prices would raise long-run rents, and so the asset value, by more than itself, and the model would have no unique bounded solution. We fix four other parameters or processes: the repricing rate $\omega = 0.35$, estimated separately from the dynamic rent equation (Section 5.1); the weight on next quarter's value, $\varpi = 0.99$, which matches a gross rental yield of about 4 percent a year; a decay of 1 percent a quarter in a deviation of the rental stock; and a construction lag of $J = 8$ quarters. Although ω is fixed during each structural matching exercise, its estimation uncertainty is propagated through the bootstrap below. The fit barely changes with $J = 4$ or $J = 12$: the distance is 11.39 and 11.28, and θ is 0.23 and 0.21. The supply response is estimated at zero with $J = 4$ and $J = 8$. With $J = 12$ it reaches its upper bound, but new supply then arrives only at the last target horizon.

The weight ϖ is consistent with the steady state of the investor's problem (Appendix A). There, $\varpi = (1-\delta)\bar{\chi}$ with $\bar{\chi} = \Lambda/(1-\bar{\lambda}\bar{\phi})$. With depreciation of 0.2 percent a quarter[3] and the

[3]Land does not depreciate, so the depreciation rate of a dwelling including its land is the structure's rate times

mean cash rate of the sample, $\varpi = 0.99$ implies a loan-to-value limit of about 0.89 at an annual discount rate of 6 percent and 0.95 at 10 percent, inside the range that Equation (6) allows. These implied limits are relatively high compared with typical observed investor loan-to-value ratios. They represent the effective collateral limit of the marginal investor in the model rather than the average leverage of investor borrowers. The state-dependence calculation in Section 6.8 uses $\bar{\phi} = 0.8$. The decay of a deviation in the rental stock is a different object. In the model the stock returns to its path through prices: while it is above its path, rents and the price are lower and investment falls below replacement (Appendix A). The fixed decay of 1 percent a quarter is a reduced-form stand-in for that return, which is faster than depreciation alone. It does not affect the estimates, because the supply response is estimated at zero. With a decay of 0.2 percent a quarter, equal to depreciation, the re-estimated supply response is 0.003, θ is 0.23, and the distance is 11.39. The model is linear, so we solve it in one step for each parameter vector. The distance is a sum of squares with several local minima, so we minimise it by a bounded Gauss–Newton search from many starting values.

The minimum-distance formula for standard errors ignores the correlation between the responses at different horizons, which is strong because the projections share the same shocks. We therefore bootstrap the whole procedure. Each draw resamples quarters in moving blocks of eight, keeping the five cities in each quarter together, as for the instruments in Online Appendix D.2. On each draw we re-estimate the price, rent and credit projections, re-estimate ω from the dynamic rent equation, and re-estimate the model with the full-sample weights. Each draw starts the search from the full-sample estimates and from nine other points, which cover a positive supply response, slow and fast tenure switching, and slower and faster price adjustment. We report 90 percent percentile intervals from 399 draws. The intervals account for sampling variation in the projections, their correlation across horizons, and uncertainty in the separately estimated ω, including its propagation into the fitted structural parameters. They also provide uncertainty measures for functions of the estimates, including the decomposition and counterfactual. They remain conditional on the maintained model, the mapping from commitments to lending conditions, the specified credit process, and the fixed calibration parameters. They do not capture uncertainty about alternative purchase-volume or expectations processes. A 74-quarter panel contains only about nine blocks of eight quarters, so we repeat the bootstrap with blocks of four and twelve quarters (120 draws each), and Section 6.4 checks how the estimator behaves in samples of this size.

6.2. Fit and the speed of price adjustment

Table 5 reports the estimates, and Figure 7 compares the model's price and rent responses with the projections. The model fits all three paths closely: the distance is 11.4 on 38 moments, and every model response lies inside the 90 percent band of the corresponding projection. The credit path is matched with a short-lived rise in the growth of credit conditions ($\rho_g = 0.38$) and a persistent level ($\rho_d = 0.96$). The rent side needs no additional friction beyond those in Section 3 to reproduce the estimated impulse responses. Lease repricing and tenure switching together generate the slow rise in rents, and the model's rent response at twelve quarters is 0.54 of its price response, as in the data.

The central estimate is the speed of price adjustment. Transaction prices close about 23 percent of the gap to the asset value each quarter ($\hat{\theta} = 0.23$, 90 percent interval $[0.12, 0.38]$). Half of a change in value therefore reaches transaction prices in about two and a half quarters, and the bootstrap interval for this half-life runs from one and a half to five quarters. The estimate does not depend on the block length: the 90 percent interval is $[0.12, 0.38]$ with blocks

the structure's share of the dwelling's value. Harding et al. (2007) estimate that housing structures depreciate at about 2 percent a year net of maintenance and 2.5 percent gross, and structures account for about 38 percent of the value of Australian households' residential land and dwellings (Australian Bureau of Statistics, 2007). Together these give 0.8 to 1 percent a year, or 0.19 to 0.24 percent a quarter. At 0.24 percent a quarter, the implied loan-to-value limit is 0.94 at an annual discount rate of 6 percent and 0.98 at 10 percent.

Table 5: Model parameters estimated by impulse-response matching

Parameter	Meaning	Estimate	90% interval
ε	Elasticity of value to credit conditions	0.011	[0.003, 0.060]
ρ_g	Persistence of credit growth	0.381	[0.156, 0.553]
ρ_d	Persistence of credit level	0.964	[0.826, 0.995]
θ	Speed of price adjustment	0.233	[0.124, 0.383]
κ	Speed of tenure switching	0.219	[0.025, 1.000]
η	Tenure response to prices	0.621	[0.005, 1.000]
ψ/α	Supply response	0.000	[0.000, 0.050]
ω	Lease repricing rate (estimated separately)	0.35	[0.244, 0.658]
Also fixed: $\varpi = 0.99$, $J = 8$, decay of a stock deviation 0.01			
Distance, 38 moments			11.39
Distance with $\theta = 1$			93.53

Parameters chosen to minimise the distance between the model's impulse responses and the local projections of the price and rent levels (horizons 0–12) and of investor credit (horizons 1–12), with each moment weighted by its inverse squared standard error. Structural matching holds the response of investor purchase volumes fixed, attributing commitments not accounted for by prices to lending conditions; estimates and intervals are conditional on this restriction. 90 percent percentile intervals from 399 draws of a moving block bootstrap over quarters (blocks of eight quarters, cities kept together); each draw re-estimates the projections, ω and the model. The intervals for κ and η reach their upper bound of 1, and the interval for ψ/α spans its whole range, $[0, 0.05]$. The interval for ω is that of the dynamic rent equation across draws; it is wider than the Driscoll–Kraay interval of Section 5.1 (s.e. 0.10) because it resamples blocks of quarters. "Distance with $\theta = 1$" re-estimates the other parameters with instantaneous adjustment to the asset value within the maintained structural specification.

of four quarters and $[0.11, 0.36]$ with blocks of twelve, and no draw at any block length gives a θ above 0.55.

Conditional on the mapping of commitments to lending conditions in Section 6.1, the credit and price targets carry different information about θ. Their joint responses discipline the implied path of credit conditions. Given that path, ε sets the scale of the change in the asset value, and θ sets how much of it reaches transaction prices on impact and how fast the rest follows. The impact price response, less than a third of the level reached by the third quarter, therefore tightly constrains θ within the maintained model, and the other parameters can offset it only in part. Figure 8 shows this. Its left panel holds θ fixed on a grid and re-estimates the other six parameters. The distance rises by 5.5 at $\theta = 0.10$, 3.2 at $\theta = 0.35$, 16 at $\theta = 0.5$, and 82 at $\theta = 1$. Because the weights ignore the correlation across horizons, these increases are not χ^2 statistics; the bootstrap intervals are the basis for inference, and they agree with the profile.

Instantaneous adjustment to the asset value is strongly disfavoured within the baseline specification. With $\theta = 1$ and the other parameters re-estimated, the distance rises from 11.4 to 93.5. The model then capitalises the expected path of credit conditions at once, so prices jump on impact and decline as the loosening decays, whereas the empirical response builds over three quarters. Under the assumed credit process and commitments mapping, re-estimating the other parameters does not reproduce that build-up. This is a conditional comparison, not a rejection of all forward-looking models with alternative expectations or purchase-volume dynamics. The result provides a quantitative account of the momentum in house prices documented by Case and Shiller (1989) and modelled through search and strategic complementarity by Head et al. (2014) and Guren (2018): in these five cities, transaction prices take most of a year to absorb a change in the financing conditions of their marginal buyers.

Fève and Moura (2024) show that momentum can arise without trading frictions when the driver of value itself follows a hump-shaped path, so momentum alone does not prove that prices are slow. Within our maintained credit-process specification, the fitted credit response disciplines that alternative explanation. The credit target constrains the path of the driver, and that path does rise for a few quarters after the innovation ($\rho_g = 0.38$). Yet with $\theta = 1$ the asset value capitalises the whole path at once and the price peaks on impact. The hump in credit conditions is too short to produce the three-quarter build-up in prices, so the estimated model attributes the build-up to gradual adjustment in the trading process.

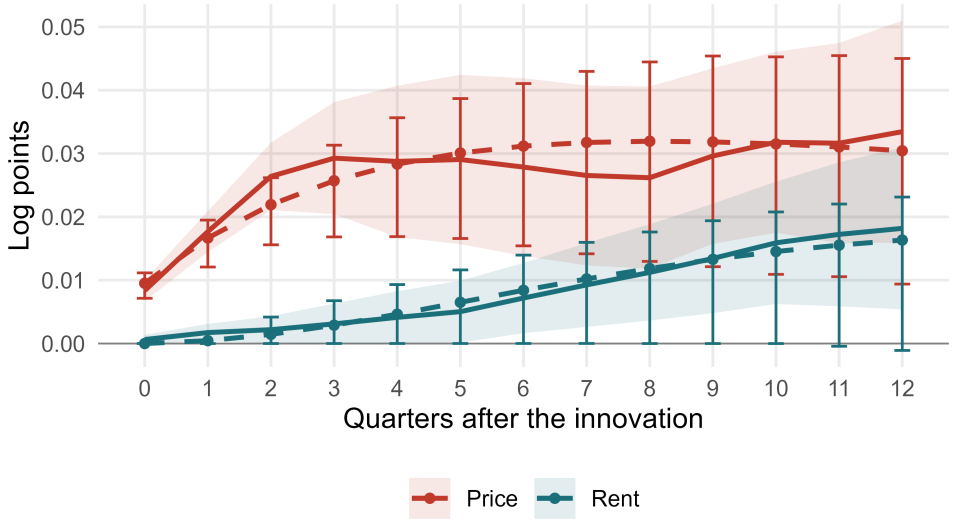


Figure 7: Model and data. Solid lines: local projections of the price and rent levels on a one-standard-deviation investor-credit innovation, with 90 percent Driscoll–Kraay bands (Figure 5). Dashed lines with points: the estimated model's impulse responses (Table 5). The model is also matched to the response of investor credit, not shown. Error bars: 90 percent bootstrap intervals for the model's responses, from re-estimating the projections and the model on 399 block-bootstrap draws.

Is θ stable enough to carry the counterfactual in Section 6.6? Three considerations bear on this. First, the estimate averages over the loosenings and tightenings of 2006–2025, including the APRA period, and the counterfactual tightening is the size of one standard deviation of credit growth, well inside the range of shocks in the sample. Second, strategic complementarity implies that adjustment may be faster after large or widely anticipated shocks, when many sellers revise at once (Guren, 2018). For a much larger tightening the model may therefore overstate the delay in prices. The timing result does not hinge on this: with the other parameters at their estimates, raising θ from 0.12 to 0.55 brings the low point in prices forward from eight quarters to two, but the low point in rents still comes six to eight quarters later, and even with $\theta = 1$ the low point in rents comes nine quarters after the tightening begins. What a faster price adjustment would change is when prices bottom out, not that rents lag them. Third, θ may vary with market conditions, for instance if dwellings sell faster in hot markets. A 74-quarter panel cannot estimate that variation, and we leave it to work with transaction-level data.

The speed of price adjustment is similar to the speed of lease repricing ($\omega = 0.35$), so prices and rents are subject to comparable frictions in how fast they reprice. What separates them is not stickiness. At the estimated parameters, credit enters the asset value directly and reaches rents primarily through tenure switching, since the estimated supply response is zero over the matching horizon. This is why rents lag prices even though neither market reprices at once.

6.3. Allowing investor purchase volumes to respond

Measured commitments combine lending conditions, transaction prices and the volume of investor purchases. To test whether the baseline estimate of θ relies on fixing purchases, we retain the same 38 targets, weights, calibration and parameter bounds, but allow purchase volumes to respond to the innovation. Each scenario re-estimates all seven parameters, and then re-estimates the other six with $\theta = 1$.

Write the unscaled lending-condition path as $\widetilde{\phi}_h$, normalised to $\widetilde{\phi}_0 = 1$, and let s denote the share of the non-price commitments response on impact assigned to purchases. In proportional scenarios, $\widetilde{X}_h = [s/(1-s)]\widetilde{\phi}_h$. In separate-decay scenarios,

$$\widetilde{X}_h = \frac{s}{1-s}\rho_X^h, \qquad s \in \{0.25, 0.50, 0.75\}, \quad \rho_X \in \{0.25, 0.75, 0.95\}.$$

The latter allow purchases to decay at a different rate from lending conditions, which can initially build after the innovation. We scale each joint path so that measured credit on impact remains

Table 6: Price adjustment with alternative investor purchase responses

Purchase response	s	ρ_X	$\hat{\theta}$	Distance	Distance with $\theta = 1$
Fixed purchases	0.00	—	0.233	11.39	93.53
Proportional	0.25	—	0.233	11.39	93.53
Proportional	0.50	—	0.233	11.39	93.53
Proportional	0.75	—	0.233	11.39	93.53
Separate decay	0.25	0.25	0.213	12.60	96.79
Separate decay	0.50	0.25	0.209	22.14	110.60
Separate decay	0.75	0.25	0.268	59.82	150.41
Separate decay	0.25	0.75	0.234	11.51	93.70
Separate decay	0.50	0.75	0.221	12.98	96.65
Separate decay	0.75	0.75	0.259	23.10	109.60
Separate decay	0.25	0.95	0.229	11.46	93.60
Separate decay	0.50	0.95	0.221	12.00	94.63
Separate decay	0.75	0.95	0.223	14.81	98.91

The baseline fixes investor purchases. s is purchases' share of the non-price commitments response on impact, not their share of total credit. Proportional scenarios assign the same fraction at every horizon; separate-decay scenarios impose the displayed persistence. Each fit uses multiple starting values and the original weighted least-squares criterion. The last column re-estimates the remaining parameters under instantaneous adjustment. Distances are descriptive and are not χ^2 statistics. The baseline bootstrap interval is not a confidence interval for these scenarios.

0.1006 log points: $\hat{\iota}_h = \hat{\phi}_h + \hat{p}_h + \hat{X}_h$. The scenarios span smaller and larger purchase responses with short and persistent adjustment; they are sensitivity assumptions, not estimates of purchase volumes.

Table 6 distinguishes scale from timing. A proportional purchase response leaves the fitted paths, distance and $\hat{\theta} = 0.233$ unchanged. It rescales ε by $1/(1-s)$: the elasticity rises from 0.011 to 0.015, 0.022 and 0.044. Investor loan numbers suggest that s is close to one: from 2019:Q4 to 2025:Q4, they account for nearly all of the movement in the non-price part of commitments.[4] The baseline estimate $\varepsilon = 0.011$ is then best read as the elasticity of the asset value with respect to the non-price part of commitments, not with respect to the loan-to-value limit. With these targets, a proportional purchase response therefore cannot be told apart from a larger credit elasticity; the separate-decay scenarios are the informative check. The counterfactual in Section 6.6 and the decomposition in Section 6.5 are the same under either reading of ε.

In the separate-decay scenarios, $\hat{\theta}$ ranges from 0.209 to 0.268. Where the fit stays close to the baseline (distance below 15), $\hat{\theta}$ lies between 0.213 and 0.234; it leaves that range only in the three scenarios with distances of 22 to 60. Instantaneous adjustment fits worse in every scenario, with a minimum increase in distance of 82.1. Gradual adjustment is therefore supported beyond the fixed-purchase baseline across this grid. The result shows robustness to the specified positive purchase responses; it does not identify purchase dynamics or test all alternative expectations processes. The baseline bootstrap continues to describe sampling precision conditional on its mapping; the scenario comparison shows how the point estimate changes when that mapping is relaxed.

6.4. What identifies the tenure parameters?

The rent side is identified less sharply. The middle and right panels of Figure 8 profile the distance over κ and η. Both profiles are flat near the estimates. Fixing κ anywhere between 0.1 and 0.4 raises the distance by less than 0.5, and fixing η anywhere between 0.5 and 1 raises it by about 0.2 or less. The full-sample projections do penalise the extremes. An η of 0.2 raises the distance by 11 and an η of 0.05 by 39, so, given these projections, the long-run tenure response cannot be small. A κ of 0.05 raises it by 11 and a κ of 0.02 by 32, so tenure switching cannot

[4]The ABS Lending Indicators report the number of new investor loans by state from 2019:Q3; we use the seasonally adjusted series excluding refinancing. With city fixed effects, we regress quarterly growth in loan numbers on growth in the non-price part of commitments, that is, commitment growth minus price growth. Over 125 city-quarters, the slope is 1.07 and the within R^2 is 0.94; city by city, the slope ranges from 1.04 to 1.19.

be very slow: with η at its bound, slow switching cannot reach the rent path. The bootstrap intervals in Table 5 reach lower values of both parameters because they also carry the sampling error in the rent projections, as we show below. Between these bounds, the projections do not separate fast switching with a small long-run response from slow switching with a large one. Along the κ profile, the re-estimated η falls from 1 at $\kappa = 0.1$ to 0.46 at $\kappa = 0.4$, and θ stays between 0.22 and 0.23.

The reason lies in how each parameter shapes the rent response within twelve quarters. A 10 percent rise in κ raises the model's rent response by about 0.0004 log points at four quarters and 0.0006 at eight, and the effect stops growing in the third year, because faster switching moves the same long-run response forward in time. A 10 percent rise in η raises the rent response by about 0.0005 at four quarters, 0.0014 at eight and 0.0020 at twelve, and the effect keeps growing, because a larger long-run response raises the level that rents approach. Both parameters raise the rent path over the window, and they differ mainly in where it levels off, which lies beyond twelve quarters. A longer window would separate them, but the projections at longer horizons are too imprecise, and too dependent on the pandemic years, to help.

What the data identify is the rent path over twelve quarters, not the product $\kappa\eta$. Proposition 4(ii) links the rent response one quarter after impact to $\omega\kappa\eta$ times the impact price response, but that rent response, about 0.001 log points, is small relative to its standard error and carries little weight in the distance. A calculation from the first two horizons alone gives $\kappa\eta \approx 0.37$, against 0.14 when all twelve horizons are matched. Holding $\kappa\eta$ at its estimate while moving κ from 0.22 to 0.4 raises the distance from 11.4 to 18.5, because the rent path then levels off too early. The bootstrap tells the same story. Across the 399 draws, the 90 percent interval for $\kappa\eta$, $[0.000, 0.62]$, is no narrower relative to its median than those for κ and η. The two parameters are negatively related across draws (correlation -0.52), but that trade-off does not narrow their product. Much of the bootstrap spread therefore reflects the imprecision of the rent projections themselves: across draws, the model's rent response at twelve quarters ranges from about zero to 0.023 log points.

The two other rent-side parameters behave differently. The supply response is at zero in the full sample and in 67 percent of the draws, because within twelve quarters the projections show no supply offset (Section 5.3). The projections carry little information about ω: fixing it anywhere between 0.075 and 1 and re-estimating the other parameters leaves the distance between 10.4 and 11.4 and θ between 0.22 and 0.24, although an ω of 0.05 raises the distance to 13.8. We therefore fix ω at the estimate from the dynamic rent equation, which uses the full rent series. Its bootstrap interval, $[0.24, 0.66]$, is wider than the Driscoll–Kraay standard error of Section 5.1 implies and skewed upward. The bootstrap resamples whole blocks of quarters, and a 74-quarter panel holds only about nine of them, so the interval reflects how much the persistence of rent growth varies across subperiods. Rents on new leases, which we observe for four of the cities, do not narrow it (Online Appendix F). The timing of CPI rents behind rents on new leases points to slower repricing, about 0.12 to 0.15 a quarter, if CPI rents absorb new-lease rents one for one. They absorb only about 60 to 80 percent, however, and once that share is estimated as well, the estimates range from 0.11 to 0.58. Because the fit and θ barely change for ω between 0.075 and 1, the choice matters mainly for how the delay in rents divides between leases and tenure switching (Section 6.5).

Five cities and 74 quarters are a small sample for a structural model, so we check how the estimator behaves at this sample size. We treat the estimated model as the truth and build 200 pseudo-samples. Each adds to the model's impulse responses the sampling error of one bootstrap draw, that is, the draw's local projections minus the full-sample projections, so the pseudo-samples carry both the size of the sampling error in this panel and its correlation across horizons. We then re-estimate the model on each, with that draw's estimate of ω, by the same search as in the bootstrap. The estimator recovers θ with little bias: the median estimate is 0.228 against a true 0.233, 78 percent of the estimates lie within 0.10 of the truth, and 3 percent exceed 0.5. The median of κ is also close to the truth (0.21 against 0.22), but the estimates

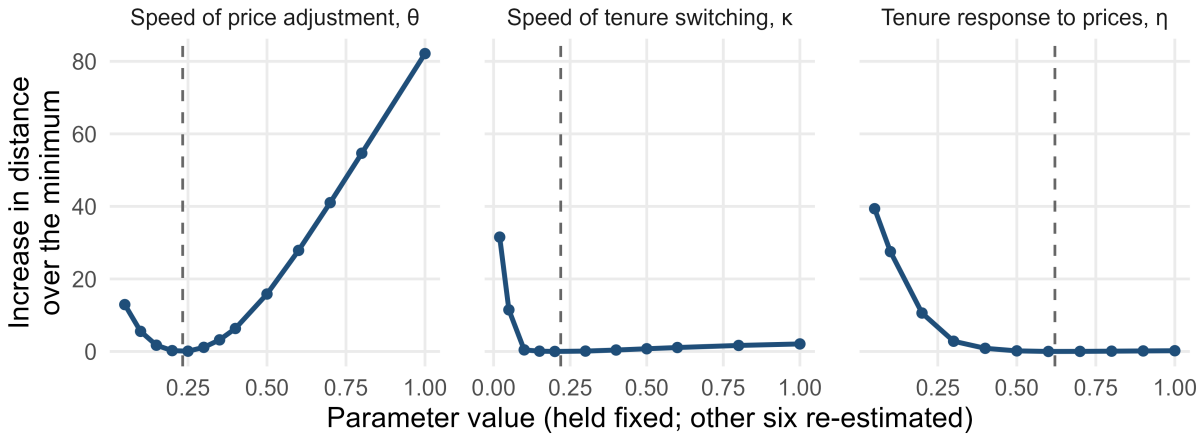


Figure 8: Profiles of the matching distance. Each panel holds one parameter fixed at the values shown, re-estimates the other six, and plots the increase in the distance over its minimum of 11.39. Dashed lines: the estimates in Table 5. The distance weights each moment by its inverse squared standard error and ignores the correlation across horizons, so the increases are descriptive, not χ^2 statistics.

are widely spread, from 0.03 to 0.66 for 90 percent of them. The estimates of η are spread over its whole range and are biased upward, with a median of 0.71 against a true 0.62. The estimator recovers the true ranking of the two frictions in 80 percent of the pseudo-samples, but it understates the share of the lease friction (median 0.06 against a true 0.11). The wide bootstrap intervals for the tenure parameters therefore reflect weak identification more than bias, and the interval for θ is a fair guide to its precision. The exercise assumes the model is correctly specified, and it inherits any error in how the block bootstrap represents sampling variation with only nine blocks.

These identification results set what the model can and cannot deliver. The speed of price adjustment is tightly estimated within the baseline, and the projections establish the ordering of the responses. The size of the rent response over three years is estimated less precisely: its 90 percent interval at twelve quarters runs from about zero to 0.023 log points (Table 7). The split of the delay in rents between the speed and the size of the tenure response is not determined, and the decomposition below inherits that imprecision. Data on tenure flows, such as transitions of renter households into ownership or the share of first-home buyers in purchases, would identify κ directly and are the natural next step.

6.5. Accounting for the delay in measured rents

The estimated model lets us switch each friction off and measure its contribution to the wedge between the price and rent responses. This exercise depends on the split between the speed and the size of tenure switching, which Section 6.4 shows the data identify only weakly, so its results are less secure than the speed of price adjustment and the timing of the two responses. Table 7 reports the rent response, and the wedge at twelve quarters, when leases reprice at once ($\omega = 1$), when tenure adjusts at once ($\kappa = 1$), and when both do. At the estimates, slow tenure switching is the larger friction. Removing it raises the rent response at four quarters from 0.005 to 0.011 log points and lowers the twelve-quarter wedge by about 22 percent. Removing the lease friction alone lowers the wedge by about 11 percent. Both shares are imprecise, because they depend on κ, η and ω, which the data identify only within wide bounds (Section 6.4). The ranking is more robust but not decisive: removing slow tenure lowers the wedge more than removing slow leases in 71 percent of the draws, and in 73 and 69 percent with blocks of four and twelve quarters. In 26 percent of the draws κ is at its upper bound, so slow tenure contributes nothing. In about two-fifths of these, letting leases reprice at once lowers the rent at twelve quarters, because the price response has already peaked and rents on older leases sit above the market rent; slow leases then narrow the wedge, and slow tenure ranks first. Among the other

draws, removing slow tenure lowers the wedge more in 81 percent (81 and 75 percent with blocks of four and twelve quarters). The ranking also depends on where on the ridge of Figure 8 the estimate lies. Fixing κ and re-estimating the other parameters, slow tenure removes more of the wedge than slow leases for κ up to 0.3, and the ranking reverses from $\kappa = 0.4$, where the fit is almost as good (distance 11.8). With such fast switching, however, neither friction removes more than 5 percent of the wedge, because the long-run tenure response is then small ($\eta \approx 0.46$) and the gap between prices and rents at twelve quarters is mostly a lasting shortfall rather than a delay. The data therefore establish that rents lag prices; they do not settle whether the gap that remains after three years reflects slow tenure switching or a small long-run tenure response.

At the baseline estimates both frictions contribute to the delay. The direct new-lease comparison supports the lease component more clearly than the inferred tenure component. Removing the lease friction raises the rent response by about 0.003 log points at four quarters, and by less at twelve, whether κ is 0.1, 0.22 or 0.4. Its baseline contribution is disciplined by the separately estimated ω from the dynamic rent equation, and it is concentrated in the first two years. Removing slow tenure switching raises the rent response at four quarters by 0.003 log points at $\kappa = 0.4$, 0.006 at the estimates, and 0.014 at $\kappa = 0.1$. Its contribution is therefore comparable to that of leases or larger, and larger at the estimates, but its size is uncertain. Removing both frictions raises the rent response at four quarters from about 0.005 log points to between 0.012 and 0.026 across these values, so together they explain much of the initial delay in rent adjustment. This is distinct from the remaining price–rent wedge: at the estimated parameters, removing both frictions reduces the twelve-quarter wedge by about 23 percent, rather than eliminating most of it. With both frictions removed, a wedge of 0.011 log points remains at twelve quarters, because in the model the long-run tenure response is incomplete ($\hat{\eta} < 1$) and there is no supply response within the window: rents rise by less than the asset value, which also capitalises the value of borrowing capacity. That part of the wedge reflects the incomplete pass-through of credit conditions to rents relative to their direct capitalisation into asset values, rather than delayed adjustment.

Rents on new leases offer a check on this split (Online Appendix F). In the model they are the market rent, which a credit innovation moves only through tenure switching. In the four cities with bond data, they rise by 0.030 log points by twelve quarters after a credit innovation, close to the price response of 0.035, while CPI rents rise by 0.017. In the estimated model the market rent rises by 0.018 (the row with $\omega = 1$ in Table 7), against 0.030 for prices and 0.016 for measured rents. Taken at face value, the data therefore put more of the twelve-quarter wedge between rents on new and on current leases (0.013, against 0.002 in the model) and less between prices and market rents (0.005, against 0.012). Two caveats apply. The gap between prices and rents on new leases is imprecise: its 90 percent bootstrap interval at twelve quarters runs from -0.009 to 0.028. And CPI rents absorb only part of the movement in rents on new leases. If they absorb 58 percent, which fits the two responses about as well at $\omega = 0.35$ as full absorption does at $\omega = 0.12$, the implied market-rent response at twelve quarters is about 0.017, close to the model's. The data on new leases therefore support the role of lease repricing in delaying measured rents, but they do not settle how the delay divides between leases and tenure switching. The quantitative decomposition is conditional on the mapping from market rents to CPI rents; the new-lease evidence does not uniquely determine the repricing rate or the relative friction contributions.

6.6. A counterfactual tightening

We simulate a four-year tightening of investor credit to quantify the estimated model's transmission mechanism. The duration mirrors the APRA period, but the exercise does not estimate the effect of a specific intervention. The non-price part of investor commitments falls by the amount implied on impact by a one-standard-deviation credit innovation, remains at that level for sixteen quarters, and then returns to its baseline path. The reversal is anticipated from the outset. The fall can come through loan-to-value limits or through the number of investor

Table 7: Contribution of each friction to the delay in rents

	Rent response, quarter			Wedge,	Share of
	4	8	12	quarter 12	wedge removed
Estimated model	0.005 [0.000, 0.009]	0.012 [0.000, 0.018]	0.016 [−0.001, 0.023]	0.0141 [0.0027, 0.0326]	
Leases reprice at once ($\omega = 1$)	0.008	0.015	0.018	0.0125	0.11 [−0.09, 0.25]
Tenure adjusts at once ($\kappa = 1$)	0.011	0.018	0.019	0.0110	0.22 [0.00, 1.21]
Both at once	0.016	0.020	0.019	0.0109	0.23
Draws in which slow tenure adds more to the wedge than slow leases: 0.71					

Responses in log points to a one-standard-deviation investor-credit innovation in the estimated model, changing one parameter at a time. The wedge is the price response less the rent response; the price response at twelve quarters is 0.030, almost unchanged across rows, since rents feed back into the asset value only through the small weight $1 - \varpi$. The supply response is estimated at zero, so switching it off changes nothing. The share of the wedge removed is one minus the ratio of the wedge in that row to the wedge in the estimated model, computed from unrounded values; a share above 1 means that rents overtake prices at twelve quarters once the friction is removed. In brackets: 90 percent intervals from 399 block-bootstrap draws, each re-estimating the projections, ω and the model (Table 5).

loans; when the two move in proportion, the split between them does not affect the results (Section 6.3).

Figure 9 shows the resulting price and rent paths. Prices fall on impact and reach a trough about 1.0 percent below baseline after six quarters. Rents barely move in the first year, then decline to a trough about 0.5 percent below baseline after twelve quarters, a year before the tightening ends. The peak decline in rents is therefore about half that in prices. Prices recover faster, leaving rents further below baseline than prices when the tightening ends and during the subsequent recovery.

The bootstrap bands in Figure 9 show which of these features the data pin down. The size of the effects is uncertain: the 90 percent interval for the fall in prices at its low point is 0.3 to 4.1 percent, and for rents 0.01 to 1.1 percent. The timing is better determined. Prices reach their low point after four to eight quarters, rents after about five to eighteen, and rents reach their low point later than prices in 94 percent of the draws (93 percent with blocks of four quarters and also with blocks of twelve). In the model, then, a macroprudential tightening lowers dwelling prices within a year, while rents adjust more gradually, reach their low point about six quarters after prices, and stay below their path after the tightening ends.

6.7. Credit conditions and measured inflation

The CPI rent index measures rents on all current leases, which is the measured rent r_t of Equation (5). The estimated responses therefore translate directly into the response of the rent component of the CPI. Rents carry 6.6 percent of the Australian CPI (Australian Bureau of Statistics, 2024). In the United States, rent of primary residence and owners' equivalent rent, both built from rents on current leases, carry about 34 percent (U.S. Bureau of Labor Statistics, 2025). The gap reflects how the two indices treat owner-occupied housing, not how many households rent. The US CPI prices it by the rent that owners' homes would fetch, while the Australian CPI prices it by the cost of new dwellings, which carry 7.6 percent. If the Australian CPI priced owner-occupied housing by imputed rents, rents would carry about 25 percent.[5] Recent work shows that official rent indices lag market rents, mainly because most tenants' rents are set in leases that reprice only occasionally (Adams et al., 2024; Ball and Koh,

[5]In the national accounts for 2021–22, owner-occupiers' imputed rents are 15.5 percent of household final consumption expenditure and actual rents 4.6 percent (Australian Bureau of Statistics, 2023). We add imputed rents to the CPI basket in that ratio to the weight of actual rents, $6.61 \times 15.5/4.6 = 22.3$ percentage points, and remove new dwelling purchases by owner-occupiers, 7.59 percent (Australian Bureau of Statistics, 2024). Rents then carry $(6.61 + 22.3)/(100 - 7.59 + 22.3)$, or 25.2 percent, of the basket.

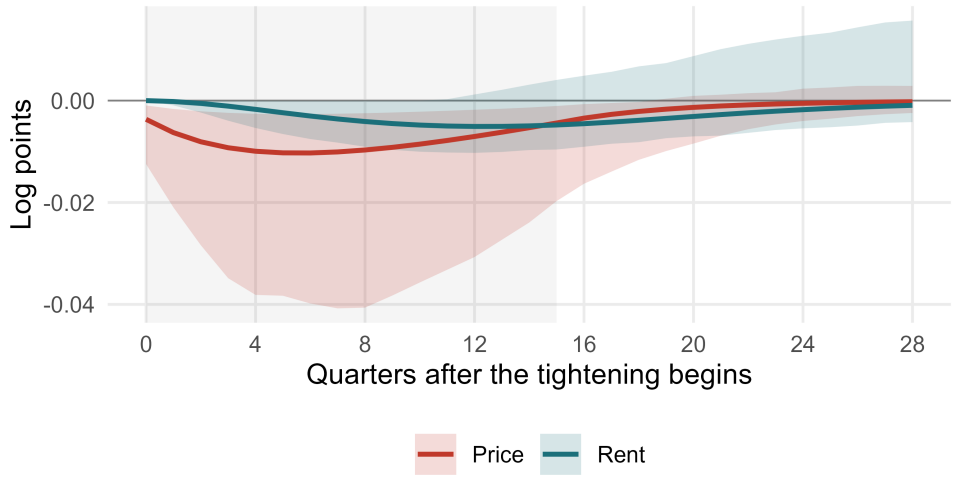


Figure 9: Counterfactual tightening in the estimated model. The non-price part of investor commitments falls for sixteen quarters (shaded) by the amount behind a one-standard-deviation credit innovation on impact and then returns, with the end foreseen. Price and rent levels in log points relative to their paths without the tightening. Bands: 90 percent pointwise intervals from 399 block-bootstrap draws, each re-estimating the projections, ω and the model.

2025). Our model adds a second source of delay, slow tenure switching, and links both to credit conditions. In the four cities with bond data, CPI rent growth is most closely correlated with growth in rents on new leases two to five quarters earlier (Online Appendix F).

Figure 10(a) shows annual inflation after a one-standard-deviation credit innovation. House-price inflation peaks at about 2.6 percent three quarters after the innovation and has largely faded by the end of the second year. Rent inflation rises slowly, peaks at about 0.7 percent seven quarters after the innovation, and is still about 0.45 percent at twelve quarters, when house-price inflation has turned slightly negative. Across the bootstrap draws, rent inflation peaks after house-price inflation in every draw in which rents rise at all, 95 percent of them, with a median lag of three quarters, and the lag is between one and seven quarters in 90 percent of the draws. House-price inflation peaks after three quarters in almost every draw, because prices rise on impact and then rise by less each quarter; the speed of price adjustment, θ, mainly sets how fast it then fades. Leases and tenure switching set the further delay in rent inflation.

Panel (b) shows the same rates during and after the counterfactual tightening of Section 6.6. House-price inflation is lowest after three quarters and turns positive after eight, while the tightening is still in force. Rent inflation is lowest after seven quarters and turns positive only as the tightening ends, after fifteen quarters. Rent inflation reaches its low point after house-price inflation in every draw in which rents fall at all, 95 percent of them. A policymaker who reads rent inflation in the CPI as a gauge of current credit conditions would therefore see conditions that had changed a year or more earlier. Rents on new leases, which respond within the first year (Section 5.4), would give an earlier reading.

In Australia the effect on headline inflation is small. After a one-standard-deviation credit innovation, rents add at most 0.05 percentage points to annual CPI inflation and 0.11 percentage points to the CPI level by twelve quarters. If owner-occupied housing were priced by imputed rents, the same responses would add about 0.18 percentage points at the peak and 0.41 percentage points to the level, and at the United States weight about 0.25 and 0.56 percentage points. The robust lesson is the timing, not the size. Two caveats apply. The shock is to credit conditions, not to the policy rate, which in the model moves prices only through the value of borrowing capacity and which we do not identify. The results therefore speak directly to macroprudential measures, such as the APRA lending benchmarks, and to monetary policy only to the extent that it works through credit conditions. And the rent responses carry the uncertainty of the rent projections, so the size of the rent inflation peak is imprecise: its 90 percent interval runs from zero to about 1.1 percent.

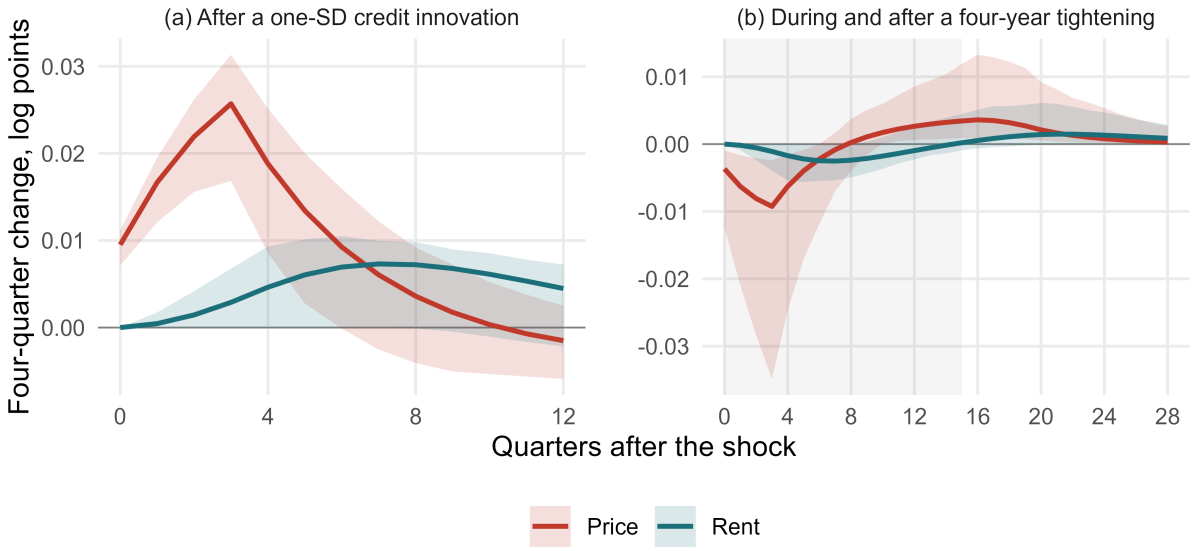


Figure 10: Annual (four-quarter) inflation of house prices and of measured rents, the rent component of the CPI, in the estimated model. Panel (a): after a one-standard-deviation investor-credit innovation. Panel (b): during and after the counterfactual tightening of Figure 9; the tightening lasts sixteen quarters (shaded). Bands: 90 percent pointwise intervals from 399 block-bootstrap draws, each re-estimating the projections, ω and the model.

6.8. How large should the state dependence be?

Proposition 3 predicts that the price response to credit is larger when interest rates are low. The reduced form does not detect this: the interaction of credit growth with the lagged cash rate is -0.005 (s.e. 0.012), with a 90 percent interval of $[-0.025, 0.015]$ (Online Appendix E). The model tells us how large the interaction should be. By Proposition 3, $\varepsilon = \bar{\lambda}\bar{\phi}/(1-\bar{\lambda}\bar{\phi})$ with $\bar{\lambda} = 1 - \Lambda(1+\bar{\imath})$, so the ratio of the credit elasticity at two cash rates depends only on the investors' discount factor Λ and the loan-to-value limit $\bar{\phi}$. We set $\bar{\phi} = 0.8$, closer to observed investor leverage than the limits the steady state implies (Section 6.1); the calculation uses only the ratio of the elasticities at two cash rates, not the matching. We scale the predicted reduced-form slope to 0.130, the price–credit coefficient at the mean cash rate with current credit in the cash-rate interaction specification (Online Appendix E). With an annual discount rate of 6 percent, the elasticity is 3.6 times larger at a cash rate of 0.1 percent than at 4.35 percent, and the implied interaction is -0.042 per percentage point, about three standard errors below the estimate. With a discount rate of 10 percent, the implied interaction is -0.019, inside the interval. The predicted interaction enters the interval once the annual discount rate exceeds about 8.1 percent. The data are therefore consistent with Proposition 3 only if investors are impatient enough that the value of borrowing capacity, $\bar{\lambda}$, is large and changes little with the policy rate. For more patient investors, the model predicts a state dependence that the panel would have detected.

7. Discussion

We interpret the results for housing economics, regional analysis, and metropolitan policy.

7.1. Two markets, two sets of drivers

The central point is that different forces govern rents and dwelling prices, which move together only while broad housing demand drives both. Rents track local fundamentals throughout the sample: the balance of migration against new completions, wage growth, and employment, with the weight on employment rising over time. Prices are most consistently associated with investor credit, and the cost of capital carries its expected sign only when the cash rate is lagged

one quarter. Even then, its estimate is not significant. In the model, the market rent depends on renter incomes and on renter households per dwelling, while the price also capitalises the value of borrowing capacity. The price–rent disconnect is therefore not an unexplained correlation. It reflects both different drivers and different adjustment speeds: investor credit is more closely associated with dwelling prices than rents, while gradual price adjustment, tenure switching, and staggered lease repricing delay the transmission of credit conditions to measured rents. The difference in drivers is the mechanical result of the two outcomes loading on different right-hand sides. When broad housing demand, the driver common to both, stops dominating, the two series separate. Prices then show the financial driver they load on throughout the sample.

This reframes the disconnect that single-city studies have documented (Egan and McQuinn, 2023; Greenaway-McGrevy and Phillips, 2023). It is not a puzzle that needs a city-specific explanation. It is the visible surface of a more general fact: rents are the price of a service and stay tied to local fundamentals, while prices value an asset and become tied to the conditions of asset finance. The disconnect is what happens when asset-finance conditions move independently of local fundamentals, as they did especially during the pandemic.

Table 8 sets each prediction of the model against the estimates. The predictions about what drives rents, and about the timing of the two responses, hold. The implication for lease repricing holds only in part: the dynamic rent equation gives $\hat{\omega} \approx 0.35$, but rents on new leases do not narrow it. Long-run pass-through of renter income is close to one, as the model implies, but the wage index alone passes through only about half. For prices, credit enters as predicted. The cash rate takes the predicted sign only when lagged, and that coefficient is not significant. The data give no support to the prediction that low interest rates raise the price loading on credit, unless investors are highly impatient (Section 6.8). The loading does not vary with the level of the cash rate (Online Appendix E). It does not rise after 2017, when rates were lower for much of the period, and where it rises, in Sydney and Adelaide, the change is not common to the five cities. The test with the cash-rate level also finds one pattern the model does not predict: the rent–credit slope, close to zero on average, is positive when the cash rate is low and slightly negative when it is high.

7.2. When and where the credit link changed

The data do not date a strengthening of the investor channel in prices to 2017. With seasonally adjusted credit, the changes in the price–credit slope after 2017 and at the two APRA dates are small and not significant (Table C.2), and the largest movements in the rolling decomposition coincide with the credit swings of the pandemic (Figure E.2). Nor is the change concentrated where state policy changed: the post-2017 rise in the slope appears in Sydney and Adelaide but not in the other three capitals (Online Appendix E). The data support a strong link between investor credit and prices across the whole sample, set by national credit conditions, not a shift that began in 2017.

7.3. Implications and external validity

The results describe a two-speed system. Settings that shape investor finance nationally, such as macroprudential limits and the tax treatment of investors, act on prices within a few quarters and reach rents only over about three years, while migration relative to new supply, wages, and employment continue to move rents directly. The counterfactual in Section 6.6 puts numbers on this lag. We do not estimate the welfare or affordability effects of any reform. The mechanism also yields predictions beyond Australia: the disconnect should be widest where investors are the marginal buyers and borrow against the dwellings they buy, and weaker where landlords price on cash flow or where public provision meets much of the demand for rental housing. Canada's foreign buyer taxes, which lowered prices temporarily (Du et al., 2022), offer one test: the model predicts a smaller and delayed effect on rents.

Table 8: The model's predictions and the estimates

Prediction	Source	Estimate	Holds?
Rent growth rises with tightness in levels, wage growth, and employment growth, with income coefficients below one	Proposition 1, Lemma 1	Tightness 0.006, wages 0.73, employment 0.14, all $p < 0.01$ (Table 3)	Yes
Measured rents close a fraction ω of the gap to market rents each quarter; long-run pass-through of wages and employment of one	Prop. 1(iv)	$\hat{\omega} \approx 0.35$; long-run wage coefficient 0.53, s.e. 0.20; wages and employment together 0.91, s.e. 0.29 (Section 5.1); CPI rents absorb only 0.6–0.8 of the movement in rents on new leases, and estimates of ω from them range from 0.11 to 0.58 (Online Appendix F)	Partly
Market rents follow their drivers at once; measured rents follow market rents with a delay	Eqs. (2), (5)	Rents on new leases: persistence 0.08, against 0.39 for CPI rents; long-run tightness and wage coefficients as for CPI rents; credit response significant from the first quarter, that of CPI rents from the sixth (Online Appendix F)	Yes
Prices rise with credit conditions and fall with the policy rate; tightness matters little	Proposition 2	Credit 0.138 current and 0.133 lagged, $p < 0.01$; cash rate −0.003 lagged, not significant; tightness not significant at 5% (Table 4)	Partly
Neither timing of credit recovers the credit elasticity; the model does not rank them	Corollary 1	Slopes close: 0.138 and 0.133 (Table 4); before 2017, 0.106 current and 0.140 lagged (Table C.1); lagged slope 0.065 with a price-momentum control	Consistent
Price loading on credit higher when interest rates are low, as for much of the period after 2017	Proposition 3	Interaction of credit with the cash-rate level −0.005 ($p = 0.70$) current and 0.004 ($p = 0.68$) lagged (Online Appendix E); post-2017 change 0.048 ($p = 0.28$) and −0.011 ($p = 0.79$) (Table C.1); no break at 2015:Q1, 2017:Q1 or 2017:Q2 (Table C.2); consistent with the model only if the annual discount rate exceeds 8.1 percent (Section 6.8)	No support
A change driven by interest rates would appear in every city	Proposition 3	Sydney-specific post-2017 term 0.136 ($p < 0.001$); the slope rises only in Sydney and Adelaide (Online Appendix E); the effective limit can differ with the investor mix, and Sydney's investor share is highest	Partly
Prices respond to a credit innovation on impact; rents do not, and then respond with a delay	Proposition 4	Prices 0.008 on impact and 0.029 by the third quarter; rents 0.004 by the fourth quarter and 0.018 by the twelfth (Figure 5); same shape with labour-market controls and after impact	Yes
Credit reaches rents only through past price growth	Prop. 1(ii)	Price lags matter for rents ($p < 0.001$); credit lags also add ($p = 0.04$) (Section 5.3)	Partly
With gradual price adjustment ($\theta < 1$), the price response builds after impact	Prop. 4(i)	$\hat{\theta} = 0.23$, 90% bootstrap interval $[0.12, 0.38]$; $\theta = 1$ raises the distance from 11.4 to 93.5 (Table 5)	Yes

8. Conclusion

This paper documents and models a clear ordering in Australian housing markets: dwelling prices move first, rents on new leases follow, and CPI rents adjust later. Investor credit comoves strongly with prices, while rent growth tracks migration relative to completions, wages and employment. The timing survives controls for labour-market conditions and the current price change. On the matched four-city sample, new-lease rents approach the price response within two years; much of the measured three-year price–rent gap lies between new and existing leases. This makes the distinction between market rents and the CPI rent stock central to understanding the disconnect.

Our semi-structural model combines collateral-based benchmark valuation with gradual transaction-price adjustment, tenure switching and lease repricing. It fits the price, rent and credit responses. In the baseline, transaction prices close about 23 percent of the gap to the

asset value each quarter, and instantaneous adjustment fits substantially worse. The purchase-volume exercise in Section 6.3 supports gradual adjustment beyond the fixed-purchase baseline: the estimated speed ranges from 21 to 27 percent across the separate-decay scenarios. The bootstrap measures sampling uncertainty within the baseline; the purchase-volume scenarios measure how much the estimate depends on the commitments mapping.

The evidence on rents is strongest about timing. The model assigns a role to both lease repricing and tenure adjustment, but it does not sharply separate the speed of tenure switching from its long-run size. New-lease evidence supports an important lease component without establishing a precise ranking of the two frictions. The decomposition quantifies these mechanisms within the model, while data on tenure flows would test the tenure channel directly. The APRA instruments provide a limited check on the response pattern and do not independently identify the speed of price adjustment.

An illustrative four-year tightening of investor credit in the baseline model lowers prices within a year and rents over about three years. CPI rent inflation lags house-price inflation by about a year in this simulation. The implication is that measured rent inflation can continue to reflect earlier financial conditions after the housing asset market has turned. These results explain a medium-run disconnect; they do not establish that rents are permanently insulated from financial conditions.

Further findings bound the result. With seasonally adjusted credit, the link between investor credit and prices does not change significantly at 2017 or at either APRA date. An unknown-date test places a break in 2021:Q1, but only with current-quarter credit, not with lagged credit. The apparent strengthening in the unadjusted series reflects seasonal noise that faded over the sample. The data are consistent with the model's prediction that lower interest rates raise the price loading on credit only if investors are highly impatient (Section 6.8). Estimated city by city, the post-2017 change is mixed: the slope rises in Sydney and Adelaide but not elsewhere. The link is therefore better understood as a lasting feature of how Australian dwelling prices are determined than as the effect of any single reform.

For policy, affordability interventions act on a rental market still anchored to fundamentals, even as the ownership market comes apart from them. The model places measures affecting investor finance, including macroprudential limits, in the asset-valuation channel; evaluating a specific policy requires identifying its effect on lending conditions and purchases. Rents continue to move with migration relative to new supply, and with wages and employment.

The mechanism yields predictions beyond Australia: where investor finance has expanded, the same separation of rent and price drivers should appear (Section 7.3). The most valuable extension would be a well-powered cross-sectional test of how the strength of this separation varies across cities, which needs sub-metropolitan (SA4/SA3) data. Data linking landlords' borrowing terms to the dwellings they let, and tenure to rents, would test the model's collateral and tenure channels directly.

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Data availability

The data and code required to reproduce the results in this paper are openly available in the replication repository at `https://github.com/Ainura-tur/price-rent-disconnect`. The repository contains the full estimation pipeline and the raw input series obtained from the Australian Bureau of Statistics (under CC BY 4.0), the Reserve Bank of Australia, and the Bank for International Settlements, each redistributed under its source's terms with attribution as documented therein. Rents on new leases are built from the rental bond reports that the governments of New South Wales, Victoria, Queensland and South Australia publish under Creative Commons Attribution licences. The repository includes the resulting indices, with the

attribution each source requires, and its README lists where to download the reports (Online Appendix F).

Declaration of competing interest

The authors report there are no competing interests to declare.

Declaration of generative AI and AI-assisted technologies in the manuscript preparation process

During the preparation of this work the authors used Claude (Anthropic) and ChatGPT (OpenAI) for language editing, code assistance, code review, and consistency checking. After using these tools, the authors reviewed and edited the content as needed and take full responsibility for the content of the published article.

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Appendix A Model Derivations and Proofs

This appendix derives the results of Section 3. Table A.1 pairs each object in the model with its counterpart in the data.

Table A.1: Model objects and their counterparts in the data

Model object	Counterpart in the data	Note
Measured rent r_t, Eq. (5)	CPI rent index (ABS 6401.0), $\Delta \ln$ rent	Rents on all current leases, not new leases
Market rent r_t^*, Eq. (2)	Rents on new bonds, four cities (Online Appendix F)	Used to test Eq. (5), not in the estimation. The gap $r_{t-1}^* - r_{t-1}$ falls into the error term of the static rent equation; lagged rent growth absorbs it in the dynamic form (Proposition 1)
Renter income $W_t e_t$	Wage growth Δw (WPI) and employment growth Δn	Employment growth proxies the growth of employment per household
Growth of renter households relative to the rental stock (Lemma 1)	Tightness, TIGHT_{t-1}	Model: single-quarter ratio M_t/C_t. Data: trailing four-quarter ratio of migration to completions, entered with a one-quarter lag.
Tenure switching $\hat{s}_t$, Eq. (4)	Lagged price–rent gap $(p - r)_{t-1}$; lagged price growth	The price–rent gap is an imperfect proxy for the affordability term in the tenure rule; lagged price growth provides a complementary test of the transmission channel (Section 5.3).
Credit conditions ϕ_t	Investor loan commitments, $\Delta\iota$	Contain Δp_t (Corollary 1); the APRA measures lower ϕ_t (Online Appendix D)
Policy rate i_t, through λ_t	Cash rate, ΔCASH	Enters in differences because the price equation is differenced
Dwelling price P_t	Spliced capital-city price index	Measures the transaction price P_t, not the asset value V_t; growth after 2021:Q4 comes from series that are not quality-adjusted
Completions C_t	Dwelling completions (ABS 8752.0)	Denominator of tightness

A.1 Renter demand

The first-order conditions of (1) give $1/c_t = \mu_t$ and $\gamma/q_t = \mu_t R_t^*$, where μ_t is the multiplier on the budget constraint. Hence $R_t^* q_t = \gamma c_t$, and the budget constraint gives $R_t^* q_t = \sigma W_t e_t$ with $\sigma = \gamma/(1+\gamma)$. Substituting $q_t = K_t/H_t^R$ gives (2).

A.2 Investor problem

With λ_t the multiplier on (6) in current-value terms, the frictionless benchmark investor's Lagrangian is

$$\mathcal{L} = \mathbb{E}_t \sum_{k \geq 0} \Lambda^k \Big\{ d_{t+k} + \lambda_{t+k} \big[\phi_{t+k} V_{t+k} K_{t+k+1} - B_{t+k} \big] \Big\}.$$

The first-order condition for B_t is $1 - \Lambda(1+i_t) - \lambda_t = 0$, which gives (7). Since $\Lambda(1+i_t) < 1$, $\lambda_t > 0$ and the constraint binds. The first-order condition for K_{t+1} is

$$-V_t + \lambda_t \phi_t V_t + \Lambda\, \mathbb{E}_t \big[R_{t+1} + (1-\delta) V_{t+1} \big] = 0,$$

which rearranges to (8). Dividing by $1-\lambda_t\phi_t$ gives $V_t = \chi_t\,\mathbb{E}_t[R_{t+1}+(1-\delta)V_{t+1}]$. Substituting forward and imposing $\lim_{k\to\infty}\mathbb{E}_t[(1-\delta)^k\prod_{m=0}^{k}\chi_{t+m}V_{t+k+1}]=0$ gives (9).

A.3 Log-linearisation of the price equation

The asset value solves (8). In the steady state $\bar{V}=\bar{P}$, and $\bar{P}(1-\bar{\lambda}\bar{\phi})=\Lambda[\bar{R}+(1-\delta)\bar{P}]$, so $\bar{R}+(1-\delta)\bar{P}=\bar{P}/\bar{\chi}$ and $\bar{P}/\bar{R}=\bar{\chi}/(1-\varpi)$ with $\varpi=(1-\delta)\bar{\chi}$. Taking logs of the equation for V_t and linearising,

$$\hat{v}_t-\frac{\bar{\lambda}\bar{\phi}}{1-\bar{\lambda}\bar{\phi}}(\hat{\lambda}_t+\hat{\phi}_t)=\frac{\bar{R}}{\bar{R}+(1-\delta)\bar{P}}\,\mathbb{E}_t\hat{r}_{t+1}+\frac{(1-\delta)\bar{P}}{\bar{R}+(1-\delta)\bar{P}}\,\mathbb{E}_t\hat{v}_{t+1}.$$

The two weights equal $\bar{\chi}\bar{R}/\bar{P}=1-\varpi$ and $(1-\delta)\bar{\chi}=\varpi$, which gives (10). From (7), $\mathrm{d}\ln\lambda_t=-\Lambda\,\mathrm{d}i_t/\bar{\lambda}$. The steady state is positive and the forward solution converges because $\varpi=(1-\delta)\Lambda/(1-\bar{\lambda}\bar{\phi})<1$ by assumption. For the calibration of Section 6.1, the steady state gives $\bar{\phi}=[1-\Lambda(1-\delta)/\varpi]/\bar{\lambda}$. With $\varpi=0.99$, $\delta=0.002$ and the mean cash rate, this limit is 0.89 at an annual discount rate of 6 percent and 0.95 at 10 percent, so the calibration lies inside the restriction $\phi_t\in(0,1)$ of Equation (6). Solving forward gives the first part of (18); iterating (11) backward gives the second.

A.4 Proof of Lemma 1

Taking log differences of (2), $\Delta r_t^*=\Delta w_t+\Delta\ln e_t+\Delta\ln H_t^R-\Delta\ln K_t$. By (14), $\Delta\ln K_t\approx-\delta+\psi C_t/K_{t-1}$. By (3), $\Delta\ln H_t^R=\nu M_t/H_{t-1}^R+\Delta\hat{s}_t$. With $K_{t-1}\approx H_{t-1}^R$, $\nu M_t/H_{t-1}^R\approx\nu(M_t/C_t)(C_t/K_{t-1})=\nu g_t\,\mathrm{TIGHT}_t$ and $\psi C_t/K_{t-1}=\psi g_t$. Setting $g_t=\bar{g}$ gives (15). □

A.5 Proof of Proposition 1

From (5), $\Delta r_t=\omega(r_t^*-r_{t-1})=\omega[\Delta r_t^*+(r_{t-1}^*-r_{t-1})]$, which is (16). Parts (i) and (ii) follow from (15) and (4): prices enter only through $\Delta\hat{s}_t$, which depends on $\hat{p}_{t-1}$ and earlier values.

For part (iii), let $G_t\equiv r_t^*-r_t$. By (5), $G_t=(1-\omega)(G_{t-1}+\Delta r_t^*)$, so in a stationary state $G_{t-1}=\sum_{k\geq1}(1-\omega)^k\Delta r_{t-k}^*$. With $\Delta r_t^*=x_t$ and $\mathrm{Cov}(x_t,x_{t-k})=\rho_x^k\sigma_x^2$,

$$\mathrm{Cov}(\Delta r_t,x_t)=\omega\sigma_x^2\Big[1+\sum_{k\geq1}(1-\omega)^k\rho_x^k\Big]=\frac{\omega\,\sigma_x^2}{1-(1-\omega)\rho_x}.$$

Dividing by σ_x^2 gives the probability limit $\omega/[1-(1-\omega)\rho_x]$. It equals ω when $\rho_x=0$, tends to one as $\rho_x\to1$, and is increasing in ρ_x. If x_t enters Δr_t^* with a coefficient b, as tightness does with $b=\nu\bar{g}$, the limit is multiplied by b.

For part (iv), difference (5): $r_t-r_{t-1}=(1-\omega)(r_{t-1}-r_{t-2})+\omega(r_t^*-r_{t-1}^*)$, which is (17). If $\Delta r_t^*=\sum_k b_k x_{kt}$, the coefficient on x_{kt} in (17) is ωb_k, and dividing by $1-(1-\omega)=\omega$ recovers b_k. □

A.6 Proof of Propositions 2 and 3 and Corollary 1

Proposition 2 follows from (18): $\varepsilon>0$ gives part (i), with the impact response of the transaction price equal to $\theta\hat{v}_t$, $\partial\hat{\lambda}_t/\partial i_t<0$ gives part (ii), and rental fundamentals appear only in the terms $(1-\varpi)\hat{r}_{t+k+1}$, which gives part (iii). Corollary 1 follows from (12), since Δp_t appears on both sides of a regression of Δp_t on $\Delta\iota_t$, while $\Delta\iota_{t-1}$ contains $\Delta\ln\phi_{t-1}$ and Δp_{t-1} but not Δp_t. The first slope can lie on either side of the price response to credit conditions. If only credit conditions move and X_t is fixed, the price change is a multiple $m>0$ of $\Delta\ln\phi_t$ (for a one-quarter change, $m\approx\theta\varepsilon$, the approximation ignoring the small feedback from expected rents), $\Delta\iota_t=(1+m)\,\Delta\ln\phi_t$, and the slope is $m/(1+m)<m$. If other shocks dominate price growth, the slope can exceed m. For Proposition 3, $\partial\varepsilon/\partial(\bar{\lambda}\bar{\phi})=(1-\bar{\lambda}\bar{\phi})^{-2}>0$, so ε rises with $\bar{\phi}$ and with $\bar{\lambda}$, and $\partial\bar{\lambda}/\partial\bar{\imath}=-\Lambda<0$. □

A.7 Proof of Proposition 4

Hold wages, employment, migration, and the policy rate at their steady-state paths, so that only $\hat{s}_t$ and $\hat{k}_t \equiv \ln K_t - \ln \bar{K}_t$ move the market rent: $\hat{r}^*_t = \hat{s}_t - \hat{k}_t$.

(i) At date t, $\hat{s}_t$ depends on $\hat{p}_{t-1} = 0$, so $\hat{s}_t = 0$. By (14), K_t depends on $C_t = I_{t-J}$, which is predetermined, so $\hat{k}_t = 0$. Hence $\hat{r}^*_t = 0$ and, by (5), $\hat{r}_t = 0$. By (18), $\hat{v}_t = \varepsilon \sum_k \varpi^k \mathbb{E}_t \hat{\phi}_{t+k} + (1-\varpi) \sum_k \varpi^k \mathbb{E}_t \hat{r}_{t+k+1}$, which is positive under the stated condition, and since $\hat{p}_{t-1} = 0$, $\hat{p}_t = \theta \hat{v}_t > 0$. By (11), $\hat{p}_{t+1} - \hat{p}_t = \theta(\hat{v}_{t+1} - \hat{p}_t) = \theta(\hat{v}_{t+1} - \theta \hat{v}_t)$, which is positive when $\hat{v}_{t+1} > \theta \hat{v}_t$.

(ii) For $1 \le h \le J-1$, K_{t+h} depends on completions $C_{t+h} = I_{t+h-J}$ with $t+h-J < t$, so $\hat{k}_{t+h} = 0$. By (4), $\hat{s}_{t+h} = \kappa\eta \sum_{m=0}^{h-1} (1-\kappa)^{h-1-m} \hat{p}_{t+m}$, which is positive when $\hat{p}_{t+m} > 0$ for $m < h$. By (5), $\hat{r}_{t+h} = \omega \sum_{m=0}^{h} (1-\omega)^{h-m} \hat{r}^*_{t+m}$, which is then positive. At $h = 1$, $\hat{r}_{t+1} = \omega \hat{s}_{t+1} = \omega\kappa\eta\, \hat{p}_t$.

(iii) By (13), $\hat{p}_t > 0$ raises I_t, so C_{t+J} and hence K_{t+J} rise, giving $\hat{k}_{t+h} > 0$ for $h \ge J$. This lowers $\hat{r}^*_{t+h}$ relative to the path in (ii). If $\hat{\phi}_{t+h} \to 0$, the direct effect of credit on the asset value vanishes. While the stock is above its path, rents are below theirs, which lowers the asset value and the price, so investment falls below replacement and the stock returns to its path; with it, $\hat{r}^*$, $\hat{s}$, and the price return to theirs. In the steady state the price is pinned by builders at $\bar{P} = 1 + \alpha\delta(1/\psi - 1)$, which credit conditions do not affect. □

Appendix B Data Sources and Variable Definitions

Table B.1 lists each variable in the analysis, with its source, geographic level, and time coverage.

Table B.1: Data sources, variable definitions, and coverage

Variable	Source	Geography	Coverage
Outcome and Price Variables			
Rent Index	ABS 6401.0 (CPI rent component); monthly CPI indicator for recent quarters	Capital city	1972:Q3–2026:Q1
New-Lease Rent, Sydney	NSW Department of Communities and Justice, Rent and Sales Report (median rents of new bonds, by bedrooms)	Greater Sydney	1990:Q1–2026:Q1
New-Lease Rent, Melbourne	Homes Victoria, Rental Report (median rents of new lettings, by dwelling type and bedrooms)	Metropolitan Melbourne	1999:Q2–2026:Q2
New-Lease Rent, Brisbane	Queensland Residential Tenancies Authority, median rents of new bonds (by dwelling type and bedrooms)	Five Greater Brisbane LGAs	2012:Q1–2026:Q3
New-Lease Rent, Adelaide	South Australian Housing Trust, Private Rent Report (new bonds, by dwelling type and bedrooms)	Metropolitan Adelaide	2008:Q2–2026:Q2
Dwelling Price Index	BIS (Sydney, all dwellings); ABS 6416.0 spliced to ABS mean dwelling price for other capitals	Capital city	2003:Q3–2025:Q4
Mean Dwelling Price	ABS Total Value of Dwellings (continuation/splice input)	State	2011:Q3–2026:Q1
Construction and Investment			
GFCF Public	ABS State Final Demand (current prices, SA)	State	2003:Q1–2026:Q1
GFCF Private	ABS State Final Demand (current prices, SA)	State	2003:Q1–2026:Q1
Investor Credit	ABS Lending Indicators (seasonally adjusted investor housing commitments, dataflow successor to 5601.0 Table 14)	State	2002:Q3–2026:Q1
National Housing Credit	RBA D2 (investor and owner-occupier credit; net loan-purpose switching)	National	1990:Q1–2026:Q1
Dwelling Completions	ABS 8752.0 (Building Activity)	State	1984:Q3–2025:Q4
Labour Market			
Employment	ABS 6202.0 (Labour Force)	State	1978:Q1–2026:Q1
Wages (WPI)	ABS 6345.0 (Wage Price Index, quarterly)	State	1997:Q3–2026:Q1
Monetary Policy			
Cash Rate	Reserve Bank of Australia	National	1990:Q1–2025:Q4
Demographic			
Net Overseas Migration	ABS 3101.0 (also OMAD by visa group)	State	1981:Q2–2025:Q3
Net Interstate Migration	ABS 3101.0	State	1981:Q2–2025:Q3

Notes: All series are quarterly. State-level variables proxy city-level measures where those are unavailable; capital cities account for 65–79 percent of state economic activity. Coverage of the ABS series ends in the last complete quarter of the June 2026 vintage used here; the dwelling price index ends with the BIS series for Sydney, the cash rate is entered to 2025:Q4, and rents on new leases run to the latest quarter each state had published by November 2026. The estimation sample runs from 2006:Q3 to 2025:Q4. Rents on new leases are chained indices of median rents of new bonds within cells of dwelling type and bedrooms; they enter the comparisons in Section 5.4 and Online Appendix F.

Online Appendix

For online publication

Appendix C Price and Rent Slopes: Divergence, Break and Predictive Tests

This appendix reports the tests summarised in Section 5.2. The investor-credit link shows no significant break at 2017 or at either APRA date once credit is seasonally adjusted, and its largest movements fall in the pandemic years (Figure E.2), so we treat 2017 as one marker of the period rather than the start of a shift.

Choice of break date. The post-2017 indicator begins in 2017:Q1. APRA announced its cap on interest-only lending at the end of that quarter, and the cap took effect in 2017:Q2; NSW announced its housing affordability package in June 2017, in 2017:Q2. Because the date is a choice, we also test for a break at the two APRA dates, 2015:Q1 and 2017:Q2, and at an unknown date. The price–rent ratio rises and the correlation between price and rent growth falls in all five capitals (Figure 3, Table 1), so the descriptive evidence does not single out 2017 either.

Specification. The central test interacts investor credit with the post-2017 break indicator in both equations:

$$\Delta y_{jt} = f_j + \pi_1 \, \Delta \iota_{jt} + \pi_2 \, (\Delta \iota_{jt} \times \mathrm{POST}_t) + \mathbf{x}'_{jt}\boldsymbol{\zeta} + \xi_{jt}, \tag{C.1}$$

estimated separately with $y = p$ and $y = r$. The controls $\mathbf{x}_{jt}$ are the post-2017 indicator, the cash-rate change, lagged tightness, and wage growth in both equations, with employment growth added in the rent equation, so that the two slopes are compared on nearly the same controls. The cash rate enters in differences, but the baseline specification does not interact it. After 2017, the near-zero rates of the COVID-19 period and the rate rises of 2022–2023 dominate the cash-rate series, so its interaction would capture broad monetary-policy regimes rather than the investor-credit channel. We therefore report the version with the cash-rate interaction only as a robustness check (Online Appendix E).

The equation answers two questions. The coefficient π_1 gives the slope before 2017 and $\pi_1 + \pi_2$ the slope after, so the two equations show whether credit co-moves more tightly with prices than with rents in each period. To test whether the price and rent slopes are equal, we stack the two equations with the same controls in both. The interaction π_2 shows whether the link changed after 2017. A large π_2 in the price equation, with a small or insignificant π_2 in the rent equation, would indicate that the price–credit link strengthened after 2017 while the rent–credit link did not. We test $\pi_2 = 0$ in each equation with a one-degree-of-freedom Wald test. Proposition 3 predicts $\pi_2 > 0$ in the price equation if the lower interest rates of much of the period after 2017 raised the price loading on credit. Proposition 4 predicts that rents do not respond to current-quarter credit in either period.

Because $\Delta \iota_{jt}$ and Δp_{jt} are partly determined within the quarter, we estimate Equation (C.1) twice: with current-quarter credit, and with credit lagged one quarter, so that it is predetermined. We compare the two estimates rather than treat either as the elasticity (Corollary 1). The current-quarter estimate may absorb within-quarter simultaneity. The lagged estimate excludes same-quarter feedback, but not feedback from earlier quarters, which the Granger tests below measure. The rent equation serves as a within-quarter comparison for reverse causality.

Break and predictive tests. Two further sets of tests ask whether the divergence result depends on the 2017 date and which way prediction runs. The first repeats the interaction in Equation (C.1) with breaks at the two APRA dates, 2015:Q1 and 2017:Q2, and searches for a break at an unknown date. For the search, we estimate the interaction at every quarter from 2010:Q3 to 2023:Q1 and compare the largest Wald statistic with the critical values of Andrews (1993, 2003), which allow for the search over dates. These tests use the same four controls in both equations,

Table C.1: Investor credit before and after 2017, price versus rent

	Price equation	Rent equation
Panel A. Current-quarter credit		
Δ ln Investor credit	0.106^{***}	-0.011
	(0.029)	(0.016)
Δ ln Investor credit $\times$ post	0.048	0.021
	(0.044)	(0.017)
Panel B. One-quarter-lagged credit		
Δ ln Investor credit	0.140^{***}	-0.006
	(0.029)	(0.013)
Δ ln Investor credit $\times$ post	-0.011	0.027^{*}
	(0.039)	(0.014)

Panel A enters investor credit in the current quarter; Panel B lags it and the cash rate one quarter. In each panel, the first row is the slope on investor credit before 2017, and the second its change after 2017 (the interaction with the post-2017 indicator). Both equations also include the post-2017 indicator, the cash-rate change, lagged tightness, wage growth and city fixed effects; the rent equation adds employment growth. Investor credit is seasonally adjusted. Driscoll–Kraay standard errors in parentheses. $^{*}p < 0.10$, $^{**}p < 0.05$, $^{***}p < 0.01$.

the cash-rate change, lagged tightness, wage growth and employment growth, so that we can stack the equations and compare their changes. The second set is Granger (1969) tests in both directions. Each regresses price, rent or credit growth on four lags of itself and four lags of the other variable, with city fixed effects and quarter-of-year dummies, so that a remaining seasonal pattern cannot pass for prediction. The tests ask whether lagged credit growth predicts price and rent growth, and whether lagged price and rent growth predict credit growth.

Results. The contrast lies in the level of the two slopes, not in a change after 2017. In the equations of Table C.1, the price–credit slope before 2017 is 0.11 and the rent–credit slope is -0.01. A stacked regression with the same controls in both equations rejects equal slopes ($p < 0.01$). When we interact investor-credit growth with the post-2017 indicator (Table C.1, Panel A), the interaction is 0.048 in the price equation and 0.021 in the rent equation, and neither is significant ($p = 0.28$ and $p = 0.24$). The link between credit and prices was already strong before 2017, and it did not strengthen significantly after. The data therefore do not show the rise that Proposition 3 predicts for a period of lower interest rates. Interacting credit growth with the level of the cash rate instead gives the same answer (Online Appendix E).

The unadjusted credit series gives a different answer: its price interaction is 0.052 ($p = 0.01$). The difference comes from seasonal noise. Quarter-of-year swings explain 63 percent of the variation in unadjusted credit growth before 2017 but 37 percent after, and noise of this kind weakens a measured slope. The unadjusted slope is 0.41 times the adjusted slope before 2017 and 0.62 times after, so the fading noise looks like a strengthening link.

Investor credit and prices are also jointly determined within the quarter. Credit is drawn as purchases settle, so a current-quarter association partly reflects prices acting on credit, not only credit acting on prices. Panel B of Table C.1 therefore re-estimates the interaction with credit lagged one quarter, so that prices cannot feed back into it within the period. The price interaction is then -0.011 ($p = 0.79$), and the rent interaction is 0.027 ($p = 0.052$). With lagged credit, the price–credit slope before 2017 is 0.14, larger than the current-quarter slope of 0.11, so the association does not rest on same-quarter feedback alone.

These results do not depend on the break date. Table C.2 repeats the test at the two APRA dates and searches for a break at an unknown date, with the same four controls in both equations. The price–credit slope does not change significantly at 2015:Q1, 2017:Q1 or 2017:Q2, with current or lagged credit (p between 0.16 and 0.81). In the stacked equations, the price and rent changes do not differ at 2017:Q1 or 2017:Q2 either. The unknown-date test does reject a constant price–credit slope, but not in the mid-2010s. With current-quarter credit, its largest

Table C.2: Break tests on the investor-credit slope

	Price equation		Rent equation		Rent minus price	
	Current	Lagged	Current	Lagged	Current	Lagged
Panel A. Change in the credit slope at a dated break						
2015:Q1, investor benchmark	0.012	−0.042	0.043***	0.044***	0.030	0.086**
	(0.040)	(0.034)	(0.014)	(0.011)	(0.040)	(0.039)
2017:Q1, break in Table C.1	0.060	−0.011	0.021	0.027*	−0.039	0.038
	(0.042)	(0.038)	(0.017)	(0.014)	(0.050)	(0.046)
2017:Q2, interest-only cap	0.059	−0.009	0.022	0.029**	−0.037	0.038
	(0.042)	(0.037)	(0.018)	(0.014)	(0.050)	(0.045)
Panel B. Break at an unknown date, 2010:Q3 to 2023:Q1						
Largest Wald statistic	13.6***	26.6***	24.6***	64.8***		
Date of the largest statistic	2021:Q1	2010:Q3[a]	2010:Q3[a]	2010:Q3[a]		
Change in the credit slope at that date	0.113	−0.112	0.054	0.055		

Main credit measure (seasonally adjusted), 370 city-quarters. Each regression adds to the city fixed effects an indicator that is one from the break quarter on and its interaction with credit growth, in the current quarter or lagged one quarter. Both equations have the same four controls: the cash-rate change (lagged with lagged credit), lagged tightness, wage growth and employment growth. The price equation therefore differs slightly from Table C.1, which omits employment growth. Panel A: the coefficient on the interaction, that is, the change in the credit slope after the break, with Driscoll–Kraay standard errors in parentheses. Rent minus price: the rent change less the price change, from the two equations stacked. Panel B: the largest Driscoll–Kraay Wald statistic for a change in the credit slope over the 51 candidate quarters, against the critical values of Andrews (1993, 2003) for one restriction and 15 percent trimming: 7.12 (10 percent), 8.68 (5 percent) and 12.16 (1 percent). [a]The first candidate date, so the test compares the first 12 quarters of the sample with the rest. $^{*}p < 0.10$, $^{**}p < 0.05$, $^{***}p < 0.01$.

statistic falls in 2021:Q1, at the start of the 2021 boom, when investor credit and prices surged together; the slope is 0.11 higher from that quarter. That quarter falls in the period of the lowest cash rate, 0.1 percent, when Proposition 3 predicts the largest price loading, but the current-quarter slope also contains the price change itself (Corollary 1). With lagged credit, which excludes same-quarter feedback, the change at 2021:Q1 is 0.04 and not significant ($p = 0.26$), and the largest statistic falls instead on the first candidate date, 2010:Q3. That date separates the first 12 quarters of the sample, which span the global financial crisis, from the rest.

In the rent equation, the credit slope rises at 2015:Q1 with current and lagged credit ($p \leq 0.002$); with current credit, it rises from −0.03 to 0.01. With lagged credit, it also rises at 2017:Q2 ($p = 0.04$), and at 2015:Q1 the rent and price changes differ ($p = 0.03$). The unknown-date test again points to the first candidate date. The rent–credit slope was slightly negative early in the sample and close to zero after. On both sides of every break, it stays far below the price–credit slope, and the stacked equations reject equal price and rent slopes before each break date ($p < 0.01$).

The Granger tests in Table C.3 show that the feedback between credit and prices runs across quarters, not only within them. Lagged credit growth predicts price growth ($p < 0.001$). Lagged price growth predicts credit growth less strongly: $p = 0.095$ with four lags, and $p < 0.05$ with two or six lags or with quarter fixed effects. Lagged credit growth also predicts rent growth ($p < 0.001$), but the sum of its lag coefficients, 0.010, is about a fifth of that for prices, 0.055. With only two lags, credit does not predict rents ($p = 0.29$), as the delay in Section 5.3 would lead us to expect. Rent growth does not predict credit growth ($p = 0.55$), although it does once quarter fixed effects remove national shocks ($p = 0.04$). Past prices thus help to predict credit. The local projections condition on four lags of price or rent growth, which absorb this feedback.

Our reading is therefore modest. Investor credit co-moves far more tightly with prices than with rents, before and after 2017, whether credit enters in the current quarter or lagged. Neither estimate is a structural causal elasticity. The current-quarter estimate absorbs same-quarter feedback, and the lagged estimate can still carry feedback from earlier quarters (Table C.3); the two are close. What survives both is a difference in co-movement, with prices tracking investor credit closely and rents hardly at all. It is not evidence that credit drives prices while leaving rents untouched. The rent equation serves as a within-quarter comparison here: if the price–credit association were only an accounting artefact of joint determination, the same force would

Table C.3: Granger tests in both directions

	Prices		Rents	
	Credit → price	Price → credit	Credit → rent	Rent → credit
Wald statistic, 4 lags	20.6	7.9	32.1	3.0
p-value	<0.001	0.095	<0.001	0.554
Sum of the lag coefficients	0.055	0.260	0.010	0.927
	(0.020)	(0.347)	(0.005)	(1.031)
p-value in other specifications				
2 lags	<0.001	0.003	0.292	0.827
6 lags	<0.001	0.033	<0.001	0.369
4 lags, quarter fixed effects	0.034	0.009	0.007	0.040
Observations	355	355	355	355

Main credit measure (seasonally adjusted). Each column regresses the growth of the predicted variable on four lags of itself and four lags of the predicting variable, with city fixed effects and quarter-of-year dummies. The test is that the lags of the predicting variable are all zero: a Driscoll–Kraay Wald test with four degrees of freedom. The sum of their coefficients has its Driscoll–Kraay standard error in parentheses. Every specification uses the same 355 city-quarters, 2008:Q2 to 2025:Q4, which hold six lags. The last row replaces the quarter-of-year dummies with quarter fixed effects, which remove every national shock.

not largely spare rents. The small rises in the rent–credit slope, at 2015:Q1 and, with lagged credit, after 2017, are instead consistent with credit reaching rents through the slower channels of the model, tenure switching and lease repricing.

Online Appendix E reports further checks on the price–credit association, including its change after 2017 in each city, its functional form, and the 2015–2017 loan-purpose reclassification. Online Appendix D then treats credit as endogenous and instruments it with two APRA lending measures in the local projections.

Appendix D Instrumenting Credit with the APRA Measures

We first instrument investor credit with two APRA lending measures and then test restrictions implied by instrument validity.

D.1 Instrumented responses

The local projections in Section 5.3 use credit innovations residualised against observed controls. Residualisation does not make these innovations predetermined or exogenous. Investors may borrow ahead of price or rent rises that they expect and that the controls do not capture. We therefore instrument credit with two measures by which APRA restrained investor lending. From the first quarter of 2015, APRA asked lenders to keep growth in investor lending below 10 percent a year, a benchmark it lifted in mid-2018. From the second quarter of 2017, it asked them to limit interest-only loans to 30 percent of new mortgage lending, a cap it lifted at the end of 2018.[6] Both measures targeted the risks in investor lending rather than housing demand, which makes them candidate instruments. In the model, both lower the effective loan-to-value limit ϕ_t in (6). Online Appendix D.2 tests whether they are valid.

We use the measures in two ways. The national-timing instrument is an indicator equal to one while either measure was in force, from 2015:Q1 to 2018:Q4, in projections with city fixed effects. It relies on timing alone, so any other national change in those years that moved prices or rents would bias it. The exposure instrument interacts the interest-only cap with each city's investor share of new lending in 2014, the year before the benchmark, in projections with city and quarter fixed effects. Quarter effects absorb national shocks, so this instrument works through credit falling more in the more exposed cities while the cap was in force. We pair

[6] APRA announced the benchmark on 9 December 2014 and the cap on 31 March 2017. We code the benchmark as in force from 2015:Q1 to 2018:Q2 and the cap from 2017:Q2 to 2018:Q4.

exposure with the cap because interest-only loans went disproportionately to investors. Paired with the benchmark, or with either measure, the exposure instrument has a first-stage F below 7 at every horizon. Sydney's investor share, 0.52 against 0.35 to 0.41 in the other capitals, carries much of the variation in exposure.

Because the instruments stay on for years, they move credit in many quarters after t, not only in quarter t. Following Ramey and Zubairy (2018), we therefore relate the cumulative change in each outcome to the cumulative change in credit over the same quarters,

$$y_{j,t+h} - y_{j,t-1} \;=\; \beta_h \sum_{s=0}^{h} \Delta\iota_{j,t+s} \;+\; \mathbf{w}'_{jt}\boldsymbol{\tau}_h \;+\; f_{j,h} \;+\; v_{j,t+h}, \tag{D.1}$$

and instrument the cumulative credit change with the APRA instrument. Here y is the log price or log rent, ι is log investor credit, and $\mathbf{w}_{jt}$ holds four lags of credit growth and of the outcome's growth, lagged tightness, and, in the national-timing projections, the lagged change in the cash rate; the exposure projections add quarter effects. The coefficient β_h is the change in the outcome per unit of cumulative credit growth. A value of 0.10 means that a 10 percent rise in investor credit over the horizon goes with a rise of about 1 percent in the price or rent level. These multipliers therefore differ in scale from the responses to a one-standard-deviation innovation in Figure 5. For comparison, we estimate OLS in the same form, and we report Driscoll–Kraay first-stage F statistics. To test for no response, we use the Anderson–Rubin test (Anderson and Rubin, 1949), which remains valid when an instrument is weak.

Figure D.1 plots the instrumented and OLS responses, and Table D.1 reports them at four, eight, and twelve quarters with the first-stage F. Both instruments lower credit. The national-timing instrument is weak in the first year, with an F near 10 at four quarters, and strong from eight quarters, with an F of 15 to 30. The exposure instrument is strong for prices at every horizon, with an F of 30 to 50 at four, eight, and twelve quarters. For rents it is strong at four quarters but weakens at eight and twelve, with an F of 10 and 9.

For prices, the two instruments tell different stories. With national timing, the price response builds slowly, from 0.01 on impact to about 0.11 by eight quarters, and stays below OLS throughout. It is also imprecise, with Anderson–Rubin p-values of 0.13 to 0.44 at four, eight, and twelve quarters. With the exposure instrument, the price response is large at once, about 0.45 to 0.50 through the first year, and falls to 0.20 by twelve quarters, below OLS. For rents, the two instruments agree. Both give rent responses that build over time, from 0.02 to 0.04 at four quarters to 0.11 to 0.19 at twelve, the delayed path of Section 5.3. The responses lie close to OLS, except that the exposure instrument gives 0.19 against 0.14 at twelve quarters.

These patterns hold with the change in investors' share of new lending as the credit measure and without the 2015–2016 reclassification quarters. Without Sydney, the exposure instrument becomes weak, with an F of 6 to 9 for prices, so that instrument rests on Sydney. With $h + 1$ Driscoll–Kraay lags in place of four, which allows for the overlap of the cumulative changes, the conclusions also hold; only the exposure instrument's first stage for rents weakens further, to an F of 9 and 6 at eight and twelve quarters. The two price responses differ widely at short horizons, so at least one instrument may be invalid. Online Appendix D.2 tests this directly.

D.2 Testing restrictions on APRA instrument validity

A strong first stage shows that an instrument moves credit. It does not show that the instrument is uncorrelated with the error of the projection, which requires the instrument to affect prices and rents only through credit and to be unrelated to other shocks. We test this condition with the correlation-restriction test of Dzhumashev and Tursunalieva (2026). Non-rejection means compatibility with the stated correlation bounds, not verification of the exclusion restriction. The test rests on an identity. The instrument's correlation with the error, ρ_{zu}, depends on four correlations. The data reveal three of them, those among credit, the outcome, and the instrument. The fourth, credit's own correlation with the error, ρ_{xu}, is unknown, and bounding it bounds ρ_{zu}. Following Dzhumashev and Tursunalieva (2026), we let ρ_{xu} lie between

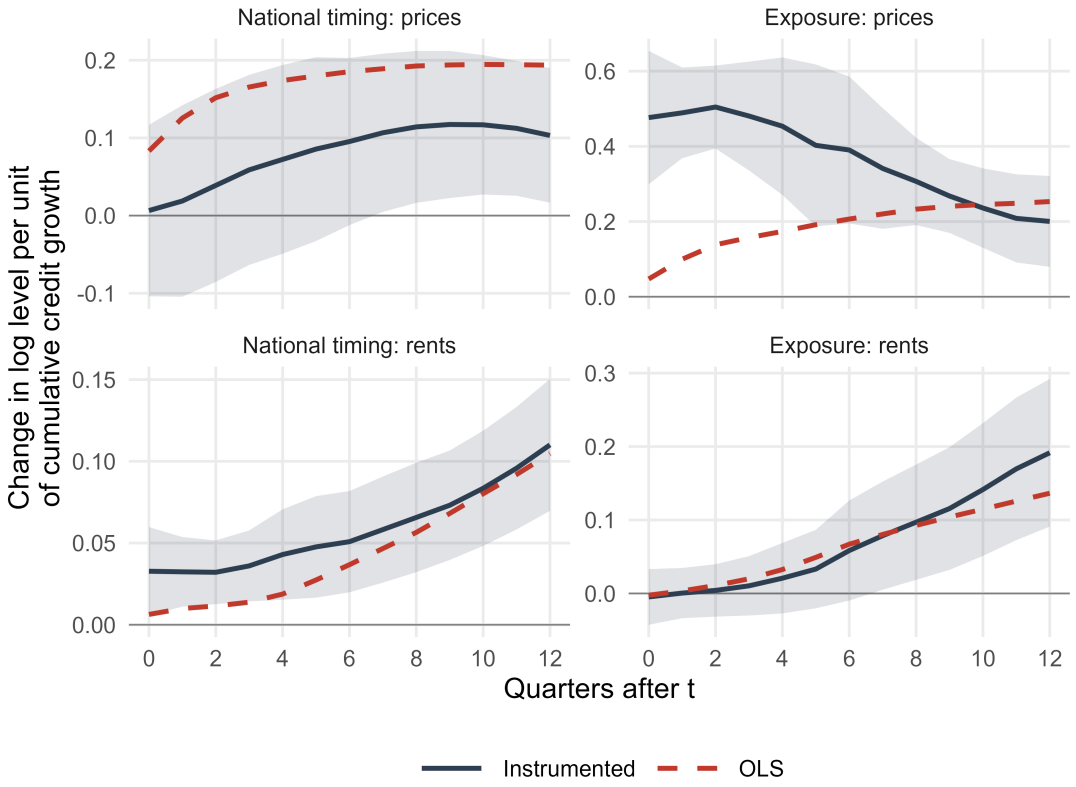


Figure D.1: Instrumented responses of the price and rent levels to investor credit. Each panel shows the change in the log price or rent level from $t-1$ to $t+h$ per unit of cumulative credit growth over the same quarters (Equation (D.1)), with credit instrumented by an APRA measure (solid line, 90 percent Driscoll–Kraay band) and estimated by OLS in the same form (dashed line). Left: national timing, an indicator that either measure is in force, with city fixed effects. Right: exposure, each city's 2014 investor share times the interest-only cap indicator, with city and quarter fixed effects. Top: prices; bottom: rents. Main credit measure, full sample; the vertical scales differ across panels.

0 and 0.8. The lower bound reflects that investors are likely to borrow more when they expect prices or rents to rise. It rules out measurement error in credit large enough to make ρ_{xu} negative, a case we return to below. The test rejects validity when zero lies outside the 95 percent modified conditional union-bound (MCUB) interval of Bei (2024) for the resulting set of ρ_{zu}. We also report p_0^{adm}, the smallest level in $(0.01, 0.5)$ at which the test rejects.

We apply the test at horizons of four, eight, and twelve quarters. Before the test, we remove the fixed effects and controls of each projection from the outcome, credit, and the instrument, and we orient each instrument so that it raises credit. A positive interval for ρ_{zu} then means that, while the APRA measure applied, the outcome grew less than credit explains. The cumulative changes in the outcome and in credit overlap from one quarter to the next, so we estimate the covariance of the three correlations with a moving block bootstrap (Künsch, 1989). It resamples blocks of eight consecutive quarters and keeps the five cities in each quarter together. For the projections in Table D.1, resampling single observations would understate the standard errors of the correlations by a factor of 1.5 to 3.8.

Table D.1 reports the MCUB intervals and p_0^{adm}, and Figure D.2 shows, for each projection, the range of assumptions about ρ_{xu} under which validity is not rejected. The results are easiest to read against OLS. At $\rho_{xu} = 0$, the implied ρ_{zu} has the sign of the gap between the IV and OLS responses. A positive ρ_{xu} makes OLS overstate the response, so a valid instrument should give a response at or below OLS, up to sampling error. Validity of the national-timing instrument is not rejected for prices at any of these horizons under the stated bounds. Its responses of 0.07 to 0.11 lie below OLS (0.17 to 0.19), the pattern a positive ρ_{xu} produces. At twelve quarters validity is not rejected only if the assumed range for ρ_{xu} reaches 0.15. The exposure instrument

Table D.1: Instrumented responses and correlation-restriction test of the APRA instruments

		Horizon h (quarters)		
		4	8	12
Panel A. National timing: either APRA measure in force				
Prices	IV response	0.072	0.114	0.103
	OLS response	0.174	0.193	0.194
	First-stage F	9.6	19.1	14.9
	MCUB interval	$[-0.46, 0.39]$	$[-0.48, 0.56]$	$[-0.44, 0.43]$
	p_0^{adm}	≥ 0.50	≥ 0.50	≥ 0.50
Rents	IV response	0.043	0.066	0.110
	OLS response	0.019	0.057	0.105
	First-stage F	9.2	22.2	29.7
	MCUB interval	$[-0.02, 0.60]$	$[-0.14, 0.72]$	$[-0.23, 0.66]$
	p_0^{adm}	0.08	0.38	≥ 0.50
Panel B. Exposure: 2014 investor share × interest-only cap				
Prices	IV response	0.454	0.307	0.200
	OLS response	0.174	0.233	0.253
	First-stage F	50.5	29.6	35.2
	MCUB interval	$[0.09, 0.58]$	$[-0.05, 0.39]$	$[-0.24, 0.28]$
	p_0^{adm}	≤ 0.01	0.17	≥ 0.50
Rents	IV response	0.021	0.097	0.192
	OLS response	0.033	0.093	0.136
	First-stage F	21.8	10.4	9.1
	MCUB interval	$[-0.16, 0.33]$	$[-0.16, 0.36]$	$[-0.11, 0.40]$
	p_0^{adm}	≥ 0.50	≥ 0.50	0.34

Instrumented local projections (Equation (D.1)) and the correlation-restriction test of Dzhumashev and Tursunalieva (2026), main credit measure, full sample. IV response: change in the log price or rent level from $t-1$ to $t+h$ per unit of cumulative credit growth over the same quarters; OLS response: the same projection estimated by OLS; first-stage F: Driscoll–Kraay Wald statistic of the instrument. MCUB interval: 95 percent modified conditional union-bound interval (Bei, 2024) for the instrument's correlation with the error, ρ_{zu}, when credit's correlation with the error, ρ_{xu}, lies in $[0, 0.8]$; validity is rejected when the interval excludes zero. p_0^{adm}: smallest level in $(0.01, 0.5)$ at which zero leaves the interval, censored at both ends. Outcome, credit and instrument are residualised on the fixed effects and controls of each projection, and the instrument is signed so that it raises credit. The covariance of the three sample correlations comes from a moving block bootstrap over blocks of eight quarters, with the five cities in each quarter kept together (Künsch, 1989). Imbens–Manski/Stoye and naive endpoint intervals (Imbens and Manski, 2004; Stoye, 2009) agree with the MCUB intervals on zero inclusion in every cell.

fails for prices at four quarters ($p_0^{\text{adm}} \leq 0.01$), where its response of 0.454 is well above OLS (0.174). While the interest-only cap was in force, prices in the more exposed cities grew less than their credit growth explains, so we set that response aside. Allowing a negative ρ_{xu}, as measurement error in credit would produce, reconciles the two responses only for values below about -0.34. At eight quarters validity of the exposure instrument is not rejected only if the assumed range for ρ_{xu} includes values below 0.18. At twelve quarters, where its response of 0.200 lies below OLS (0.253), validity is not rejected unless the assumed range lies entirely above 0.68.

For prices, then, the national-timing instrument gives a response that builds over two years and stays below OLS, and its Anderson–Rubin tests do not reject a zero response at four, eight, or twelve quarters. The fast price response comes from the exposure instrument, which fails the test at four quarters. The instrumented projections therefore do not settle how fast credit reaches prices.

For rents, the instrumented responses lie close to OLS, apart from the exposure instrument at

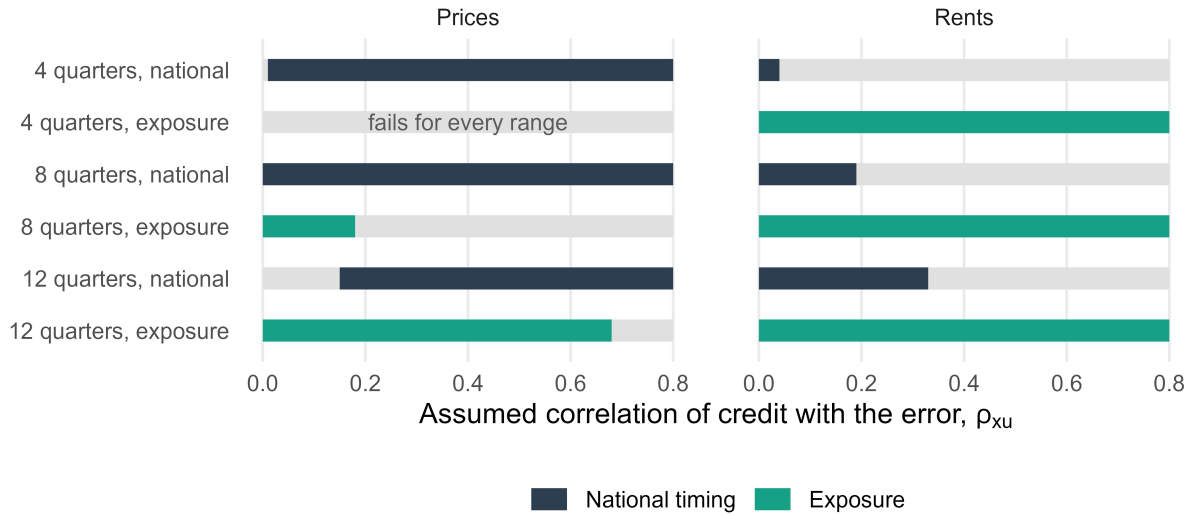


Figure D.2: Valid ranges for credit's correlation with the error, by outcome, horizon and instrument. At each horizon, the upper bar is the national-timing instrument (either APRA measure in force) and the lower bar the exposure instrument (2014 investor share times the interest-only cap). Each coloured bar shows the assumptions about ρ_{xu} under which validity is not rejected by the correlation-restriction test: restricting ρ_{xu} to $(0, r)$ yields non-rejection once r reaches the bar's left end, and restricting it to $(r, 0.8)$ fails once r exceeds its right end. Grey marks the assumed domain, $[0, 0.8]$; a grey bar with no coloured part fails for every value. Main credit measure, full sample; 95 percent MCUB intervals, with a moving block bootstrap over quarters.

twelve quarters, and the test rejects neither instrument at any horizon. Validity of the exposure instrument is not rejected for any range of ρ_{xu} we assume within $[0, 0.8]$, but with less power, since its first stage weakens for rents at eight and twelve quarters (F of 10 and 9). Validity of the national-timing instrument is not rejected only if the assumed range for ρ_{xu} includes small values: below 0.04 at four quarters, 0.19 at eight, and 0.33 at twelve. Its rent responses lie above OLS at every horizon, which a valid instrument produces only with a negative ρ_{xu}; small positive values remain compatible with validity within sampling error. Measurement error in credit could produce this pattern, and so could the high dwelling completions of 2016–2018, which coincided with the APRA measures. Completions also rose most in Sydney, the most exposed city, so this caution applies to both instruments. The rent evidence is consistent with credit being nearly uncorrelated with the rent error, and it supports the delayed rent response. The rejection and non-rejection results are the same with the change in investors' share of new lending as the credit measure, without the 2015–2016 reclassification quarters, and with blocks of four or thirteen quarters. Blocks of thirteen quarters cover the full overlap at twelve quarters. Without Sydney, the exposure instrument also fails for prices at every horizon, and the national-timing instrument fails for rents at four quarters.

Appendix E Statistical Tests and Supplementary Results

This appendix collects supplementary evidence: checks on the price–credit relationship that Online Appendix C refers to, and checks on the rent equation and the local projections.

Timing and breadth of the post-2017 change. Two interaction specifications probe the timing and breadth of the post-2017 change in the price equation. Adding a post-2020 pandemic interaction on investor credit alongside the post-2017 interaction leaves both terms small and insignificant (0.065, $p = 0.18$, and -0.028, $p = 0.65$). We also let the post-2017 interaction differ for Sydney, the city to which the 2017 NSW package applied, by adding the interaction of investor credit with the Sydney indicator D_{jt}^{NSW} (Section 4.2). The Sydney-specific term is positive and significant (0.136, $p < 0.001$ with Driscoll–Kraay standard errors), while the other

capitals' common post-2017 change is small (0.030, $p = 0.48$). Sydney's slope before 2017 does not differ significantly from the other capitals' ($p = 0.18$). Its post-2017 rise holds, and is clearer, in a stricter test that lets both Sydney's slope and its mean price growth differ before and after 2017 (0.111, $p = 0.01$). The rise is concentrated in 2017–2019: a separate Sydney term from 2020 is negative (-0.123, $p = 0.09$).

Sydney is not alone, however. Estimated separately for each city, with Newey–West standard errors, the post-2017 change in the price–credit slope is 0.149 in Sydney ($p = 0.002$) and 0.106 in Adelaide ($p = 0.02$), against 0.076 in Brisbane ($p = 0.11$), -0.015 in Melbourne ($p = 0.86$), and 0.002 in Perth ($p = 0.96$). Adelaide's rise is about 70 percent of Sydney's, and the pooled change is not significant, so the pattern does not single out the NSW package. We report it as description, not as evidence for or against a reading driven by NSW policy. With the unadjusted credit series, every city showed a positive post-2017 change and the Sydney term was negative (-0.032, $p = 0.05$), a pattern consistent with that series' fading seasonal noise (Online Appendix C).

Cash-rate interaction. The divergence test leaves the cash-rate change uninteracted (Online Appendix C). Adding its interaction with the post-2017 indicator to all four equations of Table C.1 leaves the credit results as they are. The price interaction falls from 0.048 to 0.033 with current credit ($p = 0.45$) and stays at -0.011 with lagged credit ($p = 0.78$). The rent interaction rises a little, to 0.025 ($p = 0.16$) and 0.029 ($p = 0.04$). The cash-rate coefficient in the current-quarter price equation does change after 2017, from 0.014 to -0.006 ($p = 0.001$ for the change), consistent with the shift in monetary-policy regimes that Online Appendix C describes.

Cash-rate level. Proposition 3 predicts a larger price loading on credit when interest rates are low. To test it directly, we replace the post-2017 terms of Table C.1 with the interaction of credit growth and the previous quarter's cash rate, measured from its sample mean of 2.9 percent; the cash rate also enters on its own. Because the cash rate is centred at its sample mean of 2.9 percent, the coefficient on credit growth gives the price–credit slope at that rate: 0.130 with current credit. In the price equation, the interaction is -0.005 ($p = 0.70$) with current credit and 0.004 ($p = 0.68$) with lagged credit. The price–credit slope is 0.14 at a cash rate of 0.1 percent and 0.12 at 4.35 percent with current credit, and 0.12 and 0.14 with lagged credit. The interaction stays small and insignificant when the post-2017 terms are kept (0.006, $p = 0.63$, and 0.007, $p = 0.51$) and when credit growth times a post-2020 indicator is added (-0.003, $p = 0.81$, and 0.003, $p = 0.77$). The reduced form thus gives no direct support to Proposition 3; Section 6.8 shows how large the model predicts the interaction to be. With the unadjusted series, the current-quarter price interaction is negative and significant (-0.016, $p = 0.02$). That series' seasonal noise, which weakens its price slope, was largest early in the sample, when the cash rate was high, so this result is consistent with the artefact described in Online Appendix C.

In the rent equation, the interaction is negative and significant: -0.009 ($p = 0.004$) with current credit and -0.010 ($p < 0.001$) with lagged credit. It stays significant in both variants. At the mean cash rate, the rent–credit slope is close to zero (-0.001, $p = 0.76$, with current credit and 0.005, $p = 0.23$, with lagged credit), as in Table C.1. But it is 0.02 to 0.03 at a cash rate of 0.1 percent and about -0.01 at 4.35 percent, and rent growth is higher after quarters of higher cash rates. The model does not predict this pattern: in the model, credit reaches rents only through prices, and only with a lag. We report it as a pattern the model leaves unexplained.

Owner-occupier credit. Adding the growth of seasonally adjusted owner-occupier commitments to the current-quarter price equation gives an investor-credit coefficient of 0.144 (s.e. 0.035) and an owner-occupier coefficient of -0.013 (s.e. 0.045). With both lagged and last quarter's price growth included, the investor coefficient is 0.056 (s.e. 0.012) and the owner-occupier coefficient 0.020 (s.e. 0.022). Investor credit, not housing credit in general, carries the price association, which supports treating investors as the marginal buyers (Section 3).

Stability of the rent equation. The structure of fundamentals is stable across cities, but its weights change over time. A Chow-style test that interacts all four rent regressors with the post-2017 indicator rejects equal coefficients ($\chi^2_4 = 15.4$, $p = 0.004$). Estimated separately before and after 2017, the employment coefficient doubles, from 0.07 to 0.15, while the wage coefficient falls from 0.65 to 0.28. A linear time trend points the same way for employment, but imprecisely (trend interaction +0.013, $p = 0.13$). Over 59 rolling windows of 20 quarters (19 after differencing), the employment coefficient rises from near zero in the early 2010s to about 0.15 by the 2020s, and the tightness coefficient stays positive and significant in every window. Wage pass-through is noisier. The wage coefficient is positive in all but two of the 59 windows, but it is significant at 10 percent in only 21: the window ending in 2013:Q1, those ending between 2015:Q1 and 2016:Q1, and every window from 2022:Q2. The same fundamentals therefore govern rents across the whole sample, with more weight on employment as the labour market tightened; the coefficients are not literally constant.

The rent equation has a mean-group counterpart in Table 3 because the paper claims that rents follow fundamentals city by city. The price equation makes no such claim. Its identification rests on the time variation within cities that links investor credit to prices, so we estimate the gap between the price and rent slopes on credit from the pooled time dimension, not from differences in the slope across cities.

Rent measurement. CPI rents are measured net of Commonwealth Rent Assistance. Adding indicators for the 2020 rent moratoria (2020:Q2–Q4) and for the quarters of Rent Assistance increases (2023:Q3 and 2024:Q3) to the dynamic rent equation changes the other coefficients little: $\hat{\omega}$ moves from 0.35 to 0.36. The moratorium indicator is small and insignificant (-0.001, $p = 0.47$), and so is the Rent Assistance indicator (-0.002, $p = 0.11$), although its sign is the one expected if those increases lowered measured rent growth.

Sample ending in 2021:Q4. Price growth after 2021:Q4 comes from the spliced series. On the sample ending in 2021:Q4 (290 city-quarters), the current-quarter price–credit slope is 0.126 and the lagged slope with the momentum control 0.063, both significant at the 1 percent level, and the post-2017 change in the price–credit slope, estimated with the controls of Table 4, is small and insignificant (0.018, $p = 0.72$). The rent equation gives a tightness coefficient of 0.008 and a wage coefficient of about 0.9. The local projections have the same shape as in Figure 5, with twelfth-quarter responses of 0.033 for prices and 0.021 for rents.

Local projections with labour-market controls and after impact. Figure E.1 reports the two variants of Section 4.3. Panel (a) adds four lags each of wage and employment growth to Equation (22). The rent response at twelve quarters is about 0.021 (0.018 when current wage and employment growth are added as well), against 0.018 without these controls, and the price response is almost unchanged. Panel (b) shows the post-impact responses of Equation (23), which hold the current price change fixed. A credit innovation still predicts later price growth, by about 0.005 log points after one quarter and 0.03 after twelve, and later rent growth, by about 0.0035 after four quarters and 0.015 after twelve. Both responses are significant at the 10 percent level at every horizon, so credit carries information about future prices and rents beyond its mechanical link with the current price change.

The shape is also the same with the change in investors' share of new lending as the credit measure. Prices lead, at 0.015 log points by the fourth quarter against 0.005 for rents, though the price response is less precise in the second year.

Rolling decomposition. Figure E.2 traces the current-quarter partial R^2 of investor credit in the price equation and in the rent equation over a 20-quarter rolling window. In the earliest windows, ending in 2011 and 2012, the rent series is as high as the price series. From 2020 the price series rises well above the rent series and stays near 0.3 through 2023. The rent series stays below 0.05 in most windows. Its one rise after 2013, to 0.12 in the window ending in 2021:Q4, reaches about two-fifths of the price series at that point and falls back below 0.05 in the window

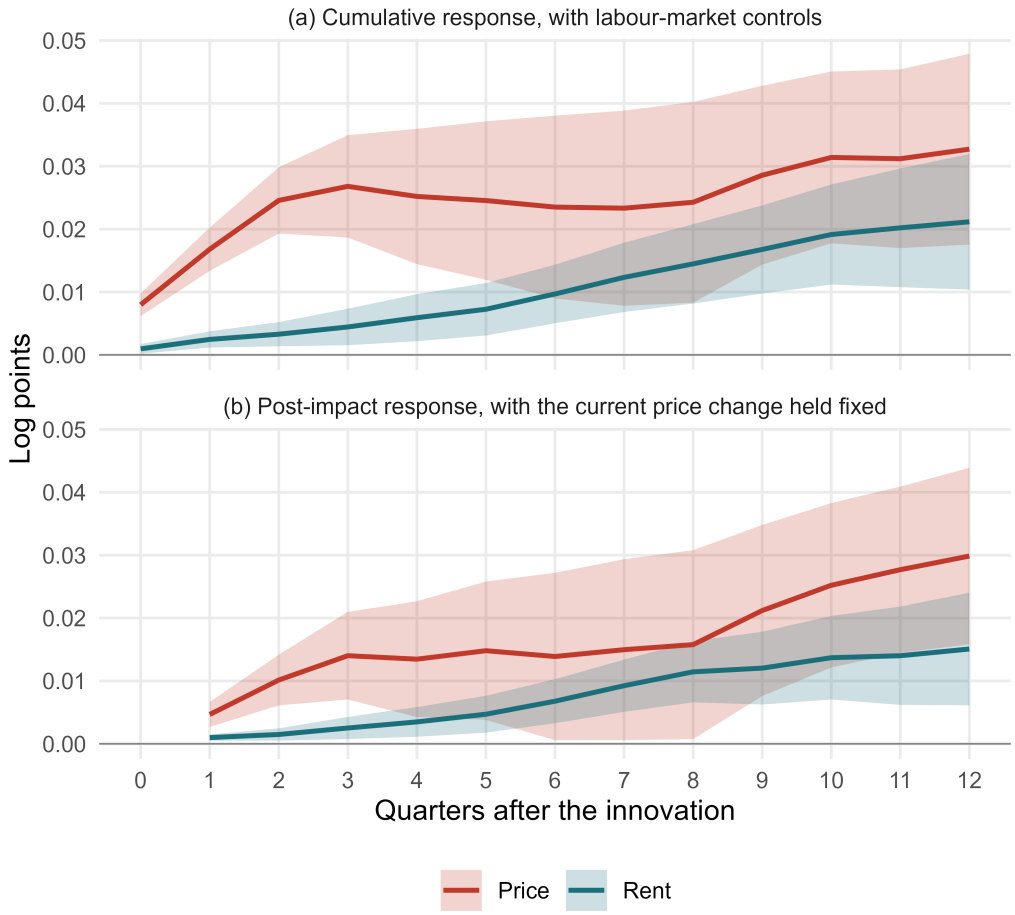


Figure E.1: Local projections with labour-market controls and after impact. Panel (a): cumulative response $y_{t+h} - y_{t-1}$ from Equation (22) with four added lags each of wage growth and employment growth. Panel (b): post-impact response $y_{t+h} - y_t$ from Equation (23), which also holds the current price change fixed (for rents, also the current rent change and four lags of price growth). One-standard-deviation innovation in seasonally adjusted investor-credit growth; city fixed effects; 90 percent Driscoll–Kraay bands.

ending in 2023:Q2. This decomposition uses current-quarter credit, so the simultaneity caveat on the current-quarter interaction applies to it too; the local projections in the main text are the cleaner dynamic evidence.

Functional form of the price–credit relationship. A linear credit term summarises ordinary quarters but not the extremes. Replacing it with a natural cubic spline in investor-credit growth raises the within-R^2 of the price equation from 0.355 to 0.395, and a joint test of the spline's curvature terms rejects linearity ($\chi^2_2 = 28.4$, $p < 0.001$). The curvature comes from the six city-quarters in which credit growth exceeds 25 percent in either direction, five of them in 2020–2021. Without them, the test does not reject linearity ($\chi^2_2 = 1.7$, $p = 0.43$). Piecewise slopes by tercile of credit growth are 0.07, 0.26, and 0.10, so the slope is steepest in the middle of the range and flatter at both ends. The linear coefficient is therefore a fair summary of ordinary quarters, and the largest credit swings move prices less than it implies.

Loan-purpose reclassification, 2015–2017. The state-level investor-credit series spans 2015–2017, when lenders reclassified a material volume of loans between investor and owner-occupier purposes under APRA supervision. In principle, this could contaminate the post-2017 interaction. We probe it with the RBA's national D2 credit series, which comes with a measure of the net volume of loans switched between purposes. Three checks point the same way. First, the post-2017 interaction re-estimated on the official national investor-credit series, raw and adjusted for cumulative switching, is small and insignificant for both prices and rents. The national series

varies only over time and, with city fixed effects, has little independent power, so we read these results as consistency checks, not as substitutes for the cross-city estimate. Second, we add the current-quarter switching flow as a control to the main price equation, over the window in which the flow is observed (from 2015:Q3). On that window, the post-2017 price interaction is 0.058 ($p = 0.20$) without the control and -0.006 ($p = 0.92$) with it. Third, dropping the 2015:Q3–2016:Q2 reclassification quarters from the full sample reduces the interaction to 0.016 ($p = 0.70$). We conclude that the reclassification does not hide a post-2017 strengthening; if anything, it accounts for part of the small change measured around 2017.

Composition of credit. A complementary national check uses the investor share of housing credit from the D2 series: the investor stock divided by the investor plus owner-occupier stock. The share is essentially flat across 2017 (0.338 before, 0.335 after), so the national composition of credit did not tilt towards investors after 2017. Adding the growth of the national share to the main price equation leaves the post-2017 interaction small and insignificant (0.043, $p = 0.31$).

Inference with five clusters. All our standard errors, including those of the instrumented local projections, inherit the limitation of a five-city cross-section. A wild cluster bootstrap with city-level six-point (Webb) weights, suited to so few clusters, gives $p = 0.02$ for the price–credit slope of Table 4 and $p = 0.11$ for the post-2017 price interaction. Both agree with the Driscoll–Kraay inference: the slope is significant, and its change after 2017 is not. With only five clusters, the number of distinct bootstrap statistics is limited, so we read these results as support, not as independent precise figures. The argument rests on the lagged-credit estimate and the local projections, which do not depend on cross-sectional resampling, not on bootstrap significance.

Construction of the tightness measure. The tightness result in the rent equation does not depend on our preferred construction of the ratio. Replacing total net migration with net overseas migration in the numerator leaves the coefficient essentially unchanged (0.0054 against 0.0058), and an unsmoothed single-quarter ratio gives a similar positive and significant coefficient (0.0043). The single-quarter ratio rejects a unit root under both lag rules (Table E.1), so the result does not depend on treating the smoothed ratio as stationary. The coefficient becomes insignificant only when the ratio is differenced (0.0018, $t = 0.63$), which removes the slow-moving variation in tightness that the levels specification uses.

Migration composition. A natural question is whether the tightness–rent relationship works through rental demand specifically. If so, it should strengthen when more of the migrant inflow rents. Using net overseas migration by visa group, we form the temporary-visa and student-visa shares of net migration in each city as trailing four-quarter ratios. We exclude the 2020–2021 border-closure quarters, when total net migration passes through zero and the ratio is undefined. The share varies nearly as much within a city over time as across the panel (the within-city standard deviation is about 0.96 of the overall), so the test has identifying variation. We find no composition effect. The interaction of lagged tightness with the student share is essentially zero (0.001, $p = 0.84$), the temporary-share version is also null, and the same interaction in the price equation, a placebo, is null too. Tightness drives rents, but the share of renters in the inflow does not measurably amplify that relationship. We report this as a null, not as evidence for a composition channel.

Nonlinearity in the tightness–rent relationship. We test whether the rent response to tightness departs from linearity, for example by passing through faster in already tight markets. A natural cubic spline in lagged tightness mildly rejects linearity in the rent equation ($\chi^2_2 = 6.7$, $p = 0.04$), but the departure is not the convex form that a rental-pressure story predicts. Splitting the tightness slope at each city's own median gives an additional high-regime slope that is small, insignificant, and of the wrong sign, and the corresponding split in the price equation is the larger of the two. We therefore keep the linear tightness term in the main specification and make no claim of convexity; the nonlinearity the spline detects is minor and has no economic interpretation. As elsewhere, these inferences inherit the five-cluster limitation discussed above.

Local-projection lag length. The delayed rent response is robust to the number of control lags in the local projections, though the horizon from which it is significant is not. Across two, four, and six lags the twelfth-quarter rent response is stable at roughly 0.018 to 0.020 log points, and the price response leads it at every horizon. The horizon from which the rent response stays significant, by contrast, is the fourth quarter with four lags, the first with six, and the seventh with two, which is why the main text states that timing as approximate.

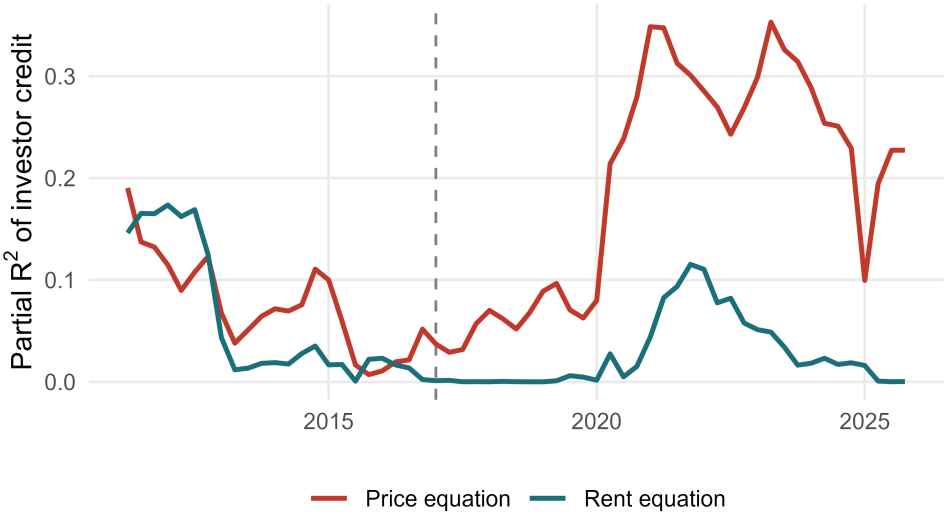


Figure E.2: Current-quarter partial R^2 of investor credit in the price and rent equations, 20-quarter rolling window, seasonally adjusted credit series. A descriptive complement to the local-projection evidence in Figure 5. The dashed line marks 2017:Q1.

Table E.1: Unit-root tests for the estimation variables

		Lags chosen by AIC				Lags chosen by SIC			
		Level		Difference		Level		Difference	
Variable	Trend	W	Cities	W	Cities	W	Cities	W	Cities
Log rent	Yes	-3.77^{***}	3	-0.90	0	-3.28^{***}	2	-1.19	0
Log dwelling price	Yes	3.43	0	-6.02^{***}	4	3.40	0	-8.71^{***}	4
Log wages (WPI)	Yes	-0.80	0	-1.85^{**}	0	-0.80	0	-1.75^{**}	0
Log employment	Yes	1.02	0	-17.07^{***}	5	0.87	0	-19.90^{***}	5
Log investor credit	Yes	0.30	0	-9.11^{***}	5	0.02	1	-10.83^{***}	5
Investor share	No	-1.29^{*}	0	-11.56^{***}	5	-1.84^{**}	0	-11.54^{***}	5
Cash rate[a]	No	-2.11	–	-5.06^{***}	–	-2.11	–	-5.06^{***}	–
Tightness	No	-1.09	0	-10.38^{***}	5	-3.89^{***}	3	-10.38^{***}	5
Tightness, one quarter[b]	No	-4.53^{***}	3	-5.04^{***}	4	-4.53^{***}	3	-5.52^{***}	5
Price–rent gap	No	3.78	0	-8.30^{***}	5	3.50	0	-9.87^{***}	5

Notes: W is the panel statistic of Im et al. (2003). It is standard normal under the null that every city's series has a unit root, and large negative values reject. Cities is the number of the five cities whose own ADF test rejects a unit root at the 5 percent level. Every test includes a constant, and the level tests of the five log series also include a linear trend, as the Trend column shows. Lag orders run from 0 to 4 and are chosen city by city, by the Akaike (AIC) or the Schwarz (SIC) criterion. The series run from 2006:Q3 to 2025:Q4, except tightness, from 2007:Q2 to 2025:Q3, and the one-quarter ratio, from 2006:Q3 to 2025:Q3. Investor credit and the investor share are seasonally adjusted. The W test assumes that the cities are independent. National shocks correlate the five series, which makes the test reject too often, so the city counts are the safer guide. [a]One national series: the row reports its ADF t statistic. [b]Net migration over dwelling completions within the quarter, the unsmoothed ratio used in the tightness checks of this appendix. $^{*}p < 0.10$, $^{**}p < 0.05$, $^{***}p < 0.01$.

Appendix F Rents on New Leases

This appendix describes our indices of rents on new leases and uses them to test the model's account of measured rents. In the model the market rent r_t^* is the rent on a lease signed in

quarter t (Equation (2)), while the CPI rent index measures rents on all current leases, the measured rent r_t of Equation (5). Rents on new bonds come as close to r_t^* as published data allow.

Data. When a tenancy starts, the landlord lodges the tenant's bond with a state authority, and the bond records the weekly rent. Four states publish quarterly median rents of the bonds lodged, by dwelling type and number of bedrooms. For Sydney we use the Rent and Sales Report of the NSW Department of Communities and Justice: median rents for Greater Sydney by number of bedrooms (one, two, three, and four or more), from its time series for March 1990 to June 2017 and from its quarterly rent tables after that (NSW Department of Communities and Justice, 2026). For Melbourne we use Homes Victoria's quarterly median rents for metropolitan Melbourne, for flats with one to three bedrooms and houses with two to four (Homes Victoria, 2026). For Brisbane we use the median rents that Queensland's Residential Tenancies Authority publishes for the five local government areas of Greater Brisbane, by dwelling type and bedrooms, from 2012 (Residential Tenancies Authority, 2026). For Adelaide we use the Private Rent Report of the South Australian Housing Trust for metropolitan Adelaide, for flats and houses with one to four or more bedrooms, from 2008 (South Australian Housing Trust, 2026). Western Australia publishes no comparable series, so Perth is excluded.

Each index chains the quarterly change in the median rent of each cell of dwelling type and bedrooms, weighting the cells by their shares of new bonds over the sample. For Sydney the weights come from the bonds lodged from September 2017, the first quarter with counts by bedrooms. Fixed weights keep shifts in the mix of dwellings let from moving the index, but within a cell the median still moves with the quality and location of the dwellings let. We remove a fixed seasonal pattern by subtracting each quarter's average change and adding back the overall average. Two checks support the indices. The annual change of the Melbourne index has a correlation of 0.98 with that of Homes Victoria's own metropolitan rent index, and for Sydney, weighting the four cells equally changes annual growth by 0.4 percentage points on average. The tests below use the quarters in which both CPI rents and rents on new leases are available: from 2007:Q1 in Sydney and Melbourne, 2008:Q4 in Adelaide and 2012:Q3 in Brisbane, to 2025:Q4. Standard errors are Driscoll–Kraay, as in the main text.

Market rents follow their drivers at once. Table F.2 estimates the static and dynamic rent equations, (19) and (20), on the same city-quarters for CPI rents and for rents on new leases. Both load on tightness, wage growth and employment growth. Their long-run coefficients on tightness are equal, 0.005, and those on wages are close, 0.57 and 0.53; the long-run employment coefficient is larger for rents on new leases. The price–rent gap enters neither significantly. The clearest difference is persistence. Growth in CPI rents carries over from one quarter to the next, with a coefficient of 0.39, while growth in rents on new leases does not (0.08, s.e. 0.09). This is the pattern the model predicts: the market rent responds to its drivers at once, and measured rents follow it as leases reprice.

How fast CPI rents follow. Table F.1 estimates the repricing rate ω in several ways. The first group imposes Equation (5) exactly, so that CPI rents absorb rents on new leases one for one in the long run. The growth form (17), with new-lease growth for Δr_t^*, gives 0.15 (s.e. 0.04), and between 0.12 and 0.25 city by city. Four-quarter changes, which damp quarterly noise, give 0.12. Matching the response of CPI rents to a credit innovation with Equation (5) applied to the response of rents on new leases (Figure 6, panel (b)) gives 0.12, with a 90 percent bootstrap interval of 0.03 to 0.21; $\omega = 0.35$ predicts a CPI response that is too large from the second year on, and it fits worse than $\omega = 0.15$ in 96 percent of the bootstrap draws. The timing of the two series agrees. Annual CPI rent growth is most closely correlated with annual new-lease growth two to five quarters earlier (three in Sydney and Melbourne, two in Brisbane and five in Adelaide), while Equation (5) implies one quarter at $\omega = 0.35$ and three or four at $\omega = 0.15$.

Two features of the data qualify these estimates. First, quarterly medians are noisy, which biases the growth-form estimates towards zero. Instrumenting new-lease growth with lagged CPI

rent growth, lagged tightness, and wage and employment growth, the drivers of the market rent in Equation (15), raises the estimate to 0.44 (s.e. 0.07). The instruments are only moderately strong, however (first-stage $F = 9.8$), and the overidentifying restrictions are rejected at the 10 percent level ($p = 0.08$). Second, CPI rents absorb only part of the movement in rents on new leases. Without the restriction, the instrumented coefficients on lagged CPI rent growth and on new-lease growth sum to 0.79 (s.e. 0.07), and a sum of one is rejected ($p = 0.002$). In log levels, CPI rents outside Sydney move about 0.78 times as much as the repricing path of rents on new leases. Medians of new bonds can move more than the rents in the CPI sample, because the dwellings let in a tight market differ from those let in a slack one. The CPI also covers public housing, whose rents the ABS estimates from the records of state housing authorities (Reserve Bank of Australia, 2019), and it is measured net of Rent Assistance (Section 4.1).

Once the absorbed share is estimated as well, the evidence on ω divides. The fit in log levels, whose slope is free, is still best at 0.11: at 0.11 in Sydney and Melbourne, 0.23 in Brisbane and 0.08 in Adelaide, with slopes of 1.07, 0.79, 0.76 and 0.77. But the coefficient on lagged CPI rent growth in the unrestricted instrumented regression, which identifies $1-\omega$ whatever the absorbed share, implies 0.58 (s.e. 0.08). The responses to a credit innovation fit almost equally well for any ω between 0.12 and 0.35, with the absorbed share falling from about one at 0.12 to 0.58 at 0.35, and the bootstrap interval for the best-fitting value runs from 0.02 to 0.93. On these four cities the dynamic rent equation itself gives 0.61 (s.e. 0.17), against 0.35 on the five-city panel. The timing of CPI rents behind rents on new leases points to slower repricing than in Table 5, while the persistence of CPI rent growth points to faster repricing. The data on new leases therefore confirm that leases delay measured rents, but they do not narrow the interval for ω.

Credit reaches rents on new leases first. Figure 6, panel (a), estimates Equation (22) for prices, rents on new leases and CPI rents on the same city-quarters. Prices take up most of their response within three quarters. Rents on new leases rise from the first quarter after the innovation, are significant at the 10 percent level from then on, and reach about 85 percent of the price response within two years: 0.024 against 0.029 log points at eight quarters, and 0.030 against 0.035 at twelve. CPI rents rise more slowly and are significant only from the sixth quarter. At twelve quarters they are 0.017, about 57 percent of the response of rents on new leases. In the block bootstrap, the response of rents on new leases exceeds that of CPI rents in 98, 95 and 94 percent of the draws at four, eight and twelve quarters, while the gap between prices and rents on new leases is significant only at four quarters. Most of the gap between prices and CPI rents after three years therefore lies between rents on new and on current leases: credit reaches the rents that new tenants pay within about two years, well before it shows in CPI rents.

Table F.1: The lease-repricing rate estimated from rents on new leases

Method	ω	Uncertainty	City-quarters
CPI rents absorb rents on new leases one for one			
Growth form, OLS	0.149	(0.036)	275
Sydney	0.131	(0.033)	76
Melbourne	0.158	(0.039)	76
Brisbane	0.254	(0.088)	54
Adelaide	0.116	(0.043)	69
Four-quarter changes, OLS	0.116	(0.009)	263
Growth form, 2SLS	0.436	(0.070)	271
Responses to a credit innovation	0.12	[0.03, 0.21]	215–263
Share absorbed estimated			
Log levels	0.11		267
Growth form, 2SLS	0.580	(0.075)	271
Responses to a credit innovation	0.19	[0.02, 0.93]	215–263
Dynamic rent equation (20)*, CPI rents only*			
Five cities (Table 5)	0.35	[0.244, 0.658]	
These four cities	0.614	(0.167)	271

Notes: Sydney, Melbourne, Brisbane and Adelaide, quarters in which both CPI rents and rents on new leases exist, to 2025:Q4; city fixed effects in pooled regressions. Parentheses: Driscoll–Kraay standard errors (four lags; six for four-quarter changes) or, city by city, Newey–West standard errors (four lags). Brackets: 90 percent intervals, from 399 draws of a moving block bootstrap over quarters (blocks of eight, cities kept together) for the responses, and the bootstrap interval of Table 5. Growth form: Equation (17) with new-lease growth for Δr_t^*, with the two coefficients restricted to sum to one or, in the second group, unrestricted, when ω is one minus the coefficient on lagged CPI rent growth. 2SLS instruments new-lease growth with lagged tightness, wage growth and employment growth, and in the restricted form also with lagged CPI rent growth; first-stage F is 9.8 and 9.6, and Hansen's J has $p = 0.08$ and $p = 0.06$, in the restricted and unrestricted forms. Four-quarter changes: the restricted growth form in four-quarter changes. Log levels: log CPI rents regressed on the path of Equation (5) computed from the new-lease index, with a city constant and trend and a free slope, for ω on a grid; the first eight quarters of each path are discarded. Responses to a credit innovation: the ω at which Equation (5), applied to the response of rents on new leases, best reproduces the response of CPI rents at horizons 0–12 (Figure 6); in the second group the prediction is scaled by a free factor.

Table F.2: Rent equations for CPI rents and rents on new leases, four cities

	Static, Equation (19)		Dynamic, Equation (20)	
	CPI rents (1)	New leases (2)	CPI rents (3)	New leases (4)
Lagged rent growth			0.386**	0.078
			(0.167)	(0.090)
Price–rent gap$_{t-1}$	0.000	0.018	0.000	0.017
	(0.003)	(0.012)	(0.002)	(0.010)
Tightness$_{t-1}$	0.0055***	0.0056***	0.0032***	0.0050**
	(0.0007)	(0.0019)	(0.0012)	(0.0020)
Δ ln WPI	0.547***	0.584**	0.350***	0.490**
	(0.125)	(0.289)	(0.070)	(0.248)
Δ ln Employment	0.123***	0.425***	0.124***	0.419***
	(0.036)	(0.091)	(0.034)	(0.097)
Long run, $b_k/(1-\varphi_r)$				
Tightness$_{t-1}$			0.005	0.005
			(0.001)	(0.002)
Δ ln WPI			0.570	0.531
			(0.194)	(0.284)
Δ ln Employment			0.202	0.454
			(0.092)	(0.115)
Observations	273	273	271	271

Notes: Dependent variable: quarterly growth of log CPI rents, columns (1) and (3), or of log rents on new leases, columns (2) and (4). Sydney, Melbourne, Brisbane and Adelaide, quarters in which both series exist, to 2025:Q4. City fixed effects; Driscoll–Kraay standard errors in parentheses. Lagged rent growth is that of the dependent variable. Long-run coefficients are $b_k/(1-\varphi_r)$, with delta-method standard errors. Tightness coefficients are shown to four decimals in the upper panel. $^{*}p < 0.10$, $^{**}p < 0.05$, $^{***}p < 0.01$.